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Comparing Black–Scholes, Lévy and
stochastic volatility models for option
pricing on the KOSPI 200

by

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Abstract

It is well known that the two-parameter Black–Scholes model does not adequately describe stock or index price behaviour, and is unsuitable for modelling or determining real-world option prices. Many models have been proposed by Schoutens (2003) to address these shortcomings, among them Lévy process models and stochastic volatility models. This dissertation examines how these models perform against a highly volatile index, the South Korean Composite Stock Price Index (KOSPI), using 212 call options quoted on 30 July 2026. The Black–Scholes model prices these options with a root-mean-square error (RMSE) of approximately 12.36. In addition, the five Lévy models examined do not improve on it, with most calibrating to a normal distribution. Analysing implied volatility, we find that it decreases from 0.89 to 0.60 as the time to expiry T increases, so a model that allows volatility to vary across expiries is essential. Introducing stochastic volatility reduces the error to 1.77, and the five best-performing models have approximately the same RMSE. The Barndorff-Nielsen–Shephard Gamma–OU model is the most parsimonious.

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1 Introduction

The KOSPI (Korea Composite Stock Price Index) has attracted significant media attention in 2026. In the year from August 2025 to June 2026, the KOSPI returned 181%, before crashing 40% into July 2026. This led to one in thirty adults in South Korea receiving margin calls* (Reuters, 2026). During this period, it was not unusual for the KOSPI to record daily gains and losses of around 2.5 percentage points. Given this unique moment in the KOSPI’s history, we attempt to price European call options† on the KOSPI.

We use Schoutens (2003) as the main source for our models. He covers the well-known Black–Scholes model, which assumes log stock returns follow a normal distribution, and then a range of models with alternative distributions in which the stock price follows a Lévy process. These cover the Meixner, variance gamma, CGMY, normal inverse Gaussian, and generalised hyperbolic distributions. He also covers modifications of all these models in which volatility itself follows a stochastic process. We follow his methodology throughout, but the calibration of each model to the KOSPI 200 chain, and the analysis of the resulting fits, are our own. Our main goal is to find a model that accurately prices options on an index exhibiting very high volatility.

2 The Data

Before suggesting specific models to fit to the KOSPI index, we describe the dataset we use. The data consist of *European call options* written on the KOSPI 200 index, recorded at the market close of 30 July 2026 and retrieved from Interactive Brokers through their Trader Workstation API. On that date the index closed at a spot price of $S_0 = 872.49$, with a risk-free rate of $r = 2.91\%$ and a dividend yield of $q = 0.79\%$ per year. For r we take the Korea three-month interbank rate (FRED series IR3TIB01KRM156N). For q we use a proxy given by the trailing yield on the KODEX 200 exchange-traded fund (069500.KS).

The raw pull returned 2,284 contracts. After restricting to monthly expiries and to calls, and after discarding contracts without a two-sided quote or with a bid–ask spread wider than half the mid price, 212 contracts remain. The full filtering sequence is set out in Appendix B.1, and the contract-level prices are tabulated in Appendix B. Each contract is priced at its bid–ask midpoint,

$$C_{\text{mkt}} = \frac{1}{2} (\text{bid} + \text{ask}) .$$

The 212 contracts span seven expiry dates, from 13 August 2026 to 11 March 2027 and strikes from $K = 700$ to $K = 1045$. Time to maturity T is measured in calendar days

*A margin call occurs when an investor has borrowed money to purchase securities and the value of those securities falls below a required threshold. The broker then demands additional funds or collateral, and if these are not supplied, sells the position to recover the loan. Because forced selling depresses prices further, margin calls tend to amplify downward moves.

†A call option gives the holder the right, but not the obligation, to buy the underlying asset at a fixed strike price K on or before a specified maturity date T . The holder exercises only when it is profitable to do so, i.e. when the asset price exceeds K , giving the payoff $(S_T - K)^+$. A *European* call may be exercised only at maturity, whereas an *American* call may be exercised at any time up to it; the options considered in this dissertation are European.

from the pricing date to the expiry date, divided by 365, so the seven expiries are at 14, 42, 70, 105, 133, 168 and 224 days. The *front month* is August 2026, which has 116 of the 212 contracts. Figure 1 plots the whole chain, and Table 36, in Appendix B, records its coverage by expiry.

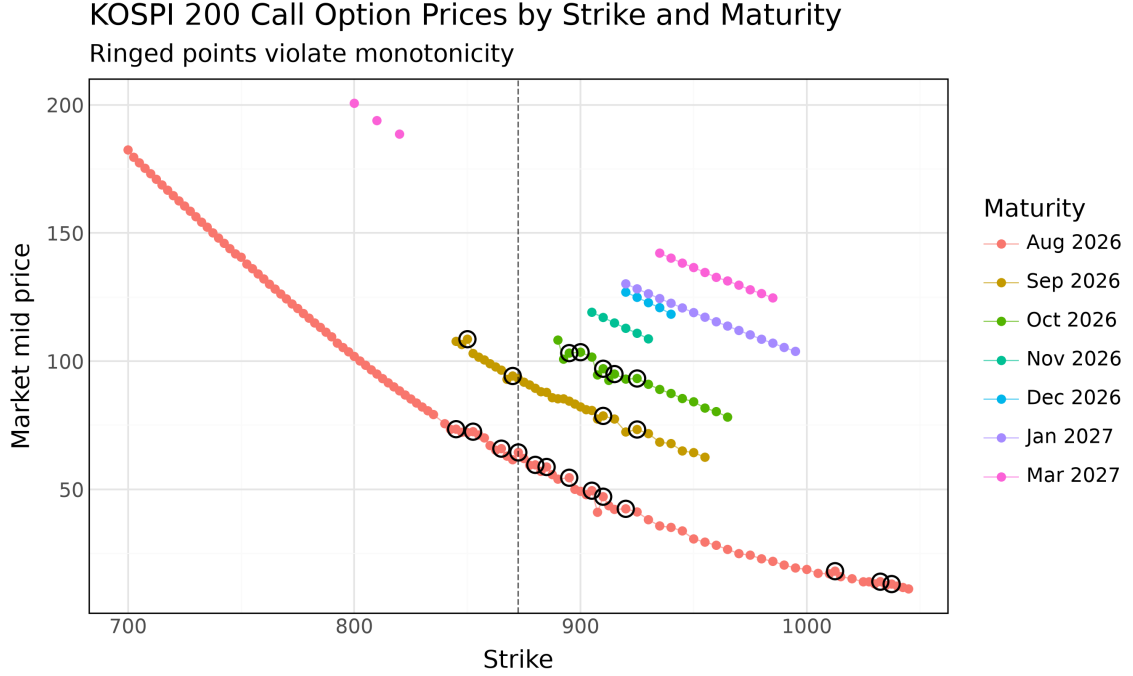


Figure 1: KOSPI 200 call option prices by strike and maturity, at the market close of 30 July 2026. Each colour denotes one expiry, and the dashed line marks the spot price $S_0 = 872.49$. Ringed points price above their next-lower strike.

Call prices should form a non-increasing curve in the strike, a condition for the absence of arbitrage. Formally, for two calls on the same index with the same maturity,

$$K_1 < K_2 \implies C(K_1, T) \geq C(K_2, T). \quad (1)$$

22 of the 212 contracts in our chain breach Equation (1), shown with rings in Figure 1. The largest breach is in the August 2026 expiry, between the calls struck at 907.5 and 910, whose quotes are given in Table 1.

Table 1: Quotes for the two August 2026 contracts forming the largest mid-price breach in the chain.

Strike	Bid	Ask	Mid	Spread	Spread (%)
907.5	35.35	46.75	41.05	11.40	27.8
910	43.30	50.95	47.13	7.65	16.2
Difference			+6.08		

In theory, we could buy the call struck at 907.5 and sell the call struck at 910 and collect $47.13 - 41.05 = 6.08$ immediately. At expiry the purchased contract pays $(S_T - 907.5)^+$

and the contract sold forces the seller to pay $(S_T - 910)^+$, so the position is worth

$$(S_T - 907.5)^+ - (S_T - 910)^+,$$

which is evaluated in Table 2.

Table 2: Value at expiry of a position long the 907.5 call and short the 910 call, for a range of terminal index levels.

S_T	Long 907.5	Short 910	Position value
900.00	0.00	0.00	0.00
907.50	0.00	0.00	0.00
908.75	1.25	0.00	1.25
910.00	2.50	0.00	2.50
920.00	12.50	-10.00	2.50
950.00	42.50	-40.00	2.50
1000.00	92.50	-90.00	2.50

This is a riskless profit, and no such quotation can persist in a functioning market. Our use of mid-prices makes it appear that this trade is available, when in fact it is not. Repricing the same trade on the basis of bid and ask prices gives Table 3. Using mid-prices can therefore be deceptive, which is worth bearing in mind as we proceed with our analysis.

Table 3: The same trade priced at the quotes actually available, against its appearance at mid prices.

Leg	Transacted at	At mid	At market
Buy the 907.5 call	its ask, 46.75	-41.05	-46.75
Sell the 910 call	its bid, 43.30	+47.13	+43.30
Net at inception		+6.08	-3.45

3 The Black–Scholes Model

3.1 Overview

This chapter examines the Black–Scholes model, the first framework in this dissertation capable of pricing options. Our first goal is to determine a stock price model for S_t , the stock price at time t . We begin with Brownian motion. A formal definition of Brownian motion is given in Definition 3.1.

Definition 3.1 (Standard Brownian Motion). *A stochastic process $X = \{X_t, t \geq 0\}$ is a standard Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ if*

- (i) $X_0 = 0$ a.s.,
- (ii) X has independent increments,

(iii) X has stationary increments,

(iv) $X_{t+s} - X_t$ is Normally distributed with mean 0 and variance $s > 0$:

$$X_{t+s} - X_t \sim \text{Normal}(0, s).$$

We denote standard Brownian motion by $W = \{W_t, t \geq 0\}$.

Definition 3.1 follows Schoutens (2003, §3.2.1) in placing no condition on the paths of W . Continuity of those paths is not implied by (i)–(iv), and is instead established separately, in his §3.2.2; we take it as given throughout. Definition 3.1 requires that a Brownian motion begin at $W_0 = 0$, which in its present form is unsuitable for simulating stock prices. Instead, we require a process that begins at S_0 , where $S_0 > 0$ denotes the spot price. Schoutens (2003, §3.3) proposes geometric Brownian motion as an alternative, see Definition 3.2.

Definition 3.2 (Geometric Brownian Motion). *Let $W = \{W_t, t \geq 0\}$ be a standard Brownian motion, let $S_0 > 0$, and let $\mu \in \mathbb{R}$ and $\sigma > 0$. The process $S = \{S_t, t \geq 0\}$ given by*

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right) \quad (2)$$

is called a geometric Brownian motion with drift μ and volatility σ .

Equation (2) is the unique solution of the stochastic differential equation

$$dS_t = S_t (\mu dt + \sigma dW_t), \quad S_0 > 0,$$

which is the form in which the model is usually motivated.

Two properties follow immediately. First, $S_t > 0$ for all t , since the exponential is positive, so the model cannot produce a negative price. Second, taking logarithms in (2),

$$\log S_t - \log S_0 = \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \sim \mathcal{N} \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t, \sigma^2 t \right),$$

so that S_t is log-normally distributed. Geometric Brownian motion has given us a stock price model; the Black–Scholes model extends this by building an option price model. The derivative price is denoted

$$V_0 = e^{-rT} \mathbb{E}_{\mathbb{Q}}((S_T - K)^+),$$

where K is the strike price, r the risk-free rate, and T the maturity. To take this expectation, we could do so over the probability space $\mathbb{P}$, but this leads to an equation that depends on μ , a highly subjective quantity that could take many values when we require an option price with a single value. Instead, we construct a new probability space over $\mathbb{Q}$ that is an equivalent martingale measure*. Using the literature provided, and as directed by Schoutens (2003, §3.4), we arrive at a process of the form

$$S_t = S_0 \exp \left(\left(r - q - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right). \quad (3)$$

*Schoutens (2003, §2.5.1) defines a probability measure to be an equivalent martingale measure if:

1. $\mathbb{Q}$ and the probability measure $\mathbb{P}$ have the same null sets, and
2. the discounted stock process $\tilde{S} = \{\tilde{S}_t = \exp(-rt)S_t, t \geq 0\}$ is a martingale under $\mathbb{Q}$.

Equation (3) is Equation (2) with the subjective drift μ replaced by the observable $r - q$, and σ reflecting volatility.

3.2 The Model

Schoutens (2003) presents and partially derives the closed-form solution for pricing a call under the Black–Scholes model, originally derived by Black and Scholes (1973). We do not reproduce the derivation, as this dissertation is primarily concerned with model fitting. The resulting price, for a European call with strike K and maturity T and with σ unknown, is given in Equation 4.

$$V_0 = C_{BS}(K, T; \sigma) = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \quad (4)$$

where N denotes the standard normal distribution function and

$$d_1 = \frac{\log(S_0/K) + (r - q + \frac{1}{2}\sigma^2) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The single free parameter is σ , which we estimate by minimising the root-mean-square error (RMSE), as shown in Equation 5.

$$\hat{\sigma} = \arg \min \sqrt{\frac{1}{n} \sum_{i=1}^n (C_{BS}(K_i, T_i; \sigma) - C_{\text{mkt},i})^2}, \quad (5)$$

where $C_{BS}(K_i, T_i; \sigma)$ is evaluated at contract i 's own strike and maturity, and $C_{\text{mkt},i}$ is its market mid price. In solving this minimisation, we use the Newton–Raphson algorithm.

3.3 Model Validation

Prior to applying this model to our KOSPI data, we first applied the Black–Scholes model to Schoutens' dataset to check whether we obtained the same results, purely as a means of model validation. Schoutens' dataset is given in Appendix A (Table 34). The parameters from our own results using Schoutens' data, alongside Schoutens' own, are given in Table 4. We obtain the same results, and can therefore trust that our model works.

The table reports the calibrated $\hat{\sigma}$ together with four measures of fit, and which we report for every model from here onwards. Write $C_{\text{mkt},i}$ for the market mid price of contract i , $C_{\text{model},i}$ for the price the calibrated model assigns to that contract, n for the number of contracts in the chain, and $\bar{C}_{\text{mkt}} = n^{-1} \sum_{i=1}^n C_{\text{mkt},i}$ for the mean market price. The four measures are then

$$\begin{aligned} \text{APE} &= \frac{1}{\bar{C}_{\text{mkt}}} \frac{1}{n} \sum_{i=1}^n |C_{\text{mkt},i} - C_{\text{model},i}|, & \text{AAE} &= \frac{1}{n} \sum_{i=1}^n |C_{\text{mkt},i} - C_{\text{model},i}|, \\ \text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (C_{\text{mkt},i} - C_{\text{model},i})^2}, & \text{ARPE} &= \frac{1}{n} \sum_{i=1}^n \frac{|C_{\text{mkt},i} - C_{\text{model},i}|}{C_{\text{mkt},i}}. \end{aligned} \quad (6)$$

Table 4: Black–Scholes calibration to the S&P 500 (2002) chain, comparing our fit against the published results of Schoutens (2003, §4.3, Table 4.3). Agreement is close on all five quantities.

	Ours	Schoutens (2003)	Difference
$\hat{\sigma}$	0.1824	0.1812	+0.0012
APE (%)	8.72	8.87	−0.15
AAE	5.3799	5.4868	−0.1069
RMSE	6.7206	6.7335	−0.0129
ARPE (%)	16.58	16.92	−0.34

APE, AAE, RMSE and ARPE are defined in Equation (6).

3.4 Results

We fitted the Black–Scholes model on our KOSPI data, obtaining results shown in Figure 2, which shows the fitted values against the data, and in Table 5, which quantifies the error. In Figure 2, the $\circ$ denotes the market price and the $+$ denotes the model’s output. There is clearly evidence of mispricing, as the $+$ and $\circ$ do not overlap: the model price falls below the market price for near-term contracts and above it for long-term contracts.

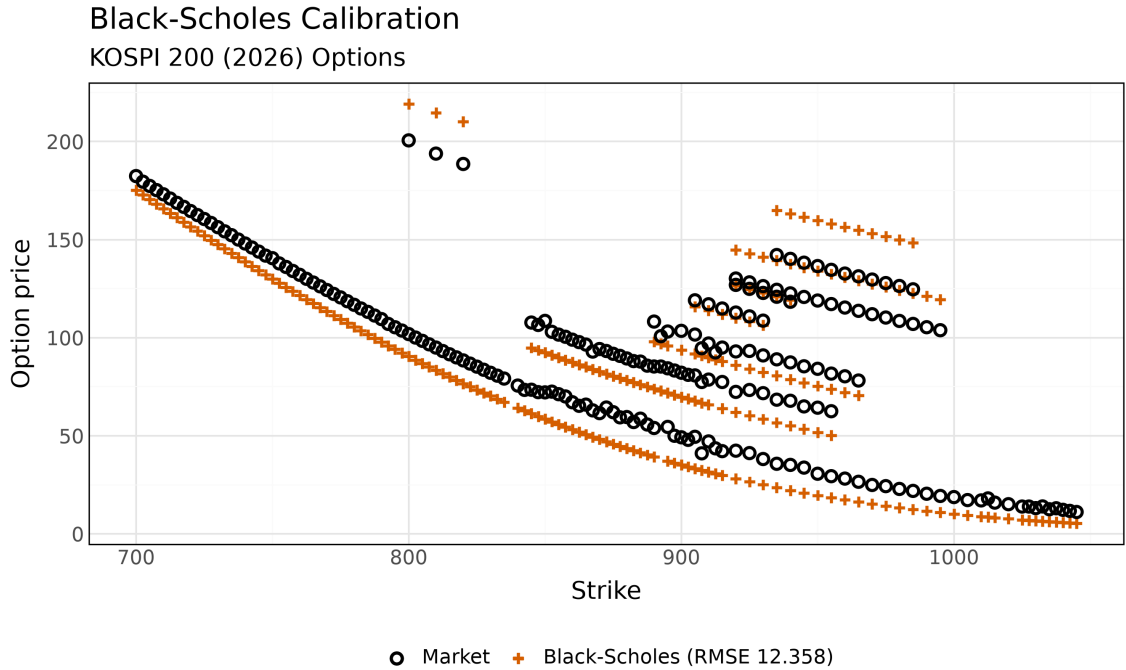


Figure 2: Black–Scholes results, with σ calibrated by minimising RMSE against market prices. Circles denote market prices, pluses denote model prices.

We obtain a $\hat{\sigma} = 0.68$. This is far higher than Schoutens’ own estimate of 0.18 on his data. This means that we estimate the KOSPI stock price to fluctuate considerably more over the life of an option. We initially obtained a much higher σ , but managed to reduce this

by removing options with limited liquidity. A major part of the remaining discrepancy is simply that the KOSPI is a considerably more volatile index, and has been particularly volatile in recent periods.

Table 5: Black–Scholes calibration to the KOSPI 200 (2026) option chain, with the error broken down by expiry.

	n	APE (%)	AAE	RMSE	ARPE (%)
$\hat{\sigma} = 0.6851$					
Whole chain	212	12.43	11.4802	12.3578	17.48
Aug 26	116	13.40	10.9892	11.3353	22.24
Sep 26	36	13.96	12.0098	12.0649	14.29
Oct 26	19	8.25	7.6430	7.8762	8.32
Nov 26	6	2.49	2.8413	2.8591	2.49
Dec 26	5	0.29	0.3582	0.4815	0.30
Jan 27	16	13.09	15.2611	15.2653	13.18
Mar 27	14	15.56	22.7474	22.7931	16.09

APE, AAE, RMSE and ARPE are defined in Equation (6).

We examine the errors more closely in Figure 3, which plots the error against log moneyness, defined as

$$\ln \left(\frac{K}{F} \right), \quad K = \text{strike price}, \quad F = S_0 e^{(r-q)T}.$$

We use log moneyness rather than the strike price on the x -axis to make the results comparable across maturities in a single chart. This chart confirms that, as maturity increases, the difference between the Black–Scholes fitted price and the market price shifts from a negative error, through no error around December 2026, to a positive error at the later maturities. This error dynamic likely reflects the model attempting to accommodate a single σ across all maturities when the maturities do not, in fact, share a common σ . We check this in the following section.

3.5 Non-Constant Volatility

We check the assumption of a constant σ across maturities by finding the optimal $\hat{\sigma}$ for a single call, and comparing it across other calls. To do this we define the implied volatility σ for a given contract as the value of σ such that the model price equals the market price, i.e.

$$C_{BS}(\sigma_{\text{imp}}) = C_{\text{mkt}}. \quad (7)$$

There is no closed-form solution to Equation (7). We therefore solve it numerically; we do so via the Newton–Raphson algorithm. The exact process for this specific task is set out in Appendix E.

As a starting value for this algorithm, Schoutens (2003, §4.3) recommends $\sigma_0 = 0.20$, described as performing “very well for most common stocks and indices” (Schoutens,

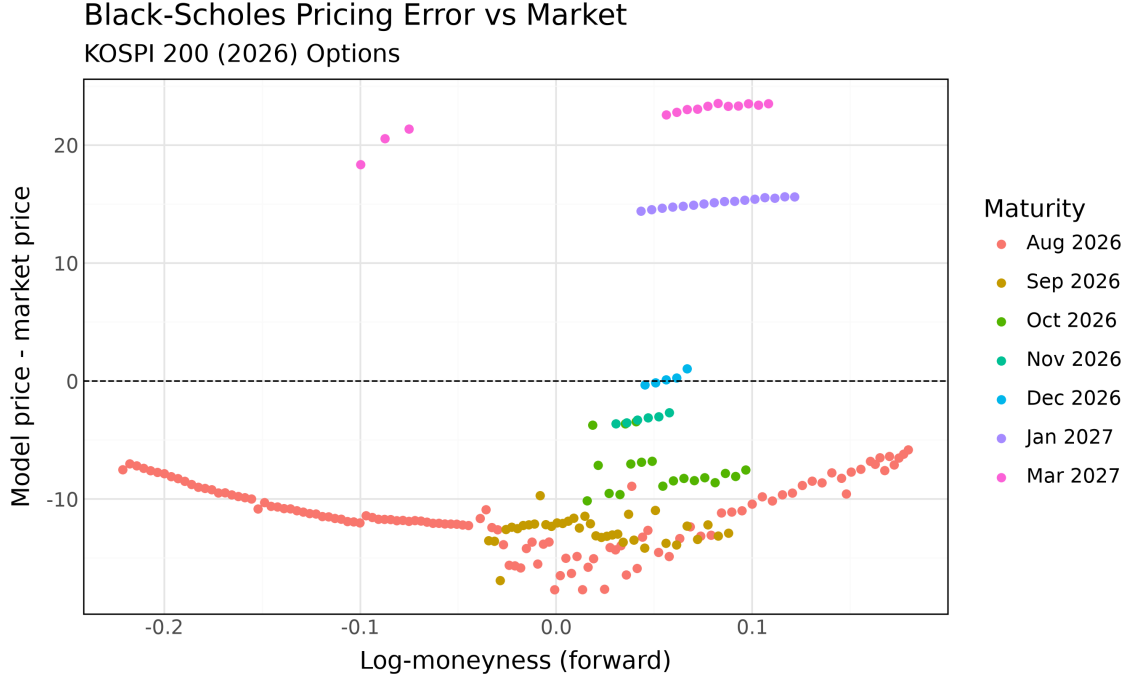


Figure 3: Black–Scholes pricing error against market prices, plotted by log-moneyness and maturity, at the calibrated σ .

2003, p. 41). On the KOSPI 200 chain it does not perform well at all. Of the 212 contracts, 62 (29.2%) fail to converge, and *every one* lies in the $T = 0.0384$ (14-day) front month, while all six longer maturities converge in full. We resolved this issue by selecting a starting value that departed from the literature. We calculated the annualised volatility specifically for the KOSPI 200 year to date, obtaining a value of 0.651, rounded to 0.66, and used this value as a starting point. Annualised volatility is calculated as $\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{N}$, given in full by Equation (8).

$$\hat{\sigma}_{\text{ann}} = \sqrt{252} \cdot \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}, \quad r_t = \ln\left(\frac{S_t}{S_{t-1}}\right). \quad (8)$$

Table 7 reports annualised volatility across different periods for the KOSPI.

Having fitted the implied volatility for each contract, we show the results in Figure 4. The figure shows that, across strikes, implied volatility is roughly constant within six of the seven maturities (from September onwards), albeit with some jaggedness. Across maturities, however, σ is far from constant, with different maturity dates having very different implied volatility. Specifically, implied volatility declines steadily as maturity moves further out. This is a breakdown of the Black–Scholes assumption of a single constant volatility for every contract.

This is an *inverted volatility term structure*. Under the original data of Schoutens (2003, p. 42), implied volatility increases with time to maturity, whereas our KOSPI data breaks the constant volatility assumption in the opposite manner, with implied volatility decreasing as time to maturity increases. This is likely because volatility is currently above its

long-run mean, so that it follows a mean-reverting process (Stein, 1989), reflecting the unusually volatile nature of the current market.

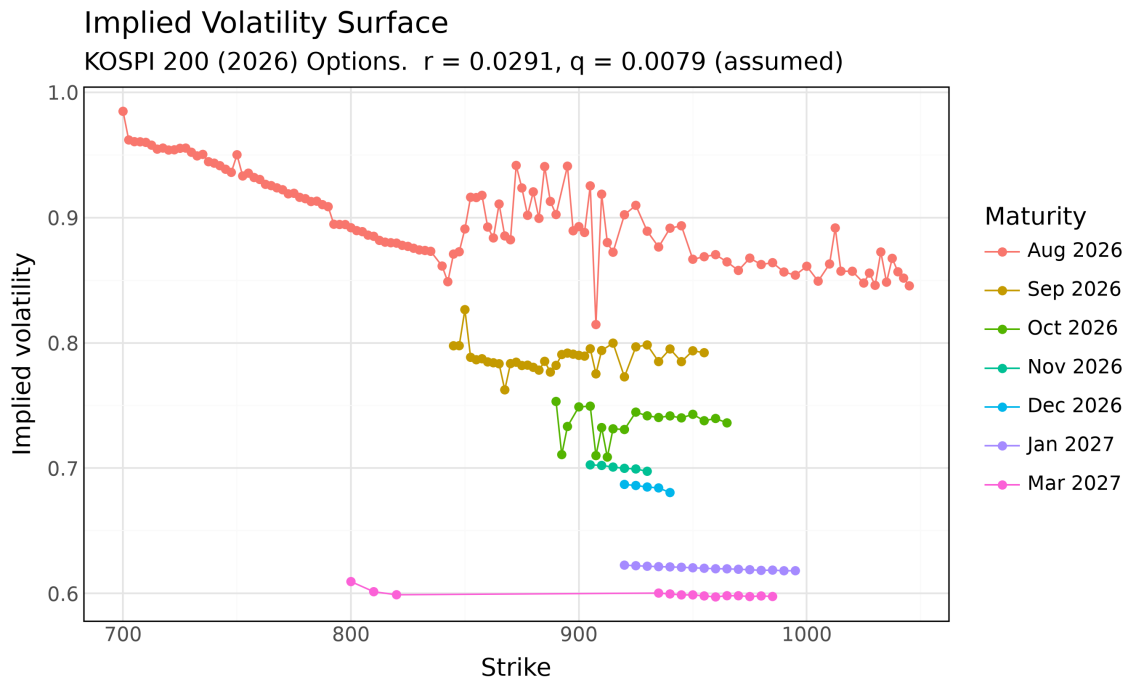


Figure 4: Implied volatility surface for call options. If the Black–Scholes model held, all points within each panel would lie on a single horizontal line.

Since implied volatility is flat across strikes for most maturities, the Black–Scholes model should fit well when calibrated separately to each maturity. We fit a separate σ per maturity and report the results in Table 6. There is substantial improvement in fit across all maturities. The highest errors are in August and October 2026, but these are still far lower than those of the single-volatility model calibrated across the whole chain. Figure 5 shows the fit, reinforcing this marked improvement. Model prices are much closer to market prices, and coincide with them in many cases. Figure 6 examines the error for each per-maturity model. The errors show a clear level shift, with the mean error across maturities reduced close to zero. The pattern of the errors, however, remains very similar to that of the original model.

Table 6: Black–Scholes calibrated separately to each expiry of the KOSPI 200 (2026) option chain, against the single- σ calibration of Table 5.

Expiry	n	$\hat{\sigma}$	APE (%)	AAE	RMSE	ARPE (%)
Aug 26	116	0.8941	1.78	1.4616	1.6811	3.48
Sep 26	36	0.7880	0.99	0.8512	1.1849	0.99
Oct 26	19	0.7355	1.57	1.4532	1.9069	1.52
Nov 26	6	0.7004	0.24	0.2777	0.3229	0.24
Dec 26	5	0.6846	0.30	0.3678	0.4686	0.30
Jan 27	16	0.6201	0.25	0.2872	0.3302	0.25
Mar 27	14	0.5993	0.32	0.4636	0.7473	0.30
Pooled, seven $\hat{\sigma}$	212		1.24	1.1433	1.4710	2.26
Single $\hat{\sigma}$, whole chain	212	0.6851	12.43	11.4802	12.3578	17.48

APE, AAE, RMSE and ARPE are defined in Equation (6).

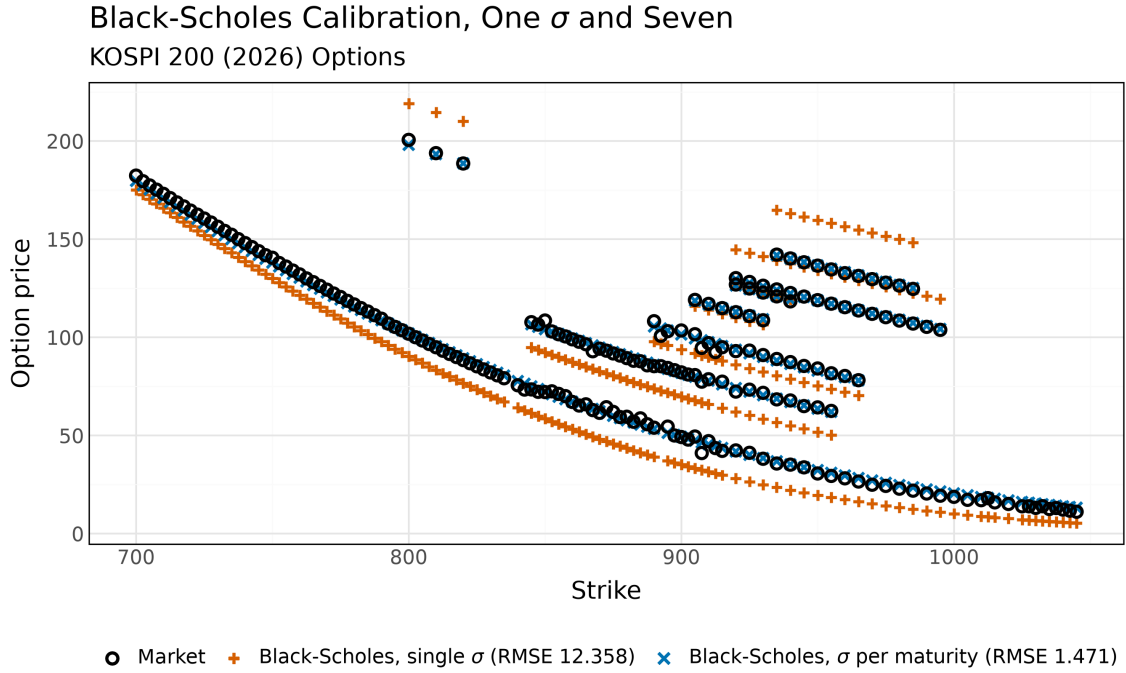


Figure 5: Market prices against two Black–Scholes calibrations on the KOSPI 200 (2026) chain.

Furthermore, Figure 7 shows the impact on implied volatility. Instead of a constant implied volatility, it now moves down with each maturity, reflecting the implied volatility levels we calculated across all maturities in Figure 4.

While this worked very well, it is not a fully appropriate fix. Our goal is to determine a single model capable of replicating the behaviour of option prices across the whole KOSPI 200 chain, whereas the per-maturity approach is really seven separate calibrations, one for each expiry. It cannot price a maturity we have not already fitted, tells us nothing

Black-Scholes Pricing Errors by Expiry

KOSPI 200 (2026) Options

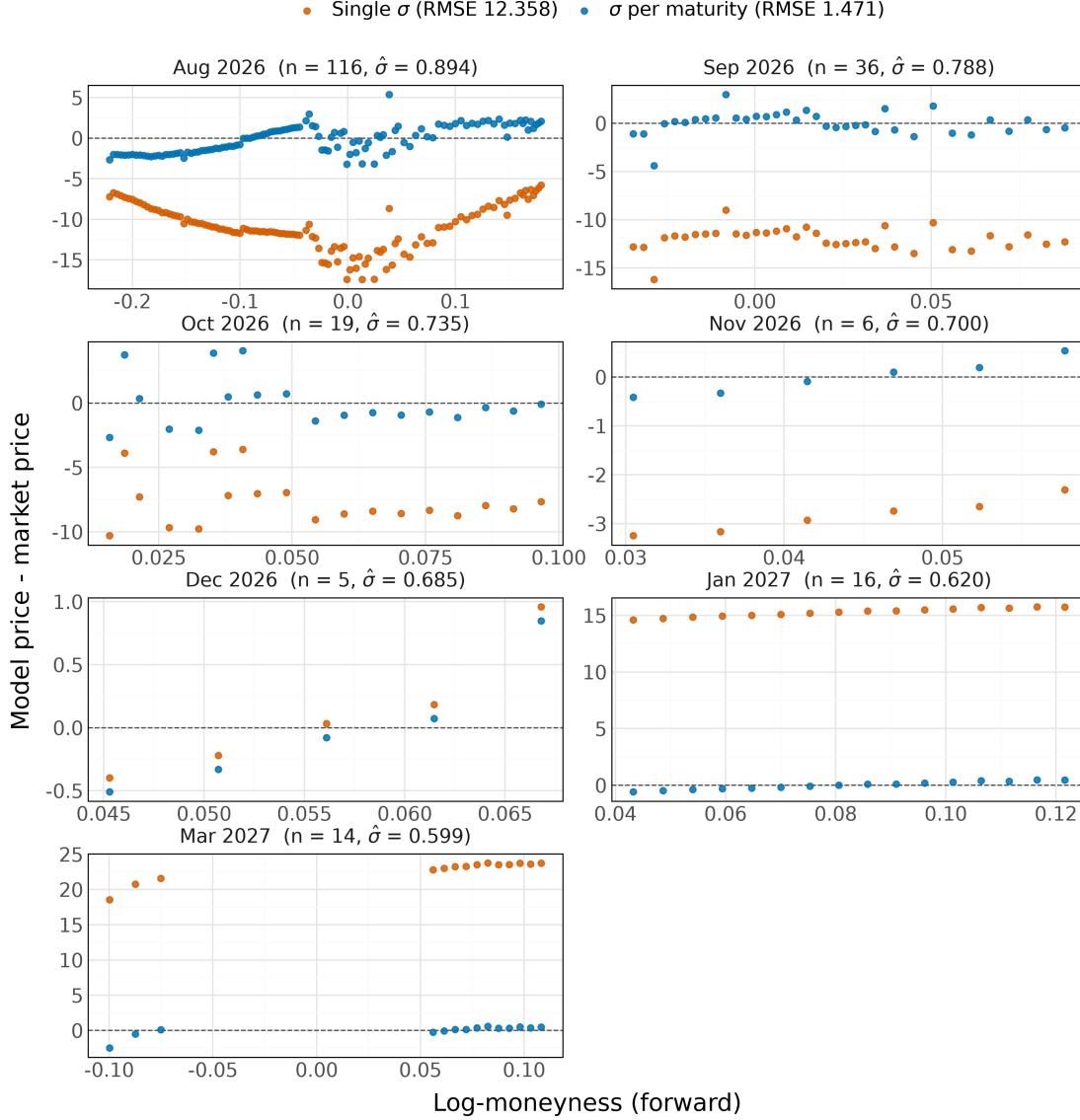


Figure 6: Pricing errors of the two Black-Scholes calibrations, plotted against forward log-moneyness and split by expiry, on the KOSPI 200 (2026) chain. Each panel carries its own vertical scale.

about how volatility behaves between expiries, and leaves us with seven volatility parameters in place of one. The most appropriate next step would be to investigate stochastic volatility models, in which the volatility parameter itself varies over time. Before doing so, however, we must first check whether the normality assumption is appropriate.

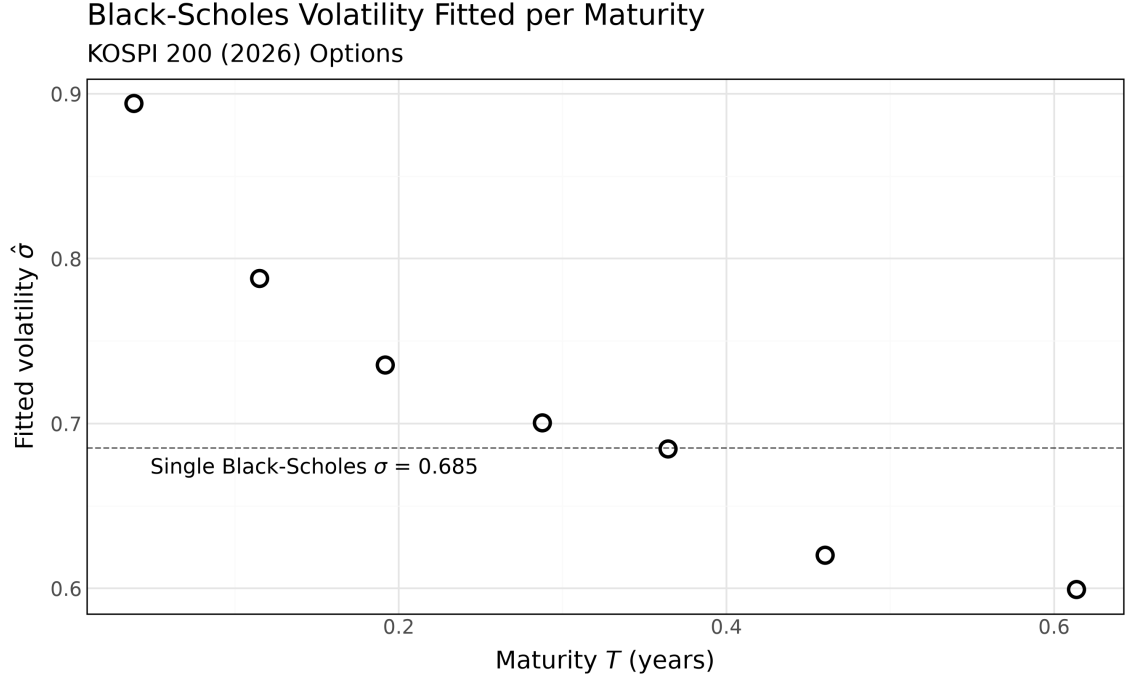


Figure 7: Black–Scholes volatility fitted separately to each expiry of the KOSPI 200 (2026) chain, against the single volatility fitted across all 212 contracts at once.

3.6 Checking Normality Assumption

When deriving the log-normality of the stock price S_t , we determined that

$$\log S_t - \log S_0 = \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \sim \mathcal{N} \left(t \left(\mu - \frac{1}{2}\sigma^2 \right), \sigma^2 t \right).$$

Setting $t = 1$, this implies that log returns over a single period are assumed to be normally distributed. This is something we can check using our chosen stock price index, the KOSPI. Table 7 shows the mean, standard deviation, annualised volatility, skewness, and kurtosis of log returns for the KOSPI across different timeframes. All timeframes examined exhibit kurtosis greater than 3, whereas under normality kurtosis should be 3. Skewness under normality should also be zero: the 2026 data exhibits clear skewness, the 2016–2025 and 2021–2025 samples exhibit mild skewness, and the 1996–2025 sample has minimal skewness.

We also plot the density of log returns using the 1996–2025 data and compare it graphically with a normal distribution, as shown in Figures 8 and 9. The density curve clearly exhibits non-normality, which is especially evident on the log scale in Figure 9, capturing the greater kurtosis. This greater kurtosis implies that rare events are more likely than a normal-distribution model would suggest. More formal tests could be run to check non-normality explicitly, but we avoid doing so, since it is graphically intuitive that log returns are not normally distributed.

Our period of analysis in these figures runs from 1996 to 2025; the departure from normality would have been more pronounced had this year’s data been included, as the KOSPI

has exhibited unusual volatility in 2026. This is evident in Table 7, which shows annualised volatility reaching 65.1%, compared with a historical average of 27.1%.

Table 7: Empirical mean, standard deviation, skewness and kurtosis of daily log returns for a set of indices. Returns are computed from unadjusted daily closing levels retrieved from Yahoo Finance with the `yfinance` package. The summary statistics are the empirical moment estimators applied to the daily log returns $r_t = \ln(S_t/S_{t-1})$; they are defined in Appendix D.6.

Index	n	Mean [¶]	SD [¶]	Annualised Volatility [§]	Skewness [¶]	Kurtosis [¶]
KOSPI 200 (2026 YTD)	132	0.0034	0.0418	0.651	−0.6625	3.77*
KOSPI 200 (1996–2025)	7287	0.0002	0.0170	0.271	−0.1045	8.88
KOSPI 200 (2016–2025)	2442	0.0004	0.0117	0.186	−0.3190	9.32
KOSPI 200 (2021–2025)	1222	0.0003	0.0124	0.197	−0.4014	7.21

* Sample too small for accurate measurement and not robust to outliers.

§ Calculated as in Equation (8).

¶ Calculated as in Equation (56), Appendix D.6.

Log-Return Density: Empirical vs. Normal KOSPI 200, 1996-2025

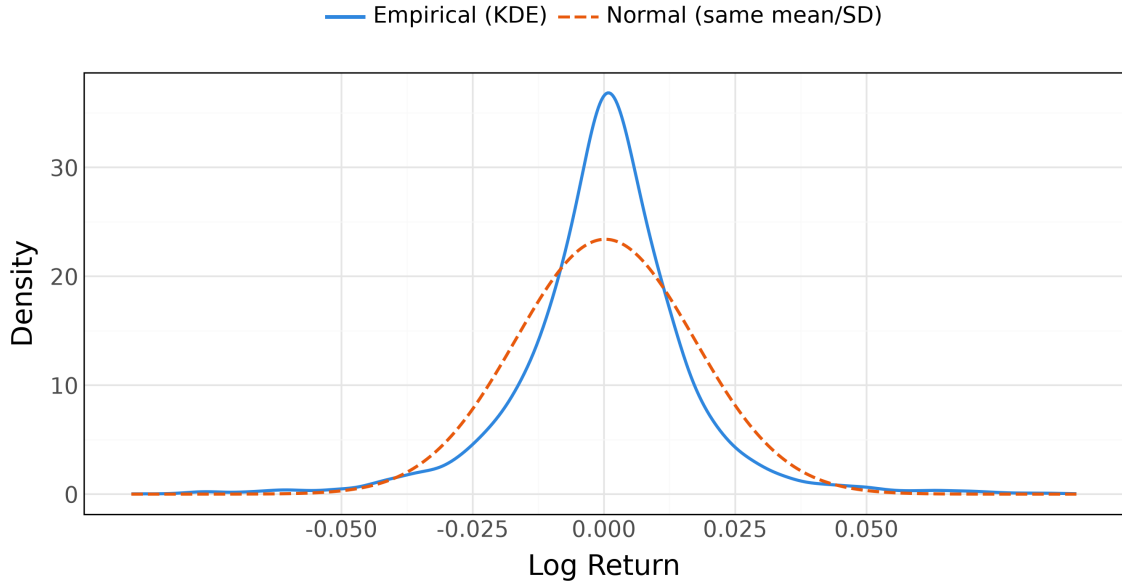


Figure 8: Density of log returns for the KOSPI 200 (1996–2025), against a fitted normal density.

3.7 Remarks

The Black–Scholes model is the first in this dissertation able to achieve our goal of pricing options. As discussed, however, there are major deficiencies in its modelling assumptions. The key issues were the breakdown of the normal distribution assumption and the

Log-Return Density, Log Scale: Empirical vs. Normal

KOSPI 200, 1996-2025

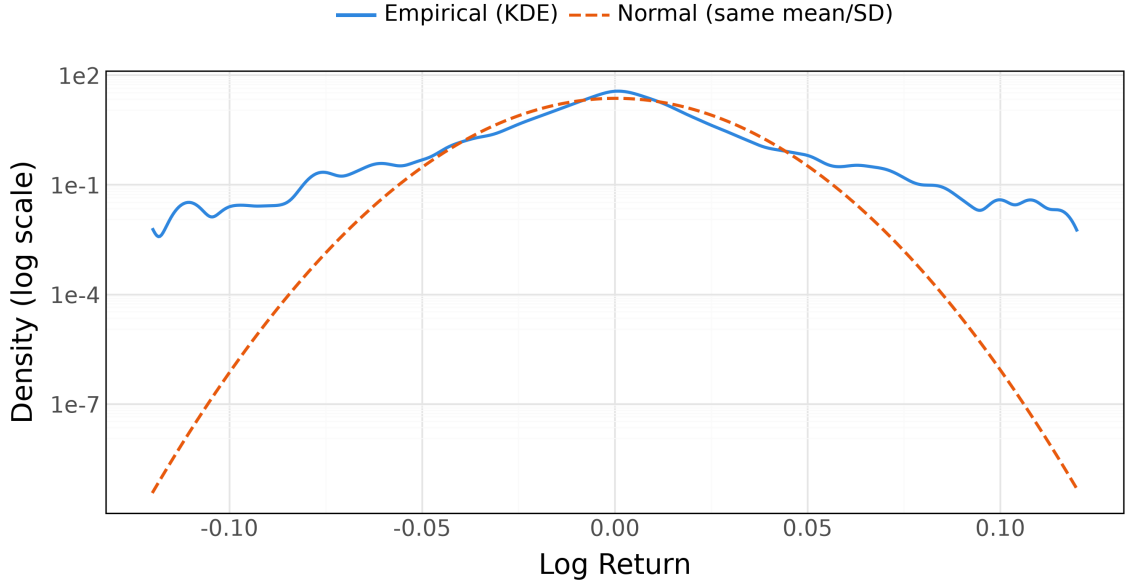


Figure 9: Density of returns for the KOSPI 200 (1996–2025), against a fitted normal density, on a logarithmic scale.

breakdown of the constant- σ assumption. We adjusted for the non-constant σ by fitting a separate Black–Scholes model per maturity, increasing from a single parameter to seven. This worked well, but it is a descriptive fix rather than a coherent model, since it fits seven independent volatilities rather than describing how volatility evolves. Before addressing this volatility directly, we address the non-normality. In the following section we discuss Lévy processes and assess their capability in handling the KOSPI data.

4 Lévy Processes

Our goal now is to build a model that retains most of the assumptions of the Black–Scholes model and Brownian motion, such as independent and stationary increments, while using a more general distribution rather than the normal distribution. To this end, we turn to Lévy processes. The material in this section draws on the definitions, models and mathematical properties described by Schoutens (2003).

4.1 Overview

We desire a process almost identical to the Brownian motion of Definition 3.1, but with part (iv) changed from normally distributed increments to an alternative distribution. To maintain the stationary and independent increments of parts (ii) and (iii), we require a distribution that is infinitely divisible. This means that the distribution must be capable of being decomposed into an arbitrary number of smaller, independent, and identically distributed pieces. This is crucial to stock price modelling, since our time measure is continuous. We define infinite divisibility in Definition 4.1.

Definition 4.1 (Infinite Divisibility). *Let X be a random variable with characteristic function $\phi(u)$. The distribution of X (equivalently, ϕ) is said to be infinitely divisible if, for every positive integer n , $\phi(u)$ can be written as*

$$\phi(u) = (\phi_n(u))^n, \quad u \in \mathbb{R},$$

for some characteristic function $\phi_n(u)$.

(Schoutens, 2003)

Processes with independent, stationary increments that are infinitely divisible are called Lévy processes. We formally define Lévy processes, following Applebaum (2009) in Definition 4.2. It provides a clear comparison with the definition of Brownian motion in Definition 3.1. The only difference lies in Point 4: here stochastic continuity is assumed, whereas the Brownian motion definition assumes normality.

Definition 4.2 (Lévy Process). *Let $X = (X(t), t \geq 0)$ be a stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We say that X has independent increments if, for each $n \in \mathbb{N}$ and each $0 \leq t_1 < t_2 \leq \dots \leq t_{n+1} < \infty$, the random variables*

$$(X(t_{j+1}) - X(t_j), 1 \leq j \leq n)$$

are independent, and that X has stationary increments if

$$X(t_{j+1}) - X(t_j) \stackrel{d}{=} X(t_{j+1} - t_j) - X(0)$$

for each such j .

We say that X is a Lévy process if:

1. $X(0) = 0$ (a.s.);
2. X has independent increments;
3. X has stationary increments;
4. X is stochastically continuous, i.e. for all $a > 0$ and for all $s \geq 0$,

$$\lim_{t \rightarrow s} \mathbb{P}(|X(t) - X(s)| > a) = 0.$$

(Applebaum, 2009)

Every Lévy process can be modified into a càdlàg process, which we formally define in Definition 4.3. This càdlàg modification is itself a Lévy process. For the remainder of this dissertation, we work primarily with the càdlàg version of the Lévy process. This version permits jumps in the process, owing to its *right-continuous with left limits* property. Every continuous function is càdlàg, but not every càdlàg function is continuous. Its jump component is particularly useful for highly volatile indices such as the KOSPI.

Definition 4.3 (Càdlàg). *Consider a real-valued function $f : [a, b] \rightarrow \mathbb{R}$. Assume that for all $t \in (a, b]$ the function f is right-continuous and has left limits.*

If f is càdlàg, we write its left limit at each point $t \in (a, b]$ as

$$f(t^-) = \lim_{s \uparrow t} f(s).$$

We stress that $f(t^-) = f(t)$ if and only if f is continuous at t . The jump at t is denoted by

$$\Delta f(t) = f(t) - f(t^-).$$

(Definition adapted from Schoutens (2003, §2.2.3).)

4.1.1 Final Stock Price Model

There are a number of well-known Lévy processes. In the following sections we model the Meixner, variance gamma, CGMY, normal inverse Gaussian, and generalised hyperbolic processes. The most significant restriction of these models is that, while the Black–Scholes model possesses market completeness, these models do not. This implies that there is no unique martingale measure, and so we must choose one. Schoutens (2003, §6.2.2) describes two distinct ways of obtaining an equivalent martingale measure. The first is the Esscher transform, and the second is the mean-correcting martingale measure. We choose to focus on the mean-correcting martingale measure, which adds a drift m to the process, chosen so that the discounted price is a martingale under $\mathbb{Q}$.

We now arrive at a stock price model. Let $X = \{X_t, t \geq 0\}$ denote the Lévy process itself, and let ϕ_X be the characteristic function of X_1 . The stock price is modelled as

$$S_t = S_0 \exp(mt + X_t), \quad \text{equivalently} \quad \log S_t = \log S_0 + mt + X_t, \quad (9)$$

with the drift given by

$$m = r - q - \log \phi_X(-i). \quad (10)$$

This choice of m makes $e^{-(r-q)t} S_t$ a martingale under $\mathbb{Q}$. The drift is not a free parameter, it depends on the others through ϕ_X , and is recomputed from Equation (10) at each trial parameter set during calibration. Its form differs from model to model. Under this specification the log returns $\log(S_{t+s}/S_t)$ follow the distribution of the increments of X over an interval of length s , shifted by ms . Choosing a Lévy process therefore amounts to choosing a distribution for log returns.

4.1.2 Pricing by the Fourier Method

Of the five Lévy models considered in this section, only the Meixner distribution has a density in closed form. The variance gamma, CGMY, NIG and generalised hyperbolic models are specified through their characteristic functions, so the expectation defining the call price cannot be integrated directly. The same is true of every stochastic volatility model that we consider in later sections.

For all of these we follow the route directed by Schoutens (2003) to Carr and Madan (1999), which recovers the call price from the characteristic function by the Fourier

method. Writing ϕ for the characteristic function of $\log S_T$ under $\mathbb{Q}$, the Carr–Madan representation gives the call price

$$C(K, T) = \frac{e^{-\zeta \log K}}{\pi} \int_0^\infty e^{-iv \log K} \varrho(v) dv, \quad \varrho(v) = \frac{e^{-rT} \phi(v - (\zeta + 1)i)}{\zeta^2 + \zeta - v^2 + i(2\zeta + 1)v}, \quad (11)$$

where $\zeta = 0.75$ is the damping value suggested by Schoutens (2003).

Only ϕ changes from model to model. Each of the following sections, apart from the Meixner model, therefore specifies its characteristic function and then prices through Equation (11).

4.2 Meixner Process

The first of five Lévy processes we choose to examine is the Meixner process.

4.2.1 Testing the Meixner Density

The Meixner distribution has a closed-form density given by

$$f_{\text{Meixner}}(x; \alpha, \beta, \delta) = \frac{(2 \cos(\beta/2))^{2\delta}}{2\alpha\pi\Gamma(2\delta)} \exp\left(\frac{\beta x}{\alpha}\right) \left| \Gamma\left(\delta + \frac{ix}{\alpha}\right) \right|^2, \quad (12)$$

where $\alpha > 0$, $-\pi < \beta < \pi$ and $\delta > 0$. In this section we check how well that density fits the log returns directly, through plots and χ^2 tests. The parameters are estimated by maximum likelihood on the daily log returns.

Writing $r_1, \dots, r_n$ for the observed log returns, the log-likelihood of the parameter vector $\theta = (\alpha, \beta, \delta, m)$ is

$$\ell(\theta) = \sum_{i=1}^n \log f_{\text{Meixner}}(r_i; \alpha, \beta, \delta, m),$$

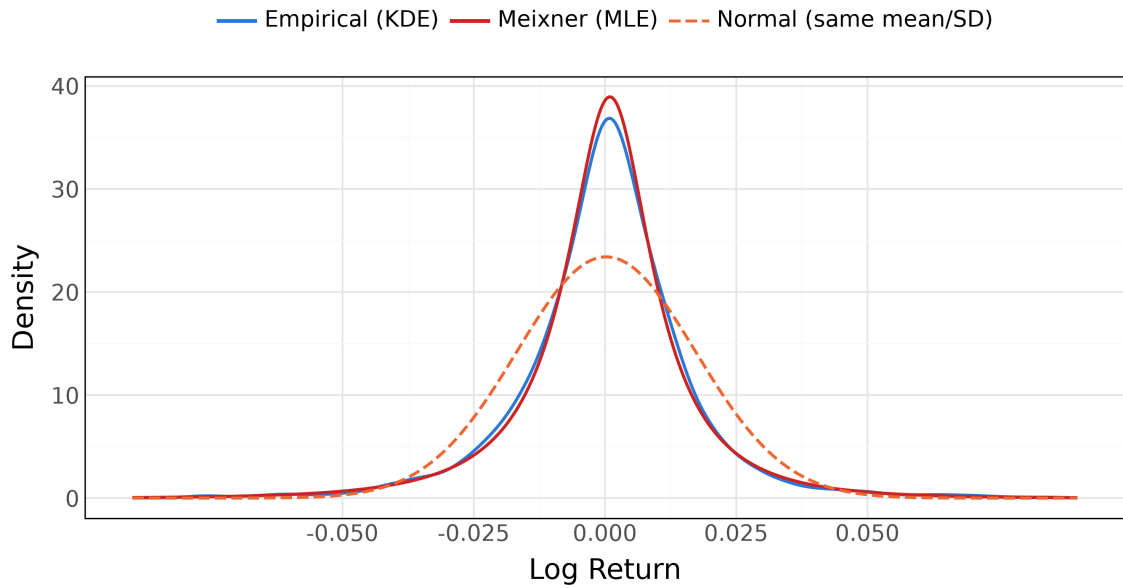
and the estimate is

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \ell(\theta) = \arg \min_{\theta \in \Theta} \{-\ell(\theta)\}, \quad \Theta = \{\alpha > 0, |\beta| < \pi, \delta > 0, m \in \mathbb{R}\}.$$

Here the parameter m is estimated alongside the other parameters. We first test the Meixner distribution by comparing its density curve with the log returns. We did this on the density and log-density scale in Figures 10a and 10b. In both figures, we see that the Meixner distribution fits considerably better than the normal distribution. The Meixner distribution is therefore a far more suitable distribution than the normal.

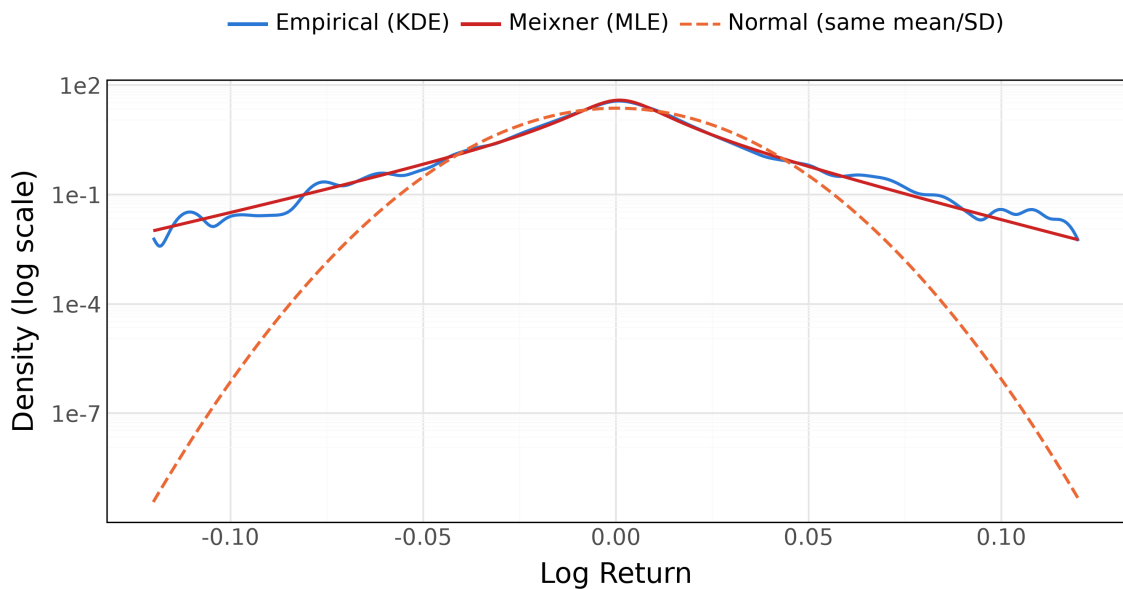
We also test the Meixner distribution via a chi-squared test, with a Meixner null hypothesis, to check whether the historical log returns do in fact follow a Meixner distribution. These results are available in Table 8, across a number of different timeframes. For the plotted data, which contains daily log returns from 1996 to 2025, we obtain a very small p -value, suggesting that the data does not follow a Meixner distribution. For other datasets, such as 2026, 2016–2025, and 2021–2025, we obtain much higher p -values, suggesting that they do follow a Meixner distribution.

Log-Return Density: Empirical vs. Normal vs. Meixner KOSPI 200, 1996-2025



(a) Density

Log-Return Density, Log Scale: Empirical vs. Normal vs. Meixner KOSPI 200, 1996-2025



(b) Log density

Figure 10: KOSPI 200 daily log-return density, 1996–2025. The Gaussian kernel estimate is compared with the fitted Meixner distribution and a normal distribution matched on mean and standard deviation. Figure (b) plots the same curves on a logarithmic density scale.

Table 8: Meixner distribution parameter estimates for daily log returns of a set of indices.

Index	n	α	β	δ	m	p -value
KOSPI 200 (1996–2025)	7287	0.0575	−0.1681	0.1737	0.0011	0.0128
KOSPI 200 (2016–2025)	2442	0.0278	−0.2467	0.3395	0.0016	0.1919
KOSPI 200 (2021–2025)	1222	0.0206	−0.1437	0.7033	0.0014	0.4421

For the Meixner distribution kurtosis is given by

$$3 + (2 - \cos \beta)/\delta.$$

We can then compute the empirical kurtosis (that of the data) and compare it with this implied kurtosis; see Table 9. For the KOSPI 200 over 1996–2025, the model-implied kurtosis closely reflects the empirical kurtosis, suggesting a good fit in this respect. This close match in kurtosis is notable, since capturing kurtosis was precisely one of the difficulties we encountered with the normal distribution assumption. For the other intervals the results were more mixed, with the implied kurtosis undershooting the empirical kurtosis in the 2016–2025 and 2021–2025 samples. Across all timeframes, however, either the chi-squared test implies the data follows a Meixner distribution or the empirical kurtosis matches the Meixner-implied kurtosis. This suggests the Meixner model is worth fitting.

Table 9: Empirical and Meixner-implied kurtosis for daily log returns of a set of indices. The empirical kurtosis is the sample quantity of Equation (56) in Appendix D.6.

Index	Empirical Kurtosis	Implied Kurtosis
KOSPI 200 (1996–2025)	8.88	8.84
KOSPI 200 (2016–2025)	9.32	6.03
KOSPI 200 (2021–2025)	7.21	4.44

4.2.2 The Model

The Meixner process has a probability density function given by Equation (12). It has characteristic function

$$\phi_{\text{Meixner}}(u; \alpha, \beta, \delta) = \left(\frac{\cos(\beta/2)}{\cosh\left(\frac{\alpha u - i\beta}{2}\right)} \right)^{2\delta}.$$

This process is infinitely divisible. We formally define a Meixner process in Definition 4.4.

Definition 4.4 (Meixner Process). *A stochastic process $X^{(\text{Meixner})} = \{X_t^{(\text{Meixner})}, t \geq 0\}$ is a Meixner process with parameters $\alpha > 0$, $-\pi < \beta < \pi$ and $\delta > 0$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ if it satisfies conditions (i)–(iii) of Definition 4.2 and, in place of (iv), $X_{t+s}^{(\text{Meixner})} - X_t^{(\text{Meixner})}$ is Meixner distributed with parameters α , β and $\delta s > 0$:*

$$X_{t+s}^{(\text{Meixner})} - X_t^{(\text{Meixner})} \sim \text{Meixner}(\alpha, \beta, \delta s).$$

We denote a Meixner process by $X^{(\text{Meixner})} = \{X_t^{(\text{Meixner})}, t \geq 0\}$.

The kurtosis of a Meixner distribution is $3 + (2 - \cos(\beta))/\delta$, so that, as long as $\delta > 0$, the kurtosis is greater than that of a normal distribution. This helps address the issues faced when fitting the Black–Scholes model, where log returns exhibited greater kurtosis than the normal distribution.

The stock price is modelled via Equation 9*, $S_t = S_0 \exp(mt + X_t)$, where

$$X_T \sim \text{Meixner}(\alpha, \beta, \delta T),$$

i.e. α and β are time-invariant, while δ accumulates linearly with T and

$$\begin{aligned} m &= r - q - \log \phi(-i) \\ &= r - q - 2\delta \left(\log \cos(\beta/2) - \log \cos\left(\frac{\alpha + \beta}{2}\right) \right). \end{aligned}$$

Since m is always recomputed from $(\alpha, \beta, \delta, r, q)$, only (α, β, δ) remain to be estimated. For a call option with strike K and maturity T ,

$$\begin{aligned} V_0 &= C_{\text{Meixner}}(K, T) = \mathbb{E}_{\mathbb{Q}}(e^{-rT}(S_T - K)^+) \\ &= e^{-rT} \int_c^\infty (S_0 e^x - K) f_{\text{Meixner}}(x; \alpha, \beta, \delta T, mT) dx. \end{aligned} \quad (13)$$

The parameters (α, β, δ) are estimated jointly, by minimising the RMSE between model and market prices across all n contracts in the chain,

$$(\hat{\alpha}, \hat{\beta}, \hat{\delta}) = \arg \min_{\alpha, \beta, \delta} \sqrt{\frac{1}{n} \sum_{i=1}^n \left(C_{\text{Meixner},i}(\alpha, \beta, \delta) - C_{\text{mkt},i} \right)^2}, \quad (14)$$

where $C_{\text{Meixner},i}$ is evaluated at contract i 's own (K_i, T_i) , with m recomputed at every trial (α, β, δ) . This is the same RMSE-minimisation principle as the single-parameter Black–Scholes calibration of Equation (5), extended to three free parameters instead of one.

The RMSE minimisation is solved computationally. In doing so, we define Equation (14) as our objective function. Python provides several algorithms with which to solve this objective function. We chose the Nelder–Mead method; it is set out in full in Appendix F. This algorithm attempts to find the global minimum over several variables. It also allows the search to be conducted from several different starting points, which we make use of. The ten starting points are listed in Table 41 of Appendix G; they are spread across the admissible region.

4.3 The Variance Gamma, CGMY and Normal Inverse Gaussian Models

The next three Lévy processes we examine are treated together. Each is specified through its characteristic function rather than a density, and each is therefore priced and calibrated in exactly the same way.

* t denotes the running time index, so the process is written in t ; T denotes the fixed maturity of the contract being priced, so payoffs, prices and the distributions entering the pricing formula are written in T .

The Variance Gamma (VG) process was introduced by Madan and Seneta (1989) as a model for stock returns. It is a three-parameter family. CGMY stands for the names of its creators, Carr, Geman, Madan and Yor. This is a four-parameter distribution. The CGMY process generalises the Variance Gamma model defined above through the addition of a fourth parameter Y . Setting $Y = 0$ recovers VG exactly, so that $\text{CGMY}(C, G, M, 0) = \text{VG}(C, G, M)$. The normal inverse Gaussian (NIG) distribution was introduced by Barndorff-Nielsen (1997). NIG is a three-parameter model. These models are defined in Definition 4.5. The parameters, time scaling, cumulant characteristic functions and m drift parameters of each model are given in Table 10.

Definition 4.5 (Lévy Processes Specified by a Characteristic Exponent). *Let $\mathcal{F}$ be one of the families of Table 10, with parameters θ in the admissible set stated there and characteristic exponent $\psi_{\mathcal{F}}(u; \theta)$. A stochastic process $X^{(\mathcal{F})} = \{X_t^{(\mathcal{F})}, t \geq 0\}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is an $\mathcal{F}$ process if it satisfies conditions (i)–(iii) of Definition 4.2 and, in place of (iv), for all $s, t \geq 0$ the increment $X_{t+s}^{(\mathcal{F})} - X_t^{(\mathcal{F})}$ is $\mathcal{F}$ distributed with parameters θ_s ,*

$$X_{t+s}^{(\mathcal{F})} - X_t^{(\mathcal{F})} \sim \mathcal{F}(\theta_s), \quad \mathbb{E} \left[\exp \left(iu (X_{t+s}^{(\mathcal{F})} - X_t^{(\mathcal{F})}) \right) \right] = \exp(s \psi_{\mathcal{F}}(u; \theta)),$$

where θ_s denotes θ with its scale parameter multiplied by s , that is $C \mapsto sC$ for the variance gamma and CGMY families and $\delta \mapsto s\delta$ for the normal inverse Gaussian family.

Table 10: The three families of Definition 4.5, with their admissible parameters, the parameter carrying the time scaling, the characteristic exponent $\psi_{\mathcal{F}}$, and the mean-correcting drift m of Equation (10).

VG	Parameters	$C, G, M > 0$
	Scaling	$C \mapsto sC$
	$\psi_{\mathcal{F}}(u; \theta)$	$C (\log(GM) - \log(GM + (M - G)iu + u^2))$
	m	$r - q + C \log \left(\frac{(M - 1)(G + 1)}{MG} \right)$
CGMY	Parameters	$C, G, M > 0, Y < 2$
	Scaling	$C \mapsto sC$
	$\psi_{\mathcal{F}}(u; \theta)$	$C \Gamma(-Y) ((M - iu)^Y - M^Y + (G + iu)^Y - G^Y)$
	m	$r - q - C \Gamma(-Y) ((M - 1)^Y - M^Y + (G + 1)^Y - G^Y)$
NIG	Parameters	$\alpha > 0, \beta < \alpha, \delta > 0$
	Scaling	$\delta \mapsto s\delta$
	$\psi_{\mathcal{F}}(u; \theta)$	$-\delta \left(\sqrt{\alpha^2 - (\beta + iu)^2} - \sqrt{\alpha^2 - \beta^2} \right)$
	m	$r - q + \delta \left(\sqrt{\alpha^2 - (\beta + 1)^2} - \sqrt{\alpha^2 - \beta^2} \right)$

In all three cases the stock price is modelled via Equation (9), $S_t = S_0 \exp(mt + X_t)$, with the drift m given by Equation (10) as $m = r - q - \log \phi_X(-i)$. The models differ

only in distribution of X_t and in the explicit form this gives m ; both are listed for each family in Table 10. Each family is closed under the time scaling of its own parameters, so the law of X_T is obtained by applying the scaling column of Table 10 with $s = T$. Explicitly,

$$X_T \sim \text{VG}(TC, G, M), \quad X_T \sim \text{CGMY}(TC, G, M, Y), \quad X_T \sim \text{NIG}(\alpha, \beta, T\delta). \quad (15)$$

For the Variance Gamma and CGMY processes only C carries the time index, the remaining parameters being time-invariant; for the Normal Inverse Gaussian process it is instead δ that accumulates linearly with T , rather than the first parameter, while α and β are time-invariant. Writing θ for the parameters of whichever of these three models is in use, each gives $\phi_X(u; T\theta) = \exp(T \log \phi_X(u; \theta))$. Combining with the deterministic terms above, the characteristic function of $\log S_T$ is

$$\phi(u) = \exp\left(iu[\log S_0 + mT] + T \log \phi_X(u)\right), \quad (16)$$

where ϕ_X is the characteristic function of the relevant process. Since m is always re-computed from the remaining parameters together with r and q , the free parameters are (C, G, M) , (C, G, M, Y) and (α, β, δ) respectively[†]. For the Variance Gamma process the restriction $M > 1$ is imposed in addition to positivity, so that the logarithm in m is well defined.

For a call option with strike K and maturity T , under $\mathbb{Q}$, the price is once more the discounted risk-neutral expected payoff. Because no density is available, this expectation cannot be integrated directly as it was for the Meixner model in Equation (13). It is instead evaluated by the Fourier method of Section 4.1.2, substituting the characteristic function of Equation (16) for ϕ in the Carr–Madan representation, Equation (11):

$$\begin{aligned} V_0 = C_X(K, T) &= \mathbb{E}_{\mathbb{Q}}(e^{-rT}(S_T - K)^+) \\ &= \frac{e^{-\zeta \log K}}{\pi} \int_0^\infty e^{-iv \log K} \varrho(v) dv, \end{aligned} \quad (17)$$

where ϱ is as defined in Equation (11).

The parameters θ of each model are estimated jointly, by minimising the RMSE between model and market mid prices across all n contracts in the chain,

$$\hat{\theta} = \arg \min_{\theta} \sqrt{\frac{1}{n} \sum_{i=1}^n \left(C_{X,i}(\theta) - C_{\text{mkt},i}\right)^2}, \quad (18)$$

where $C_{X,i}$ is evaluated at contract i 's own (K_i, T_i) , with m recomputed at every trial θ . This is the same RMSE-minimisation principle as the Meixner calibration of Equation (14), applied to three further families and extending to four free parameters in the CGMY case.

[†]The parameters α, β, δ of the NIG and GH distributions are unrelated to the Meixner parameters of the same names; the meaning is determined by context throughout.

The RMSE minimisation is again solved computationally, taking Equation (18) as our objective function and the call price as V_0 , given by Equation (17). As for the Meixner model, we use the Nelder–Mead method of Appendix F, here restarted from six initial points and retaining the lowest-RMSE solution. The first start of each family is Schoutens (2003, Table 6.1)’s own estimate, with the remainder dispersed more widely and at larger scale parameters, given the higher volatility of this chain; all are listed in Tables 42 to 44 of Appendix G. Parameter restrictions are imposed as a penalty barrier that returns a large objective value outside the admissible region.

4.4 The Generalised Hyperbolic Model

The generalised hyperbolic (GH) distribution was introduced by Barndorff-Nielsen (1977). The GH distribution is not a member of its own separate family in the strict sense. It is instead an umbrella distribution containing the VG, hyperbolic and NIG distributions as special or limiting cases, obtained by fixing the additional shape parameter ν to particular values. We formally define the GH process in Definition 4.6.

Definition 4.6 (Generalised Hyperbolic Process). *A stochastic process $X^{(GH)} = \{X_t^{(GH)}, t \geq 0\}$ is a Generalised Hyperbolic process with parameters $\alpha > 0$, $|\beta| < \alpha$, $\delta > 0$ and $\nu \in \mathbb{R}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ if it satisfies conditions (i)–(iii) of Definition 4.2 and, in place of (iv), $X_1^{(GH)}$ is GH distributed with parameters α , β , δ and ν , so that for all $s, t \geq 0$*

$$\mathbb{E}\left[\exp\left(iu(X_{t+s}^{(GH)} - X_t^{(GH)})\right)\right] = \left[\left(\frac{\alpha^2 - \beta^2}{\alpha^2 - (\beta + iu)^2}\right)^{\nu/2} \frac{K_\nu\left(\delta\sqrt{\alpha^2 - (\beta + iu)^2}\right)}{K_\nu\left(\delta\sqrt{\alpha^2 - \beta^2}\right)}\right]^s,$$

where K_ν denotes the modified Bessel function of the third kind with index ν . We denote a GH process by $X^{(GH)} = \{X_t^{(GH)}, t \geq 0\}$.

The admissible parameter range for the GH distribution depends on the sign of ν : $\delta \geq 0$ and $|\beta| < \alpha$ for $\nu > 0$, $\delta > 0$ and $|\beta| < \alpha$ for $\nu = 0$, and $\delta > 0$ and $|\beta| \leq \alpha$ for $\nu < 0$. The GH distribution does not admit simple closed-form expressions for skewness and kurtosis in terms of elementary functions. With four free parameters $(\alpha, \beta, \delta, \nu)$ and the special-case relationships noted below, the GH model subsumes the VG, hyperbolic and NIG distributions as particular parameterisations rather than sitting alongside them as an independent alternative, since

- $\nu = 1$ recovers the hyperbolic distribution,
- $\nu = -\frac{1}{2}$ recovers the NIG distribution, i.e. $\text{GH}(\alpha, \beta, \delta, -\frac{1}{2}) = \text{NIG}(\alpha, \beta, \delta)$,
- the VG distribution is recovered in the limit $\delta \rightarrow 0$ with $\nu = \sigma^2/\nu_{\text{VG}}$, $\alpha = \sqrt{2/\nu_{\text{VG}} + \theta^2/\sigma^4}$ and $\beta = \theta/\sigma^2$, where $(\sigma, \nu_{\text{VG}}, \theta)$ are the parameters of the VG distribution.

This makes the GH model a natural benchmark against which the more parsimonious models can be judged. Any improvement the three-parameter models achieve over GH would suggest the optimisation over the GH model has not found a global optimum, whereas a clear advantage for GH would point to genuine extra flexibility being needed.

No single parameter of the GH distribution carries the time index, so X_T cannot be written as a member of the same family at rescaled parameters; the characteristic function of the increment over $[0, T]$ is instead obtained as $(\phi_{\text{GH}}(u))^T$. Its drift is

$$m = r - q - \log \left[\left(\frac{\alpha^2 - \beta^2}{\alpha^2 - (\beta + 1)^2} \right)^{\nu/2} \frac{K_\nu(\delta \sqrt{\alpha^2 - (\beta + 1)^2})}{K_\nu(\delta \sqrt{\alpha^2 - \beta^2})} \right].$$

Pricing and calibration proceed exactly as in Section 4.3. The characteristic function of $\log S_T$ is Equation (16) with $\phi_X = \phi_{\text{GH}}$ and m as above, the call price is Equation (17), and the four free parameters $(\alpha, \beta, \delta, \nu)$ are estimated by the RMSE minimisation of Equation (18), from the six starting points of Table 45 in Appendix G. The first start of each family is Schoutens' own estimate, with the remainder dispersed more widely and at larger scale parameters, similar to the other calibrations. Parameter restrictions are also imposed as a penalty barrier that returns a large objective value outside the admissible region.

4.5 Model Validation

Prior to implementing each model, we check that the model, when fitted to Schoutens' data, produces results similar to his. Our results are listed in Table 11, which shows that all models produced roughly similar parameters and RMSEs. We define “similar” as within $\pm 30\%$ of the original parameter value and RMSE. The only parameters of concern are the CGMY G and M parameters, which show larger discrepancies; however, given the similar RMSEs and other parameters, the agreement is adequate. We can therefore proceed to fit the models on our KOSPI data, knowing that our implementation matches Schoutens'.

Table 11: Our calibration of each Lévy model to the S&P 500 (2002) chain, set against the results published by Schoutens (2003, Tables 6.3 and 6.4). The symbols of each model are given above its block. Models whose parameters reproduce his are shaded green.

	Parameters			RMSE
<i>Meixner</i>				
	α	β	δ	
Ours	0.3801	−1.4980	0.3768	3.3035
Schoutens	0.3977	−1.4940	0.3462	3.2451
<i>Variance gamma</i>				
	C	G	M	
Ours	1.4919	6.1352	15.2171	3.6195
Schoutens	1.3574	5.8704	14.2699	3.5600
<i>CGMY</i>				
	C	G	M	Y
Ours	0.0210	0.0000	10.1571	1.3774
Schoutens	0.0244	0.0765	7.5515	1.2945
<i>NIG</i>				
	α	β	δ	
Ours	6.5527	−4.1288	0.1699	3.1617
Schoutens	6.1882	−3.8941	0.1622	3.1119
<i>Generalised hyperbolic</i>				
	α	β	δ	ν
Ours	4.1106	−4.1106	0.2492	−1.8603
Schoutens	3.8288	−3.8286	0.2375	−1.7555

RMSE is defined in Equation (6).

4.6 Goodness-of-Fit

Table 12 reports the goodness-of-fit measures for all five Lévy models calibrated to the KOSPI 200 (2026) chain, set against the Black–Scholes benchmark. It is clear that each model has failed to improve significantly on the Black–Scholes model. For the Meixner, variance gamma, CGMY, and NIG models the errors are very similar, suggesting an almost identical model fit. The GH model does improve slightly on Black–Scholes, but not by any significant margin.

Table 12: Goodness-of-fit for the Black–Scholes model and the five Lévy models calibrated to the KOSPI 200 (2026) option chain.

Model	APE (%)	AAE	RMSE	ARPE (%)
Black–Scholes	12.43	11.4802	12.3578	17.48
Meixner	12.38	11.4342	12.3251	18.05
Variance Gamma	12.43	11.4812	12.3581	17.47
CGMY	12.43	11.4737	12.3528	17.58
NIG	12.44	11.4835	12.3578	17.48
GH	11.98	11.0616	11.8135	15.55

APE, AAE, RMSE and ARPE are defined in Equation (6).

We plot each of the models' fitted values to examine the exact fit, and it is clear that the Meixner, variance gamma, CGMY, and NIG model fits are almost identical to the Black–Scholes model for every option. This is evident in Figure 11. The GH model does have a unique fit, but this is only visually noticeable in the August 2026 maturity.

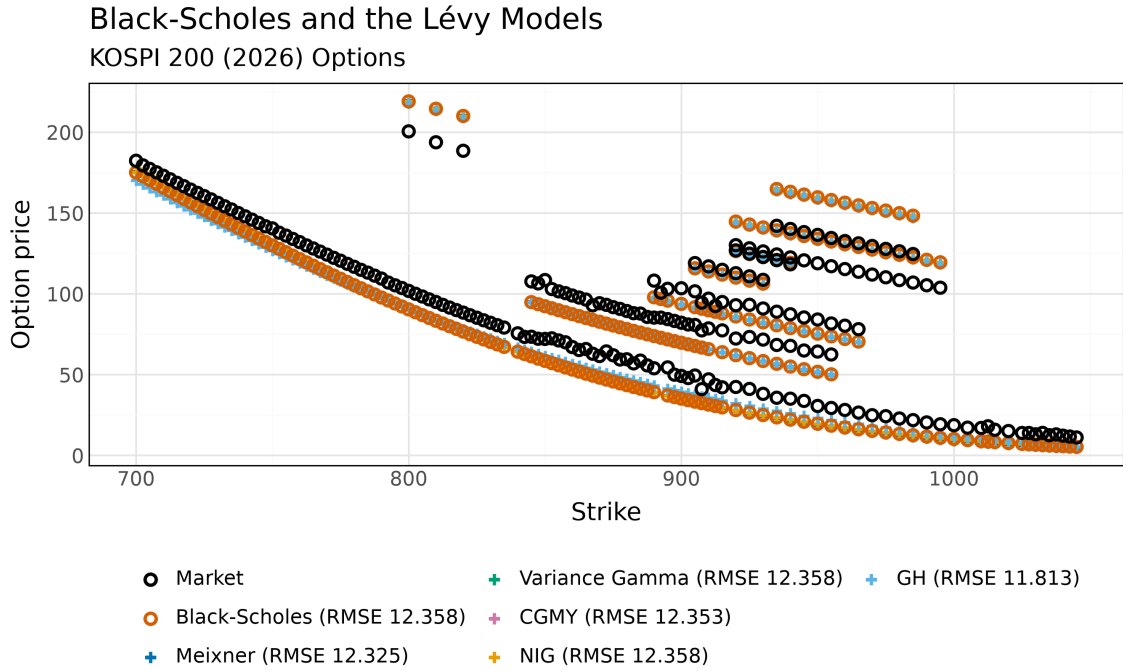


Figure 11: Black–Scholes and the five Lévy models against the market on the KOSPI 200 (2026) chain.

4.7 Examining Density Functions

In this section we generate figures comparing the density functions for each Lévy model with the Black–Scholes normal model, except the GH model. The GH model density function could not be attained due to computational difficulty. The figures are drawn on the log-return scale. All the models of this section take the form $S_T = S_0 e^{mT + X_T}$ and

differ only in the law of X_T , so comparing the densities of $\log(S_T/S_0)$ isolates exactly what distinguishes them. The method used to recover each density from its characteristic function is set out in Appendix D.7.1.

The recovered log-return densities are shown in Figure 12: Figure 12a shows the fit for the August 2026 parameterisation and Figure 12b that for the March 2027 parameterisation. The two may differ because the maturity T enters the parameters. Both charts clearly indicate that, for every model, the fitted density converges back to the normal distribution. This suggests that our calibrated models are not fat-tailed, despite the fat tails present in the log-return data itself. The Lévy models have therefore failed to distinguish themselves from the Black–Scholes model and its assumption of normally distributed log returns.

4.8 Examining Parameters

We examine exactly what went wrong in the parameterisation, causing the fit either to become identical to the Black–Scholes model or to fail to improve substantially on it.

4.8.1 Meixner Parameterisation

The Meixner fit converges on the boundary of our parameter space. These results are shown in Table 13. The α and β parameters lie right on the edge of the feasible region, and δ is exceptionally large relative to the Schoutens calibration in Table 11.

Table 13: Calibrated Meixner parameters on the KOSPI 200 (2026) option chain.

Model	Parameter	Estimate
Meixner	α	0.0013
	β	−3.0425
	δ	1350.91

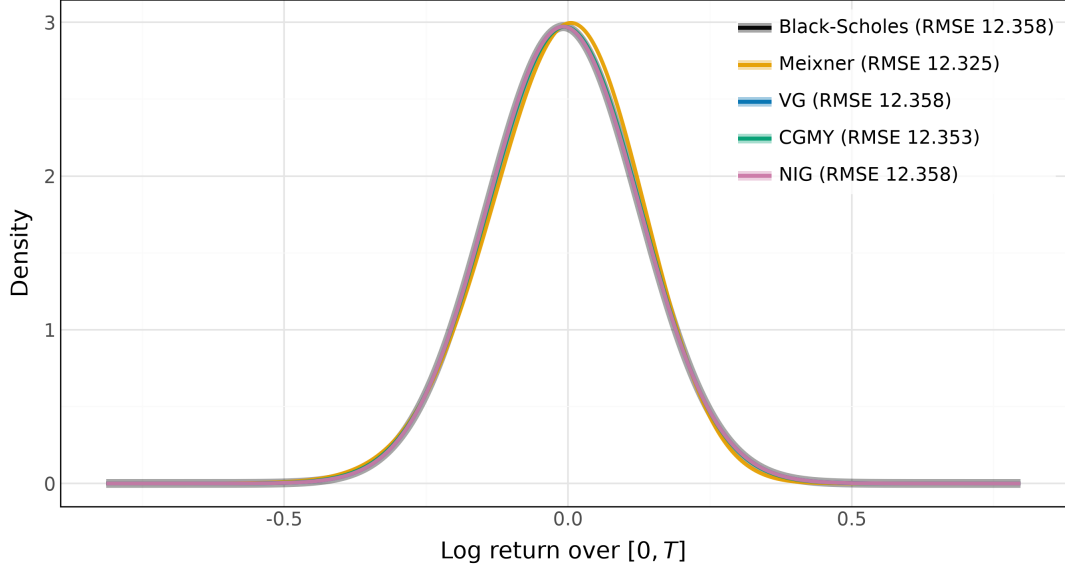
The magnitude of $\hat{\delta}$ is what drives the result. The Meixner skewness and kurtosis are

$$\text{skewness} = \sin\left(\frac{\beta}{2}\right) \sqrt{\frac{2}{\delta}}, \quad \text{kurtosis} = 3 + \frac{2 - \cos \beta}{\delta},$$

and δ appears in the denominator of both, so that as $\delta \rightarrow \infty$ the skewness tends to zero and the kurtosis to three whatever β does. The calibration drives it to $\hat{\delta} \approx 1351$, and the fitted law has skewness -0.038 and kurtosis 3.002 . The boundary value $\hat{\beta} = -3.0425$ is doing its best to counteract this, since $\sin(\beta/2) = -0.999$ is as close to -1 as the constraint $|\beta| < \pi$ permits, but the $\sqrt{2}/\delta = 0.038$ multiplying it leaves the skewness negligible regardless.

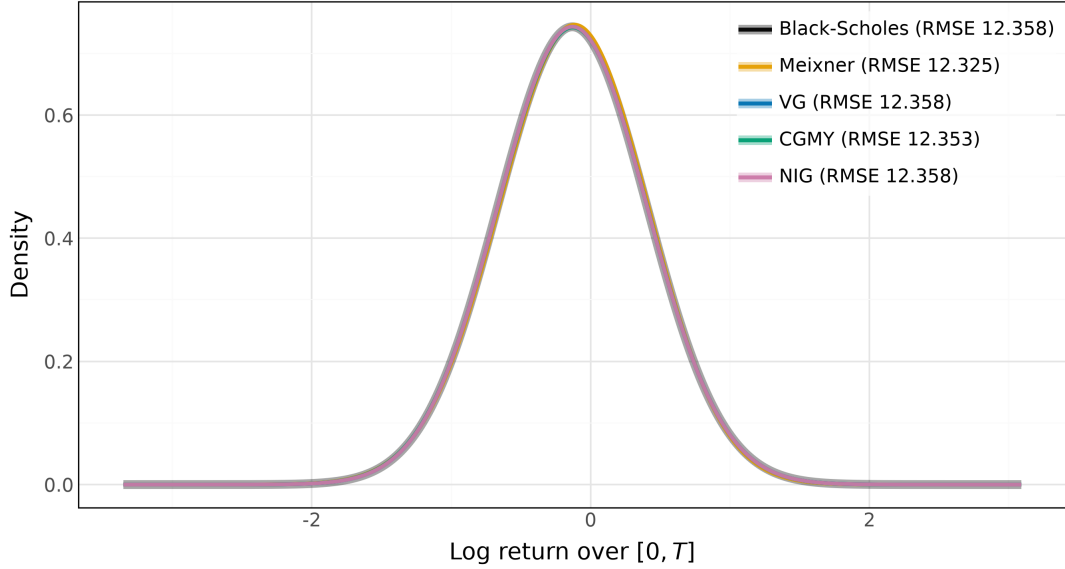
Eight of the ten starts agreed on the final RMSE, but all differed on the parameter estimates. One area of concern in our parameterisation is that the model gets stuck at the boundary. We enforce the boundary condition by setting the RMSE to 10^6 at the boundary, which could be forcing the simplex into a local minimum.

Fitted Lévy and Normal Log-Return Densities
August 2026



(a) Front month August 2026, $T = 0.038$.

Fitted Lévy and Normal Log-Return Densities
March 2027



(b) Longest maturity March 2027, $T = 0.614$.

Figure 12: Risk-neutral log-return densities of the four calibrated Lévy models against the Black–Scholes normal, on the KOSPI 200 (2026) chain. All five models take the form $S_T = S_0 e^{mT + X_T}$ and differ only in the law of X_T .

4.8.2 Variance Gamma Parameterisation

The parameters for the variance gamma model are in Table 14. The magnitude of $\hat{C}$ is what drives the result, and it eliminates both higher moments simultaneously.

Table 14: Calibrated parameters on the KOSPI 200 (2026) option chain.

Model	Parameter	Estimate
Variance Gamma	C	1.0519×10^8
	G	9.2905×10^7
	M	1.4971×10^4

The Variance Gamma skewness and kurtosis are

$$\text{skewness} = \frac{2}{\sqrt{C}} \frac{G^3 - M^3}{(G^2 + M^2)^{3/2}}, \quad \text{kurtosis} = 3 \left(1 + \frac{2}{C} \frac{G^4 + M^4}{(M^2 + G^2)^2} \right),$$

so that as $C \rightarrow \infty$ the skewness tends to zero and the kurtosis to three. The calibration drives it to $\hat{C} \approx 10^8$, so that the fitted values match the normal distribution in both moments. This is a particularly problematic outcome for the kurtosis, given that the empirical distribution the model is asked to describe has daily log returns with a kurtosis of 8.88 over 1996–2025.

In addition, three of the six starting points converged independently to this region, with a further two terminating at the penalty barrier; the only start to remain at plausible values ($C = 0.92$, $G = 0.29$, $M = 1.52$) attained a materially *worse* fit, with an RMSE of 13.80. The additional parameters are failing to improve the fit.

4.8.3 CGMY Parameterisation

The parameters for the fitted CGMY model are in Table 15.

Table 15: Calibrated parameters on the KOSPI 200 (2026) option chain.

Model	Parameter	Estimate
CGMY	C	2.9389
	G	60.0430
	M	130.2595
	Y	1.3556

We can also see the convergence to normality via its moments. We see this mathematically in the skewness and kurtosis:

$$\text{skewness} = \frac{C(M^{Y-3} - G^{Y-3}) \Gamma(3 - Y)}{[C(M^{Y-2} + G^{Y-2}) \Gamma(2 - Y)]^{3/2}},$$

$$\text{kurtosis} = 3 + \frac{C(M^{Y-4} + G^{Y-4}) \Gamma(4 - Y)}{[C(M^{Y-2} + G^{Y-2}) \Gamma(2 - Y)]^2}.$$

Since $Y < 2$, the exponents $Y - 2$, $Y - 3$, and $Y - 4$ are all negative, so the terms M^{Y-k} and G^{Y-k} shrink rapidly as k increases when G and M are large. Consequently both the excess kurtosis and the skewness are close to zero. Just as in the variance gamma and

Meixner models, the fit is therefore converging back to a normal distribution. Despite the additional parameters, we have again failed to distinguish the results from those of the Black–Scholes model.

Three of the six starting points terminated at $\hat{Y} = 2.0000$, the upper boundary of the admissible region, with $\hat{C} \rightarrow 0$; all three returned an RMSE of 12.3578, matching the Black–Scholes value to four decimal places. A further two terminated at the penalty barrier. The parameters reported above, being the lowest-RMSE solution of the six, differ wildly from every other start while scoring almost identically, indicating that the objective is near-flat across a wide region of the parameter space rather than possessing a well-located optimum. As with the Variance Gamma model, the additional parameters are being used to remove the very features that motivated the model.

4.8.4 NIG Parameterisation

The NIG parameters are in Table 16. The calibrated parameters are unusually large.

Table 16: Calibrated parameters on the KOSPI 200 (2026) option chain.

Model	Parameter	Estimate
NIG	α	1.4232×10^5
	β	3.6011×10^4
	δ	6.0471×10^4

Explicitly, we can see convergence to the normal distribution in the parameter estimates and the analytical form of skewness and kurtosis, which are

$$\text{skewness} = \frac{3\beta}{\alpha\sqrt{\delta}\sqrt{\alpha^2 - \beta^2}}, \quad \text{kurtosis} = 3 \left(1 + \frac{\alpha^2 + 4\beta^2}{\delta\alpha^2\sqrt{\alpha^2 - \beta^2}} \right),$$

both converge to their Gaussian values as α and δ grow. At the fitted values, with $\alpha \sim 10^5$ and $\delta \sim 10^4$, the denominators are of order 10^9 or larger, and both excess terms are therefore negligible. The calibration has located the Gaussian limit of the NIG family, which is precisely why the RMSE coincides with the Black–Scholes value rather than merely approaching it. Once again, we have failed to produce a model that distinguishes itself from the Black–Scholes model.

Two of the six starting points reached this limiting region independently, though at quite different magnitudes ($\alpha \approx 1.4 \times 10^5$ and $\alpha \approx 5.5 \times 10^3$), and both returned the same RMSE of 12.3578. One further start terminated at the penalty barrier, and the remaining three converged to plausible parameters at materially worse fits, with RMSE values of 13.24, 20.07 and 32.50. The pattern is consistent across all four Lévy models considered. Parameter regions exhibiting genuine skewness and excess kurtosis fit this chain substantially worse than the Gaussian corner of the same families.

4.8.5 Generalised Hyperbolic Results

The GH model is the only one of the five to improve materially on the Black–Scholes model when calibrated to the KOSPI 200 chain, reducing the RMSE by 4.40%. It is also the only model to improve where every other Lévy specification performed roughly the same as the one-parameter benchmark.

Table 17: Calibrated parameters on the KOSPI 200 (2026) option chain, with the corresponding S&P 500 estimates for comparison.

Model	Parameter	KOSPI 200	S&P 500
GH	α	36.4861	4.1106
	β	3.0902	−4.1106
	δ	18.8285	0.2492
	ν	−83.6559	−1.8603

The estimate $\hat{\nu} = -83.66$ lies far from the hyperbolic and NIG special cases of $\nu = 1$ and $\nu = -\frac{1}{2}$ respectively, so our parameterisation does not correspond to any recognised member of the class. The calibration is also not reproducible across starting points. The six multi-start runs returned RMSE values of 11.81, 11.93, 11.97, 12.02, 12.14, and 17.42, and the associated estimates of ν ranged from -83.66 to $+75.00$.

Figure 13 shows the pricing error by maturity. Error distribution among August 2026 contracts is unique to the Black–Scholes model, but this converges with later maturities, which show approximately the same error distribution.

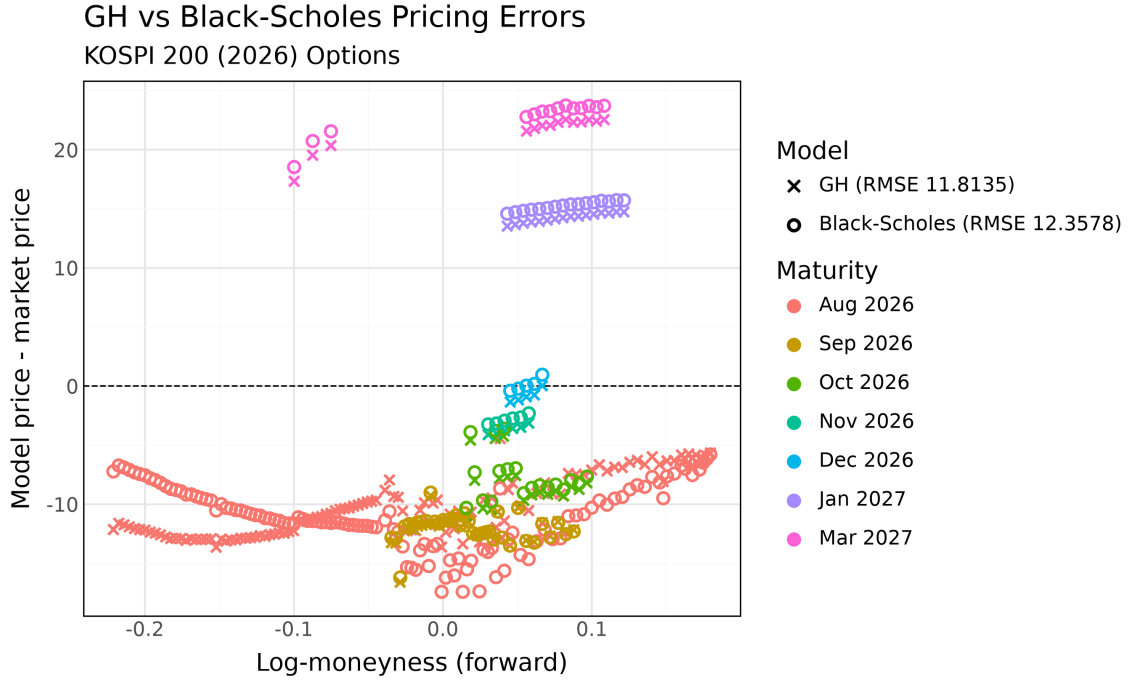


Figure 13: Generalised hyperbolic pricing error against forward log-moneyness on the KOSPI 200 (2026) chain compared to the Black–Scholes model, coloured by maturity.

4.9 Modelling by Maturity

Another approach, as we did for the Black–Scholes model, is to fit each maturity independently and examine the results.

Table 18: The five Lévy models and Black–Scholes calibrated separately to each expiry of the KOSPI 200 (2026) chain. We use RMSE as the error measure, as defined in Equation (6).

Model	Expiry							Pooled	Whole chain
	Aug 26	Sep 26	Oct 26	Nov 26	Dec 26	Jan 27	Mar 27		
Contracts	116	36	19	6	5	16	14	212	212
Black–Scholes	1.6811	1.1849	1.9069	0.3229	0.4686	0.3302	0.7473	1.4710	12.3578
Meixner	1.2649	1.1019	1.8533	0.0559	0.1610	0.0297	0.4931	1.1859	12.3251
VG	1.2664	1.0985	1.8435	0.0547	0.1597	0.0279	0.4935	1.1848	12.3581
CGMY	1.2382	1.1010	1.8508	0.0553	0.1605	0.0279	0.4935	1.1699	12.3528
NIG	1.2629	1.1019	1.8534	0.0553	0.1601	0.0279	0.4928	1.1847	12.3578
GH	1.2576	1.0985	1.8531	0.0561	0.1601	0.0279	0.4929	1.1810	11.8135

Table 18 contains these results. Here we see that modelling by maturity has significantly improved the fit of each Lévy model. The models generally converge to the same errors, and it is likely they are converging to the same fit. We check this in Figure 14[‡] and it is clear that the Lévy models all converge to the same values, as they have the same pricing error. They are, however, distinct from the Black–Scholes per-maturity model, which suggests that the fat-tailed distributions may have some merit.

We plot the new fitted volatility measurements in Figure 15. This is clear evidence that we need to account for volatility in our model, as the by-maturity volatility differs so much for each maturity date. The Lévy models produce a volatility that is consistently above the Black–Scholes estimates, most markedly at the September and October 2026 maturities. The fitted volatility also does not decline steadily as T increases for the Lévy models; instead it is more jagged. For example, for all models except the CGMY, August volatility is lower than September and October, and all models have higher volatility in December than November. This contradicts the fitted per-contract implied volatility, which declined as T increased (Figure 4). The disagreement between the Black–Scholes model and the Lévy models is largest where the number of contracts is smallest, for example six contracts at November against 116 at August. This suggests that the parameters are weakly identified at those maturities, rather than that the extra flexibility is being used.

4.10 Summary of Lévy Processes

Our Lévy models as a whole failed to substantially differentiate themselves from the Black–Scholes model; the parameters show that most converged to the normal distribution, and those that did not improved only minimally. When modelling by maturity,

[‡]The corresponding fitted values are plotted in Appendix C, Figure 24.

Pricing Errors of the Per-Maturity Calibrations KOSPI 200 (2026) Options

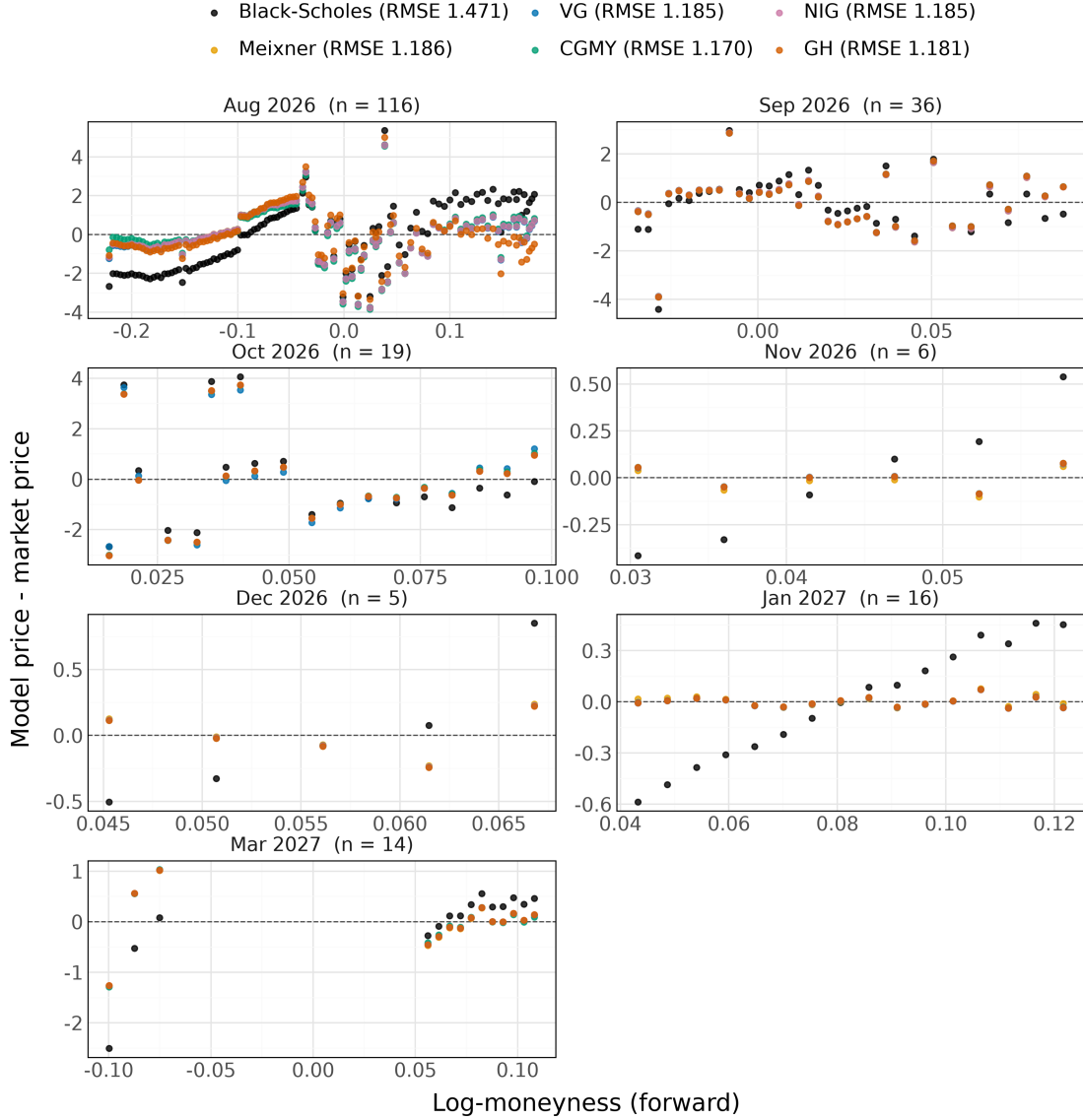


Figure 14: Pricing errors of the six per-maturity calibrations, plotted against forward log-moneyness and split by expiry, on the KOSPI 200 (2026) chain.

however, we see a clear improvement over both the Black–Scholes model and the models calibrated across all maturities at once. This suggests that the distributional assumption may be appropriate, but that, before adjusting for any non-normal distribution within each maturity, we must first allow volatility to vary across maturities. This leads us to stochastic volatility models, in which the volatility parameter itself follows a stochastic process. We consider two such approaches: a stochastic volatility extension of the Black–Scholes model, described in Section 5, and its Lévy-process counterpart, described in Section 6.

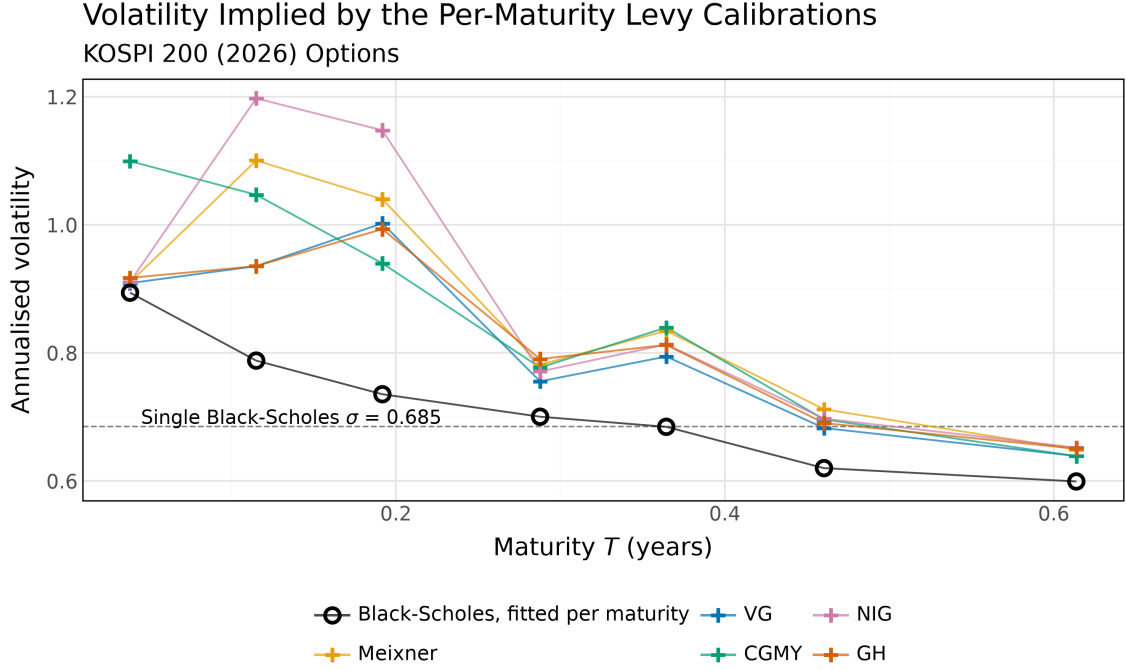


Figure 15: Annualised volatility $\sqrt{\text{Var}(\log(S_T/S_0))/T}$ implied by the per-maturity Lévy calibrations.

5 Stochastic Volatility Models

5.1 Overview

In this section we turn to *stochastic volatility*. As in the previous section, we rely on Schoutens (2003) for the models and their mathematical underpinnings. The primary alteration in these models compared the Black–Scholes model is that σ^2 is replaced by a random process $y = \{y_t, t \geq 0\}$ evolving in time. Following Schoutens (2003), we model y as an Ornstein–Uhlenbeck process driven by a Lévy subordinator, defined in Definition 5.1.

Definition 5.1 (Ornstein–Uhlenbeck Process). *Let $z = \{z_t, t \geq 0\}$ be a subordinator, i.e. a Lévy process with no Brownian component, nonnegative drift and only positive increments. Let $\lambda > 0$. A stochastic process $y = \{y_t, t \geq 0\}$ is an Ornstein–Uhlenbeck (OU) process driven by z if it satisfies the stochastic differential equation*

$$dy_t = -\lambda y_t dt + dz_{\lambda t}, \quad y_0 > 0. \quad (19)$$

The process z is called the Background Driving Lévy Process (BDLP) and λ the rate parameter. Equivalently, the solution admits the explicit representation

$$y_t = e^{-\lambda t} y_0 + \int_0^t e^{-\lambda(t-s)} dz_{\lambda s} = e^{-\lambda t} y_0 + e^{-\lambda t} \int_0^{\lambda t} e^s dz_s.$$

(Definition adapted from Schoutens (2003, §5.2).)

The two terms of Equation (19) have a direct interpretation. The stochastic differential equation states that the change in volatility over a small time step, dy_t , decomposes into

two parts. The first, $-\lambda y_t dt$, is always negative, since $\lambda > 0$ and $y_t > 0$, and pulls the process downwards. The size of this pull is proportional to the current level, so that when y_t is large the decay is rapid, and as y_t approaches zero the decay slows. The parameter λ therefore controls how quickly volatility decays: a large λ means volatility subsides quickly, while a small λ means it persists. The second term, $dz_{\lambda t}$, delivers shocks. Since z is a subordinator, it has only positive increments, so the process moves upwards only by jumps, and then tails off. The resulting pattern is a sharp upward spike followed by gradual decay, repeated irregularly. This is the behaviour observed in historical volatility, and is the component that both the Black–Scholes model and the pure Lévy models of the previous sections fail to capture.

It is our choice which distribution σ^2 should follow. We propose two different distributions with which to model the volatility. We can take σ^2 to follow a Gamma(a, b) distribution, giving the *Gamma–OU* case, or an inverse Gaussian IG(a, b) distribution, giving the *IG–OU* case. Both distributions are self-decomposable, see Definition 5.2, which is the condition required for an OU process with that specific distribution to exist.

Note that it is the stationary law of σ^2 that is being specified, not the law of the driving subordinator. Once the stationary law is fixed the BDLP z is determined by it. In the Gamma–OU case z is a compound Poisson process with jumps arriving at rate a and exponentially distributed with mean $1/b$, giving Lévy density $w(x) = ab e^{-bx}$ and cumulant function

$$k(u) = -au(b + u)^{-1}. \quad (20)$$

In the IG–OU case the corresponding BDLP has cumulant function

$$k(u) = -uab^{-1} (1 + 2ub^{-2})^{-1/2}. \quad (21)$$

In both cases k is the cumulant function of z_1 under $\mathbb{Q}$, defined by $k(u) = \log \mathbb{E}_{\mathbb{Q}}[\exp(-uz_1)]$ (Schoutens, 2003, p. 87, 88).

Definition 5.2 (Self-Decomposability). *Quoting Schoutens (2003, p. 47): “Let ϕ be the characteristic function of a random variable X . Then X is self-decomposable if*

$$\phi(u) = \phi(cu) \phi_c(u)$$

for all $u \in \mathbb{R}$ and all $c \in (0, 1)$ and for some family of characteristic functions $\{\phi_c : c \in (0, 1)\}$.” In this case the law of X is said to belong to Lévy’s class L . A random variable with law in L is infinitely divisible.

5.2 The Models

The Barndorff-Nielsen and Shephard model (BNS) is our main stochastic volatility model. We take the log stock price to follow

$$d \log S_t = \left(\mu - \frac{1}{2} \sigma_t^2\right) dt + \sigma_t dW_t + \rho dz_{\lambda t}, \quad (22)$$

where W is a standard Brownian motion and $\sigma^2 = \{\sigma_t^2, t \geq 0\}$ is an OU process driven by a subordinator z ,

$$d\sigma_t^2 = -\lambda \sigma_t^2 dt + dz_{\lambda t}.$$

The parameter ρ captures the *leverage effect*. Since z has only upward jumps, a jump in the variance moves the log price by ρ times the jump size. A negative ρ means an up-jump in volatility coincides with a down-jump in the stock price, as is typically observed in equity markets, while a positive ρ pairs it with an up-jump.

Now let

$$f_1 = iu\rho - \frac{1}{2}(u^2 + iu)(1 - e^{-\lambda t}), \quad f_2 = iu\rho - \frac{1}{2}(u^2 + iu).$$

The characteristic function of the log stock price under $\mathbb{Q}$ for the Gamma–OU case is available in closed form. Writing

$$\begin{aligned} \phi(u; t, \rho, \lambda, a, b, S_0, \sigma_0^2) &= \mathbb{E}_{\mathbb{Q}}[\exp(iu \log S_t) \mid S_0, \sigma_0] \\ &= \exp\left(iu\left(\log S_0 + (r - q - a\lambda\rho(b - \rho)^{-1})t\right)\right) \\ &\quad \times \exp\left(-\frac{1}{2}\lambda^{-1}(u^2 + iu)(1 - e^{-\lambda t})\sigma_0^2\right) \\ &\quad \times \exp\left(a(b - f_2)^{-1}\left[b\log\left(\frac{b - f_1}{b - iu\rho}\right) + f_2\lambda t\right]\right) \end{aligned} \quad (23)$$

The characteristic function for the IG–OU case is

$$\begin{aligned} \phi(u; t, \rho, \lambda, a, b, S_0, \sigma_0^2) &= \exp\left(iu\left(\log S_0 + (r - q - \rho\lambda ab^{-1}(1 - 2\rho b^{-2})^{-1/2})t\right)\right) \\ &\quad \times \exp\left(\frac{1}{2}\lambda^{-1}(-u^2 - iu)(1 - e^{-\lambda t})\sigma_0^2\right) \\ &\quad \times \exp\left(a\left(\sqrt{b^2 - 2f_1} - \sqrt{b^2 - 2iu\rho}\right)\right) \\ &\quad \times \exp\left(\frac{2af_2}{\sqrt{2f_2 - b^2}}\left(\arctan\sqrt{\frac{b^2 - 2iu\rho}{2f_2 - b^2}}\right.\right. \\ &\quad \left.\left.- \arctan\sqrt{\frac{b^2 - 2f_1}{2f_2 - b^2}}\right)\right). \end{aligned} \quad (24)$$

(Schoutens, 2003, §7.1.1)

We derived the variance of the log return, which separates into three contributions,

$$\text{Var}(\log(S_T/S_0)) \approx \underbrace{\lambda^{-1}(1 - e^{-\lambda T})\sigma_0^2}_{(i)} + \underbrace{\frac{\lambda a}{b}(T - \lambda^{-1}(1 - e^{-\lambda T}))}_{(ii)} + \underbrace{\rho^2 \lambda T \text{Var}(z_1)}_{(iii)}, \quad (25)$$

where the variance of the BDLP over one unit of time follows from the derivatives of the cumulant function k at the origin,

$$\text{Var}(z_1) = 2a/b^2 \quad (\text{Gamma–OU}), \quad \text{Var}(z_1) = 2a/b^3 \quad (\text{IG–OU}).$$

The derivation is set out in full in Appendix D.3.

5.3 Model Fitting Process

Pricing follows the same route as for the Lévy models. The characteristic function of Equation (23) or Equation (24) is substituted for ϕ in the Carr–Madan representation of Section 4.1.2, Equation (11). No separate mean-correcting term m is required, as long as we choose the risk neutral μ . Parameters are estimated by minimising the root-mean-square error between model and market mid prices across all n contracts simultaneously,

$$(\hat{\rho}, \hat{\lambda}, \hat{a}, \hat{b}, \hat{\sigma}_0^2) = \arg \min_{\rho, \lambda, a, b, \sigma_0^2} \sqrt{\frac{1}{n} \sum_{i=1}^n \left(C_{\text{BNS},i}(\rho, \lambda, a, b, \sigma_0^2) - C_{\text{mkt},i} \right)^2}.$$

The minimisation is solved by the Nelder–Mead simplex method of Appendix F, restarted from seven (Gamma–OU) and four (IG–OU) initial points and retaining the lowest-RMSE solution. The values of those starts were not stored; Tables 46 and 47 of Appendix G instead list the eight starts of a subsequent re-calibration, which recovers the same Gamma–OU optimum. Parameter restrictions are imposed throughout. We require $a > 0$ and $b > 0$ by the restrictions of the gamma distribution, $\lambda > 0$ because it is the mean-reversion rate parameter of an OU process, and $\sigma_0^2 > 0$ because it is a variance parameter.

5.4 Model Validation

Both BNS models are taken from Schoutens (2003), so before fitting either to the KOSPI 200 chain we checked them against his own S&P 500 (2002) data. Table 19 sets our calibration against the values he publishes in his Table 7.1, shading in green the parameters we reproduce and in red those we do not, on the same 30% criterion applied earlier for the Lévy models.

The Gamma–OU model reproduces cleanly. Every parameter, and the RMSE, agrees with Schoutens’ published values well inside the 30% band, the largest departure being 7.4% in a . For the IG–OU model, however, we did not recover Schoutens’ parameters, and so our implementation appears to differ slightly from his. Pricing his chain at his own published IG–OU parameters returns an RMSE of 3.548 against his 1.3225, whereas the Gamma–OU gives 1.656 against his 1.444. Four of the five parameters — ρ , λ , a and b — fall outside the 30% band, three of them by more than a factor of two; only σ_0^2 and the RMSE, ours being 1.66 against Schoutens’ 1.32, agree within it. We proceed to fit the model regardless, but any result should be flagged as differing from the Schoutens model, since if the parameters cannot be reproduced, the two implementations are not strictly the same model.

Table 19: Our calibration of each BNS model to the S&P 500 (2002) chain, against the values published in Schoutens (2003, Tables 7.1 and 7.2). Each cell is shaded by how far our value departs from Schoutens’, green where the two agree to within 30% in absolute terms, and red where they do not.

	ρ	λ	a	b	σ_0^2	RMSE
<i>Gamma-OU</i>						
ours	−1.2304	0.5976	1.3274	10.9913	0.01509	1.5156
Schoutens	−1.2606	0.5783	1.4338	11.6641	0.01450	1.4440
<i>IG-OU</i>						
ours	−1.1868	0.5223	0.3656	2.4599	0.01552	1.6576
Schoutens	−0.1926	0.0636	6.2410	0.7995	0.01560	1.3225

RMSE is defined in Equation (6).

5.5 Model Fit and Performance

The KOSPI 200 results in Table 20 show that both models vastly improve the fit, and that we have finally escaped the original Black–Scholes parameterisation. The BNS models reduce the RMSE to between 1.7998 and 1.8833, an improvement of roughly 85% over the Black–Scholes model. The Gamma-OU model achieves a lower error than the IG-OU model. Figure 16 shows market and model prices plotted against strike, a marked improvement on the Black–Scholes model. In this figure the two BNS models appear to give the same fit, but plotting the pricing errors in Figure 17 reveals that the fits are slightly different.

Table 20: Goodness-of-fit on the KOSPI 200 (2026) option chain.

Model	APE (%)	AAE	RMSE	ARPE (%)
Black–Scholes	12.43	11.4802	12.3578	17.48
Meixner	12.38	11.4342	12.3251	18.05
Variance Gamma	12.43	11.4812	12.3581	17.47
CGMY	12.43	11.4737	12.3528	17.58
NIG	12.44	11.4835	12.3578	17.48
GH	11.98	11.0616	11.8135	15.55
BNS Gamma–OU	1.41	1.2994	1.7998	1.67
BNS IG–OU	1.55	1.4311	1.8833	2.02

APE, AAE, RMSE and ARPE are defined in Equation (6).

Under a Lévy process the increments are stationary and independent, so a single parameter set must account for the entire implied volatility term structure. This is why the pricing errors of the pure Lévy models moved systematically with maturity: under the Black–Scholes model, the mean error ranges from −10.99 in August 2026 to +22.75 in March 2027, a spread the constant-volatility models could not correct. Table 21 shows

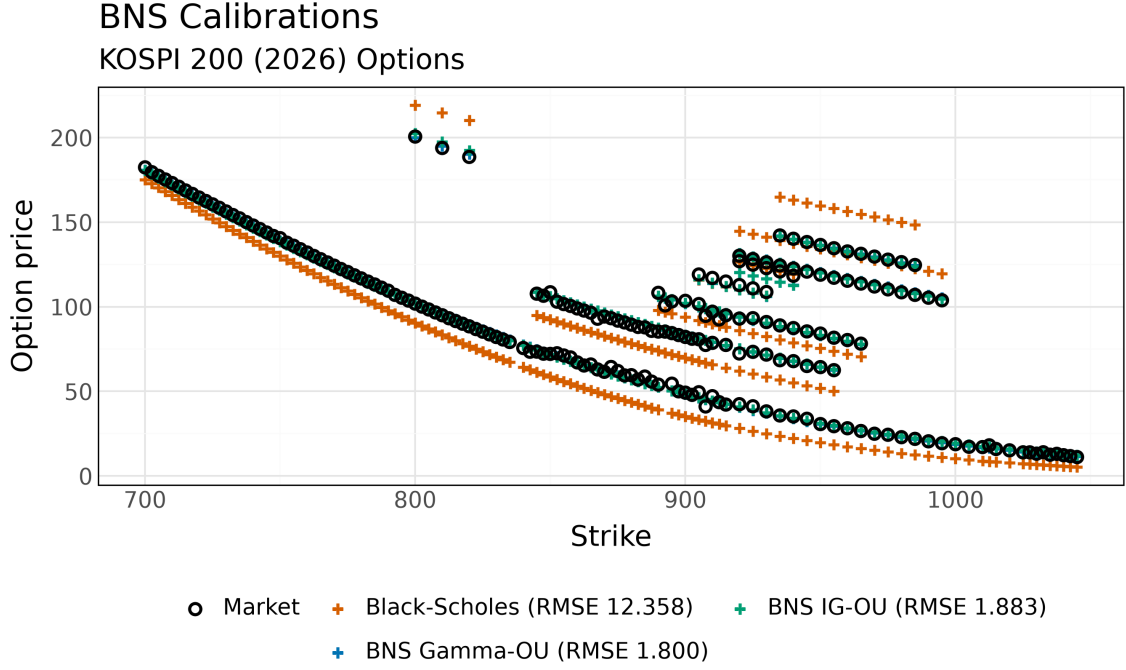


Figure 16: BNS IG–OU and BNS Gamma–OU calibration of the KOSPI 200 (2026) chain, market mid prices (circles) against fitted model prices (crosses).

that this spread largely disappears under the BNS models. The mean error on the August 2026 contracts falls to between -0.42 and -0.56 , and on the March 2027 contracts to between $+0.11$ and $+0.31$, with the Gamma-OU model lying closer to the market at both ends. Because near-term options are priced off the current variance σ_0^2 while longer-dated options revert towards the long-run level, the two ends of the term structure are no longer forced to share a single volatility parameter. The one clear exception to the improvement is the December 2026 expiry, where the Black–Scholes model, at $+0.11$, substantially outperforms both BNS models, which sit near -6 . The per-maturity models for the Black–Scholes model and Lévy models, also performed relatively worse for December 2026 expiries.

Table 21: Mean pricing error by maturity on the KOSPI 200 chain, the two BNS models against the Black–Scholes model. Best model for each row in bold.

Expiry	n	Black–Scholes	BNS Gamma–OU	BNS IG–OU
Aug 26	116	-10.99	-0.42	-0.56
Sep 26	36	-12.01	$+1.35$	$+1.47$
Oct 26	19	-7.64	$+0.20$	$+0.45$
Nov 26	6	-2.84	-2.85	-2.67
Dec 26	5	$+0.11$	-6.23	-6.24
Jan 27	16	$+15.26$	$+1.50$	$+1.15$
Mar 27	14	$+22.75$	$+0.11$	$+0.31$

Figure 17 plots the same errors against forward log moneyness, one panel per expiry, revealing the structure within each maturity that the means conceal. The mean errors

do not oscillate cleanly around zero. For the August 2026 expiry, the BNS-Gamma-OU model produces higher prices than the BNS-IG-OU model at negative log moneyness; the two coincide around the money, after which the ordering reverses, and the errors rise with log moneyness before scattering more widely in the positive tail. For September and October the two models are almost indistinguishable, with errors largely above zero in September and straddling zero in October. In November and December every contract is underpriced, most severely in December, where all five contracts cluster near -6 . In January 2027 both models sit well above zero, and the BNS-IG-OU error falls steadily as log moneyness increases, opening a clear gap below the roughly constant BNS-Gamma-OU error. In March 2027 the errors return close to zero, and the August ordering is inverted, with the BNS-IG-OU model producing the lower prices at positive log moneyness.

5.6 Model Parameterisation

The starting points of the re-calibration described in Appendix G are in Tables 46 and 47; the results reported here are those of the original seven-start run for the Gamma-OU model and four-start run for the IG-OU model. The parameters of both models are given in Table 22. For the Gamma-OU model they are far more extreme than Schoutens', and only weakly identified. Of seven dispersed starting points only one reached the reported optimum, the remainder terminating between 1.99 and 3.24. Table 23 orders them by RMSE: the best fits agree closely on λ and σ_0^2 but disagree wildly on ρ , a and b , with ρ ranging over many orders of magnitude.

However, prices respond to ρ and b almost entirely through the ratio ρ/b , as shown in Equation (25), in which ρ and b influence volatility only through that ratio. This tells us the model is overparameterised, a single ρ/b parameter would be just as effective. In the ρ/b column of Table 23, the solutions separate into three clear cases.

The best fit has $\rho/b = -0.050$, corresponding to negative leverage. The solution at $\rho = 1.13 \times 10^4$ has $b = 1.01 \times 10^9$, and hence $\rho/b \approx 10^{-5}$, corresponding to *no* leverage; it attains exactly the 1.9933 of a constrained refit at $\rho = 0$. The solution at $\rho = 1.83 \times 10^{12}$ is the same case, at $\rho/b = 0.0095$. The solutions at $\rho/b = 0.57$ and 0.49 carry positive leverage, and both fit worse. The variance can only jump upwards, so the sign of ρ , and of ρ/b , will alter the direction of the related price move. A negative ρ means volatility spikes with price falls, while a positive ρ means volatility spikes with rallies. The ordering of fits, 1.7998 for negative leverage, 1.9933 for no leverage, and 2.0406 and above for positive leverage, shows that volatility is more likely to be paired with negative price moves.

Table 22: Calibrated BNS parameters on the KOSPI 200 (2026) chain.

	ρ	λ	a	b	σ_0^2
Gamma-OU	-210.7703	9.2976	5.5663	4226.0556	0.65821
IG-OU	-0.7067	8.3587	0.0749	1.9974	0.84648

In the IG-OU case, only one of the four starts reached the reported optimum, and the different convergence points are shown in Table 24. In the best run, λ and σ_0 are similar to those of the Gamma-OU model. For two of the four successful starts $b \approx 0$, though

BNS Pricing Errors by Expiry

KOSPI 200 (2026) Options

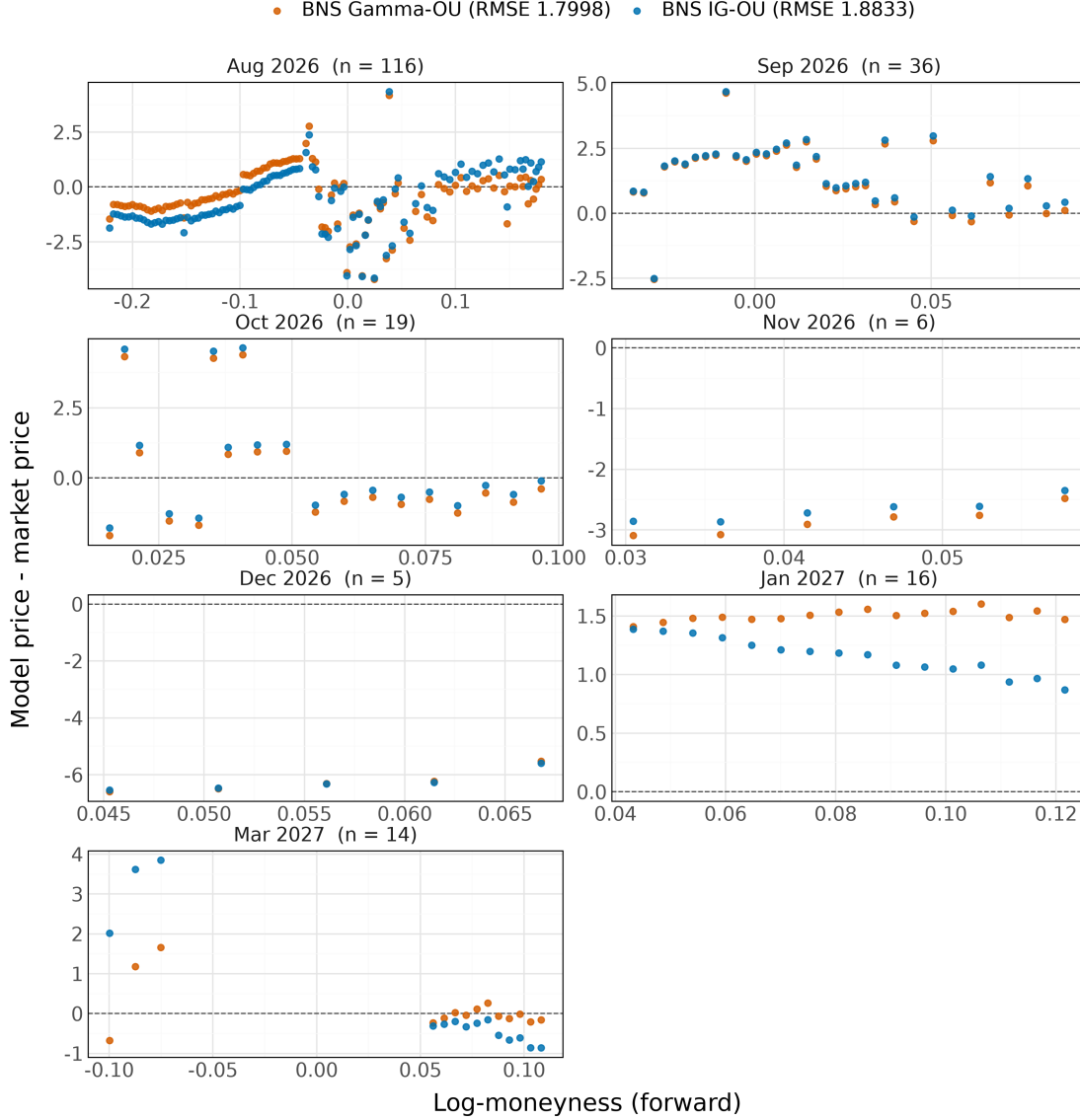


Figure 17: Pricing errors of the two BNS models, plotted against forward log-moneyness and split by expiry, on the KOSPI 200 (2026) chain. Each panel carries its own vertical scale.

still strictly greater than zero. At this value the model can still fit, but it corresponds to a degenerate inverse Gaussian distribution. The RMSE remains reasonably low, since the optimisation algorithm prioritises minimising it. By contrast, a can be zero, as in fit 1, but this produces a relatively high RMSE compared with the best fit. In one case ρ is also positive, but refitting at $\rho = 0$ gives an RMSE of 1.99, suggesting that a zero ρ is better than a positive one. The cost of removing leverage is smaller here, 5.8% against 10.8% for Gamma-OU, consistent with the fitted $\rho/b = -0.35$. With $\rho = 0$ the two BNS models converge to the same fit, RMSE 1.9933 with $a/b = 0.026$.

Table 23: BNS Gamma–OU calibration to the KOSPI 200 (2026) chain, showing terminal parameter values from each of seven dispersed starting points, ordered by goodness of fit.

Start	ρ	λ	a	b	σ_0^2	ρ/b	RMSE
1	−210.77	9.2976	5.5663	4.23×10^3	0.65821	−0.0498	1.7998
2	1.13×10^4	9.6970	2.11×10^7	1.01×10^9	0.84321	1.1×10^{-5}	1.9933
6	5.6568	2.5685	0.0019	9.9609	0.70692	0.5679	2.0406
7	1.83×10^{12}	9.7306	140.71	1.93×10^{14}	0.64478	0.0095	2.0793
4	−0.3268	4.0353	0.0206	0.0293	0.74683	−11.15	2.6134
5	−19.5336	3.2437	0.0005	0.0018	0.77404	-1.09×10^4	3.0832
3	0.0346	0.4019	0.2985	0.0706	0.64285	0.4901	3.2426

RMSE is defined in Equation (6).

Table 24: BNS IG–OU calibration to the KOSPI 200 (2026) chain, showing terminal parameter values from each of four dispersed starting points, ordered by goodness of fit.

Start	ρ	λ	a	b	σ_0^2	ρ/b	RMSE
4	−0.7067	8.3587	0.0749	1.9974	0.84648	−0.3538	1.8833
3	−2.2982	4.9911	0.0119	0.0000	0.79978	—	2.5130
1	2.9550	3.2227	0.0000	4.1886	0.77415	0.7055	3.0928
2	−0.1843	0.3056	0.0000	0.0000	0.49702	—	11.0860

RMSE is defined in Equation (6).

5.7 Model Interpretation

In this section we examine the model parameters and how they affect the variance, kurtosis, and the fitted density of log returns. These models do not have their exact moments and variance available in closed form, so instead we derive them from the characteristic function, using the method of Appendix D.7.2. The respective densities are derived following the same process as for the Lévy distributions, described in Appendix D.7.1.

5.7.1 Fitted Volatility

We plot the fitted volatility, $\sqrt{\text{Var}(\log(S_T/S_0))/T}$, in Figure 18. The figure shows why the Gamma–OU model performs well: it tracks the implied volatility structure from the Black–Scholes model per-maturity modelling of Section 3.5. The IG–OU alternative performs less well because it fails to do this, consistently setting the implied volatility levels too high.

This can also be seen in the fitted values, using Equation (25). We estimate the components of this equation, (i), (ii), and (iii), and report them in Table 25. The parameters λ and σ_0^2 govern term (i). For the Gamma–OU fit $\sigma_0^2 = 0.658$, an initial volatility of $\sigma_0 = 81.1\%$. This is the instantaneous volatility at $t = 0$, not the annualised volatility of Figure 18, which is $\sqrt{\text{Var}(\log(S_T/S_0))/T}$ and accumulates all three terms over $[0, T]$: at August the initial variance alone would give 74.5%, the leverage term lifts the total to the 90.2% plotted. The mean-reversion rate is $\lambda = 9.30$. At the August 2026 maturity

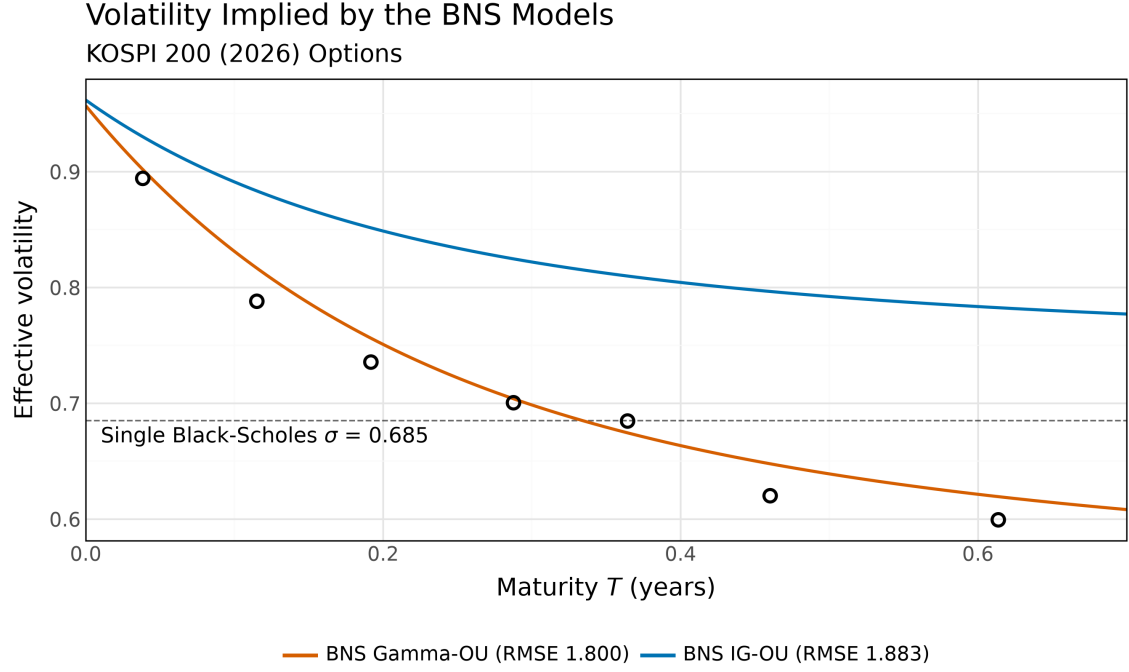


Figure 18: Annualised volatility implied by the reported BNS Gamma–OU and IG–OU best fits of Table 22. Rings mark the per-maturity Black–Scholes volatilities of Table 6 and the dashed line the single volatility fitted to the whole chain.

$e^{-\lambda T} = 0.70$, so 70% of the initial variance has yet to decay, and this term supplies 0.0212 of a total variance of 0.0312*, some two thirds. By March 2027 the factor is approximately zero and the other two terms have grown linearly in T .

Table 25: Contributions to $\text{Var}(\log(S_T/S_0))$ under Equation (25), for the calibrated parameters of Table 22. Columns (i)–(iii) are the three terms of that equation, in order.

Model	Expiry	(i)	(ii)	(iii)	Total	$\hat{\sigma}$
Gamma–OU	Aug 26	0.0212	0.0001	0.0099	0.0312	0.902
	Mar 27	0.0706	0.0062	0.1580	0.2348	0.619
IG–OU	Aug 26	0.0278	0.0017	0.0030	0.0325	0.921
	Mar 27	0.1007	0.1551	0.0482	0.3039	0.704

What separates the two models is terms (ii) and (iii). This is noticeable particularly in the full decomposition of Equation (25) in Table 25. The Gamma–OU fit reaches $b = 4226$, which makes the jump level $\lambda a/b = 0.0122$ almost negligible. Term (ii) contributes 0.0062 at March, under 3% of the total. Its long-dated variance is controlled by term (iii), with $\rho = -210.77$, since $\text{Var}(z_1) = 2a/b^2$ is itself tiny at that b and only a large ρ^2 can compensate. The IG–OU fit works the other way round. Its $b = 1.997$ leaves $\lambda a/b = 0.3134$, so term (ii) supplies 0.1551 at March, over half the total, while its modest $\rho = -0.71$ contributes only 0.0482.

*The fitted volatility is $\sqrt{\text{Var}(\log(S_T/S_0))/T}$, which accounts for the small value of the variance.

5.7.2 Fitted Densities

Finally, using the method of Appendix D.7.1[†], we compare the fitted densities against the Black–Scholes model in Figure 19. We had difficulty producing the BNS-IG-OU fit, so we present only the BNS-Gamma-OU fit. There is a clear departure from the Black–Scholes model at both the near-term August 2026 contracts and the long-term March 2027 contracts. For August 2026 the distribution has wider variance than the Black–Scholes model, indicated by its greater spread and lower density at the mean. For March 2027 the distribution has less variance, clearly shown by its lesser spread and increased density at the mean. This suggests that the stochastic volatility models are doing what they were intended to, since the distribution now differs noticeably across maturities.

Given kurtosis is difficult to diagnose using the charts, as it is most pronounced at the tails, we compute these values in Table 26. They were computed via the method of Appendix D.7.2, which shows that the BNS-Gamma-OU model has increased kurtosis that declines as the maturity T increases. The BNS-IG-OU model has very high kurtosis, which may explain its slightly higher errors.

Table 26: Excess kurtosis of the risk-neutral log-return law for Black–Scholes, fitted separately to each expiry, and the two BNS models on the KOSPI 200 (2026) chain, by maturity. Best model highlighted in yellow.

	Black– Scholes	BNS Gamma–OU	BNS IG–OU
Aug 26	0.0000	0.304	7.748
Sep 26	0.0000	0.152	8.537
Oct 26	0.0000	0.125	9.501
Nov 26	0.0000	0.116	9.888
Dec 26	0.0000	0.112	9.767
Jan 27	0.0000	0.107	9.340
Mar 27	0.0000	0.096	8.449
RMSE	12.358	1.800	1.883

These stochastic volatility models have been sufficient to significantly reduce the RMSE, providing a much better fit. The next step is to apply them to Lévy processes via stochastic time models.

6 Stochastic Time Models

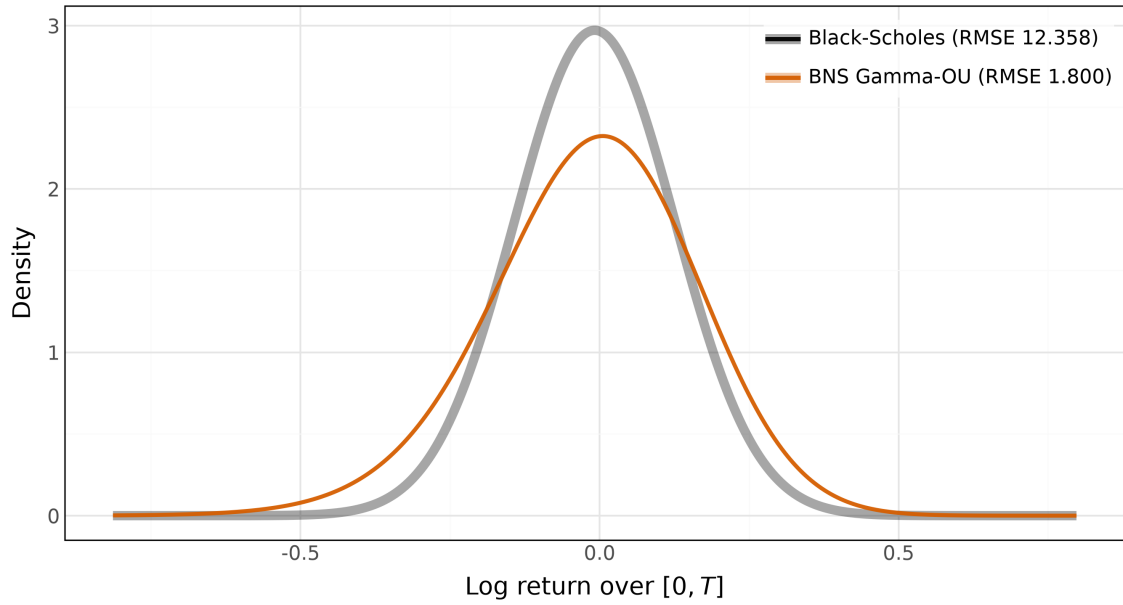
6.1 Overview

An alternative way of making volatility stochastic is to make time itself stochastic. The intuition is that, in periods of high volatility, time runs faster than in periods of low volatility

[†]Except we replace the exponential form $\phi_T(u) = \exp(T\psi_X(u) + iumT)$ with Equation (23). Since it returns the characteristic function of $\log S_T$ rather than of the log return, it is shifted by $e^{-iu \log S_0}$ before inversion so that the densities are drawn on a common scale.

Log-Return Densities: BNS Gamma-OU

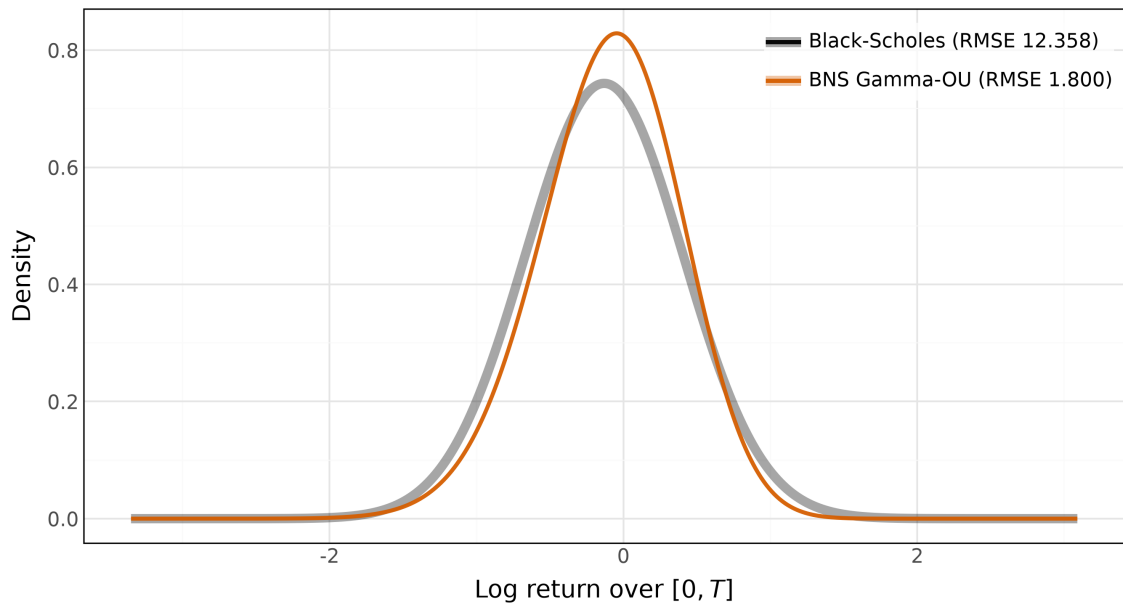
August 2026



(a) Front month August 2026, $T = 0.038$.

Log-Return Densities: BNS Gamma-OU

March 2027



(b) Longest maturity March 2027, $T = 0.614$.

Figure 19: Log-return densities of the calibrated BNS Gamma-OU model against the Black-Scholes normal, drawn as the heavy grey line, on the KOSPI 200 (2026) chain.

(Schoutens, 2003). In this section we rely on Schoutens (2003) for the models and their mathematical components, applied here to our KOSPI data. He describes two ways of formulating stochastic time-change models, namely the integrated CIR process and the integrated OU time change.

6.1.1 Integrated CIR Process

The CIR process refers to the Cox–Ingersoll–Ross process for y , which has the stochastic differential equation

$$dy_t = \kappa(\eta - y_t) dt + \lambda y_t^{1/2} dW_t,$$

where W_t is a standard Brownian motion, κ is the rate of mean reversion, η is the long-run rate of time change, and λ is the volatility of time change. The Integrated CIR process is denoted

$$Y_t = \int_0^t y_s ds,$$

where y_t is the CIR process; note that y is a positive process. The characteristic function of Y_t , given y_0 , is the following

$$\begin{aligned} \phi_{\text{CIR}}(u; t, \kappa, \eta, \lambda, y_0) &= E[\exp(iuY_t) | y_0] \\ &= \frac{\exp(\kappa^2 \eta t / \lambda^2) \exp(2y_0 i u / (\kappa + \gamma \coth(\gamma t / 2)))}{(\cosh(\gamma t / 2) + \kappa \sinh(\gamma t / 2) / \gamma)^{2\kappa \eta / \lambda^2}}, \end{aligned}$$

where

$$\gamma = \sqrt{\kappa^2 - 2\lambda^2 i u}.$$

6.1.2 Integrated OU Time Change

The integrated OU time change consists of the Gamma–OU method and the IG–OU method. When describing these methods together we call them the intOU models. These methods are OU processes driven by a subordinator, and so contain jumps in the rate of time. The rate of time change is now a solution to the SDE

$$dy_t = -\lambda y_t dt + dz_{\lambda t}. \quad (26)$$

In this instance, z is our subordinator, a non-decreasing Lévy process. The time elapsed is given by $Y_t = \int_0^t y_s ds$. There exists a Gamma(a, b) distribution that is self-decomposable, which implies that there exists an OU process following the dynamics of Equation (26). This gives the characteristic function

$$\begin{aligned} \phi_{\text{Gamma-OU}}(u; t, \lambda, a, b, y_0) &= E[\exp(iuY_t) | y_0] \\ &= \exp\left(iu y_0 \lambda^{-1} (1 - e^{-\lambda t}) + \frac{\lambda a}{iu - \lambda b} \left(b \log\left(\frac{b}{b - iu \lambda^{-1} (1 - e^{-\lambda t})}\right) - iut\right)\right). \end{aligned}$$

The IG–OU method follows the same idea but with an IG distribution, which in turn gives the characteristic function

$$\begin{aligned}\phi_{\text{IG-OU}}(u; t, \lambda, a, b, y_0) &= E[\exp(iuY_t) \mid y_0] \\ &= \exp\left(\frac{iu y_0}{\lambda}(1 - e^{-\lambda t}) + \frac{2aiu}{b\lambda}A(u, t)\right),\end{aligned}$$

where

$$\begin{aligned}A(u, t) &= \frac{1 - \sqrt{1 + \kappa(1 - e^{-\lambda t})}}{\kappa} \\ &\quad + \frac{1}{\sqrt{1 + \kappa}} \left(\operatorname{artanh}\left(\frac{\sqrt{1 + \kappa(1 - e^{-\lambda t})}}{\sqrt{1 + \kappa}}\right) - \operatorname{artanh}\left(\frac{1}{\sqrt{1 + \kappa}}\right) \right), \\ \kappa &= -2b^{-2}iu/\lambda.\end{aligned}$$

6.2 The Model

We model the stock price as

$$S_t = S_0 \frac{\exp((r - q)t)}{\mathbb{E}(\exp(X_{Y_t}) \mid y_0)} \exp(X_{Y_t}), \quad (27)$$

with X as our Lévy process with a Y_t process attached to it. Write ψ_X for the characteristic exponent of the Lévy process X , so that $E[\exp(iuX_t)] = \exp(t\psi_X(u))$, and $\phi_Y(u; t, y_0)$ for the characteristic function of the integrated time change Y_t given y_0 . The log stock price under $\mathbb{Q}$ then has characteristic function

$$\begin{aligned}\phi(u) &= \phi(u; t, S_0, y_0) \\ &= \mathbb{E}(\exp(iu \log(S_t)) \mid S_0, y_0) \\ &= \exp(iu((r - q)t + \log S_0)) \frac{\phi_Y(-i\psi_X(u); t, y_0)}{\phi_Y(-i\psi_X(-i); t, y_0)^{iu}}.\end{aligned} \quad (28)$$

Here y_0 is the initial rate of time change, fixed at $y_0 = 1$. The characteristic exponent ψ_X carries the parameters of the underlying Lévy model, and ϕ_Y , the characteristic function of the integrated time change $Y_t = \int_0^t y_s ds$, carries those of the chosen time process.

Under each stochastic time model, we can model the stock price process as a Lévy process. This means that, for each Lévy process, we have three potential stochastic time processes, giving fifteen potential models in total. We implement twelve of these, covering the Meixner, variance gamma, CGMY and NIG families; the generalised hyperbolic case is set aside for the reasons given in Section 6.3.

We again adopt the Carr and Madan (1999) approach of Section 4.1.2, which requires only the characteristic function, and price through Equation (11) with the characteristic function of Equation (28),

$$\phi(u) = \exp(iu((r - q)t + \log S_0)) \frac{\phi_Y(-i\psi_X(u); t, y_0)}{\phi_Y(-i\psi_X(-i); t, y_0)^{iu}},$$

in which ψ_X is the characteristic exponent of the Lévy process. See Appendix D.1 for a list of characteristic functions for each model.

We fitted each of these twelve models by the same Nelder–Mead search of Appendix F, from eight starting points, retaining the lowest-RMSE solution. The starting points combine Schoutens (2003, Table 7.3) estimates with dispersed alternatives, the first three of each set taken from his published fits; they are listed in Tables 48 to 59 of Appendix G. The retained optimum is only weakly identified: for ten of the twelve models no other start came within 1% of the retained RMSE, and for the remaining two, CGMY-CIR and VG-IG-OU, only one other did. We can compare the models on the basis of RMSE, but this weak identification must be borne in mind when interpreting the parameter values themselves.

6.3 Model Validation

Every model in this section is taken from Schoutens (2003), so before fitting anything to the KOSPI 200 chain we checked each one against his own S&P 500 (2002) data. Table 39, reproduced in Appendix C, sets our calibration against the values he publishes in his Table 7.3, parameter by parameter, shading in green the parameters we reproduce and in red those we do not. There are some models of concern, containing parameters that are 30% smaller or larger than Schoutens', such as the Meixner-IG-OU, the CGMY models, and the NIG intOU models. However, all of these models produce an RMSE very close to Schoutens', so we are not unduly concerned about our calibration process and choose to fit them.

The main model of concern is the generalised hyperbolic model, where the difficulty is of a different kind. Even when we fix Schoutens' given parameters, rather than estimating them ourselves, the model produces results very different from his, with significantly higher RMSEs. We therefore choose not to fit this model. This leaves us with twelve models: four families of Lévy process — Meixner, VG, CGMY, and NIG — each combined with three stochastic time components.

6.4 Model Fits

We obtained results for every model discussed. Table 27 collects the goodness-of-fit measures for every model calibrated in this dissertation, from the Black–Scholes benchmark through the pure Lévy models to the stochastic volatility and stochastic time-change models.

The stochastic time models, like the earlier BNS models, improve considerably relative to both the Black–Scholes model and the pure Lévy models. This improvement holds for all the models we attempted. The Meixner-IG-OU and VG-Gamma-OU models are the best performing, on (APE, AAE) and (RMSE, ARPE) respectively. However, they belong to a collection of models that perform approximately equally well. These best-performing models are the BNS-Gamma-OU, Meixner-Gamma-OU, Meixner-IG-OU, VG-Gamma-OU, and VG-IG-OU. If we had to choose a single model, it would likely be the BNS-Gamma-OU from Section 5, as it has the fewest parameters, with only five as opposed to six for the other best models. Moreover, this model is already over-parameterised,

since a single ρ/b parameter would suffice, further reducing the count. Across all models, the CIR stochastic time change was noticeably weaker, although less so for the CGMY-CIR model, which was the best-performing CIR model. Neither the NIG nor the CGMY family had a model in the best-performing group.

The CGMY process contains the variance gamma process as the special case $Y = 0$, so under any clock the CGMY optimum can be no worse than the VG optimum. Yet the CGMY-Gamma-OU reports 2.6367 against 1.7746 for the VG-Gamma-OU. It is clear that the CGMY-Gamma-OU search did not locate its true optimum. As shown in Table 32, the calibrated $Y = 1.96$ sits close to the boundary $Y < 2$. To prevent Y exceeding 2, we impose an RMSE of 10^6 whenever $Y > 2$, and it is likely the simplex became stuck against this boundary during its search. The CGMY-IG-OU fit an RMSE of 1.8038 against VG-IG-OU's 1.8034 shows the same failure on a much smaller scale. This is partially why the CGMY-IG-OU model is not included in the best model list despite very similar error metrics, it is not the true optimum and is outperformed by its VG counterpart. We report the CGMY figures as obtained, rather than replacing them with the VG results, but they should be read as a failed optimisation.

Table 27: Goodness-of-fit for every model calibrated to the KOSPI 200 (2026) option chain. The best value in each column is in bold, and the best-fitting models are highlighted. A model is counted as best-fitting if it satisfies all four thresholds: $APE \leq 1.45$, $AAE \leq 1.31$, $RMSE \leq 1.82$, and $ARPE \leq 1.8$.

Model	APE (%)	AAE	RMSE	ARPE (%)
Black–Scholes	12.43	11.4802	12.3578	17.48
Meixner	12.38	11.4342	12.3251	18.05
Variance Gamma	12.43	11.4812	12.3581	17.47
CGMY	12.43	11.4737	12.3528	17.58
NIG	12.44	11.4835	12.3578	17.48
GH	11.98	11.0616	11.8135	15.55
BNS Gamma–OU	1.41	1.2994	1.7998	1.67
BNS IG–OU	1.55	1.4311	1.8833	2.02
Meixner–CIR	2.91	2.6839	3.3936	5.55
Meixner–Gamma–OU	1.41	1.3044	1.8177	1.74
Meixner–IG–OU	1.39	1.2873	1.8127	1.74
VG–CIR	2.78	2.5686	3.3218	4.86
VG–Gamma–OU	1.41	1.3050	1.7746	1.65
VG–IG–OU	1.42	1.3095	1.8034	1.68
CGMY–CIR	1.79	1.6514	2.0492	2.96
CGMY–Gamma–OU	1.81	1.6732	2.6367	3.11
CGMY–IG–OU	1.45	1.3386	1.8038	1.88
NIG–CIR	3.05	2.8153	3.7479	5.21
NIG–Gamma–OU	1.50	1.3841	1.8776	2.16
NIG–IG–OU	3.54	3.2672	3.8426	4.66

APE, AAE, RMSE and ARPE are defined in Equation (6).

Table 28 examines how each model performs per maturity. Interestingly, for five of the seven maturities the best-performing model was not one of our group of best-performing models overall. It is likely that our best models perform a good balancing act across all maturities, rather than minimising the error at any single one. The CGMY-Gamma-OU model was the only model to perform best at more than one expiry, being especially strong on the near-term August and September 2026 contracts. The best model for the December 2026 expiry was a non-stochastic-volatility model, the NIG model. The CIR models also perform best at two maturities, Meixner-CIR for October 2026 and VG-CIR for November 2026. This is despite the CIR clock being, on average, a weaker stochastic time change model than the intOU models.

Table 28: Mean pricing error by maturity for every model calibrated to the KOSPI 200 (2026) option chain. The model closest to zero in each column is in bold, and the best-fitting models of Table 27 are highlighted.

Model	Expiry						
	Aug 26 ($n = 116$)	Sep 26 (36)	Oct 26 (19)	Nov 26 (6)	Dec 26 (5)	Jan 27 (16)	Mar 27 (14)
Black–Scholes	−10.99	−12.01	−7.64	−2.84	+0.11	+15.26	+22.75
Meixner	−10.87	−12.04	−7.77	−2.85	+0.10	+15.21	+22.87
Variance Gamma	−10.99	−12.01	−7.65	−2.85	+0.11	+15.26	+22.74
CGMY	−10.97	−12.02	−7.67	−2.84	+0.11	+15.25	+22.77
NIG	−10.99	−12.02	−7.66	−2.86	+0.09	+15.23	+22.72
GH	−10.30	−12.14	−8.25	−3.69	−0.83	+14.23	+21.56
BNS Gamma–OU	−0.42	+1.35	+0.20	−2.85	−6.23	+1.50	+0.11
BNS IG–OU	−0.56	+1.47	+0.45	−2.67	−6.24	+1.15	+0.31
Meixner–CIR	+0.56	+1.02	+0.08	−3.46	−6.80	+1.45	+0.73
Meixner–Gamma–OU	−0.45	+1.43	+0.36	−2.64	−6.13	+1.33	+0.04
Meixner–IG–OU	−0.36	+1.33	+0.15	−3.00	−6.33	+1.59	+0.10
VG–CIR	−1.22	+1.77	+2.73	−0.13	−4.24	+2.02	−1.94
VG–Gamma–OU	−0.41	+1.25	+0.20	−2.72	−6.08	+1.52	+0.08
VG–IG–OU	−0.41	+1.30	+0.19	−2.91	−6.24	+1.66	−0.01
CGMY–CIR	−0.77	+1.16	+1.59	−1.69	−5.55	+1.28	−0.10
CGMY–Gamma–OU	+0.05	−0.60	+1.13	−0.42	−3.06	+4.97	−3.34
CGMY–IG–OU	−0.40	+1.31	+0.15	−2.92	−6.28	+1.47	+0.20
NIG–CIR	−2.10	+0.80	+3.26	+0.42	−3.71	+2.21	−0.17
NIG–Gamma–OU	−0.42	+1.49	+0.33	−2.92	−6.43	+1.17	+0.37
NIG–IG–OU	−3.13	+1.48	+4.34	+2.85	−0.99	+4.46	−4.47

Entries are mean signed errors, so a positive value indicates the model overprices that expiry on average.

We compare the best models against the fitted data in Figure 20. It is clear that these models improve considerably on the Black–Scholes model and on the Lévy models whose fitted values are shown in Figure 11. It is also evident that the models obtain very similar fits.

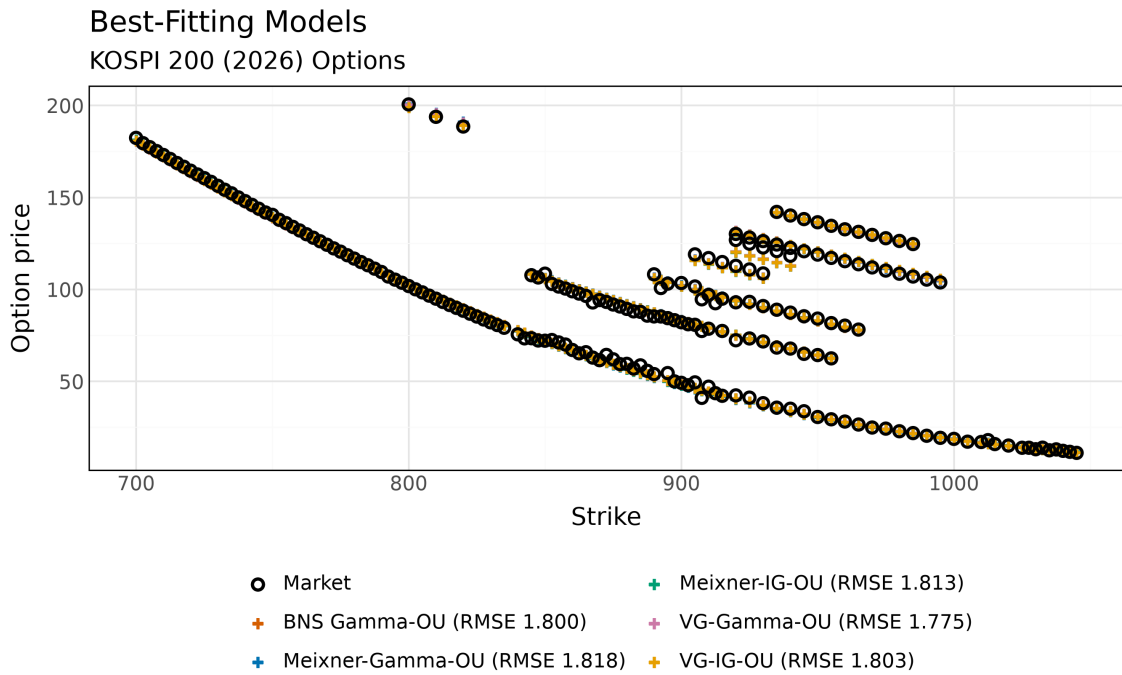


Figure 20: The five best-fitting models of Table 27 against the fitted values, on the KOSPI 200 (2026) chain.

The pricing error chart in Figure 21 shows the pricing error by each one of the best models for each maturity. For almost every month the five best models produce near-identical errors, their points overlapping closely. The exceptions are January and March 2027. For January 2027 the models fan out at large log moneyness, and hence large strike: the Meixner-Gamma-OU model drifts well below the others, while the VG-IG-OU model sits highest, a spread of roughly 0.8 between them at the largest strikes. The March 2027 contracts separate at negative log moneyness. Even for these best-fitting models, the errors do not oscillate around zero within each maturity; they sit largely above zero at some expiries and below at others. This suggests that, while these models outperform all those tried so far, some systematic bias remains and the fit could be improved further.

Pricing Errors of the Best-Fitting Models by Expiry KOSPI 200 (2026) Options

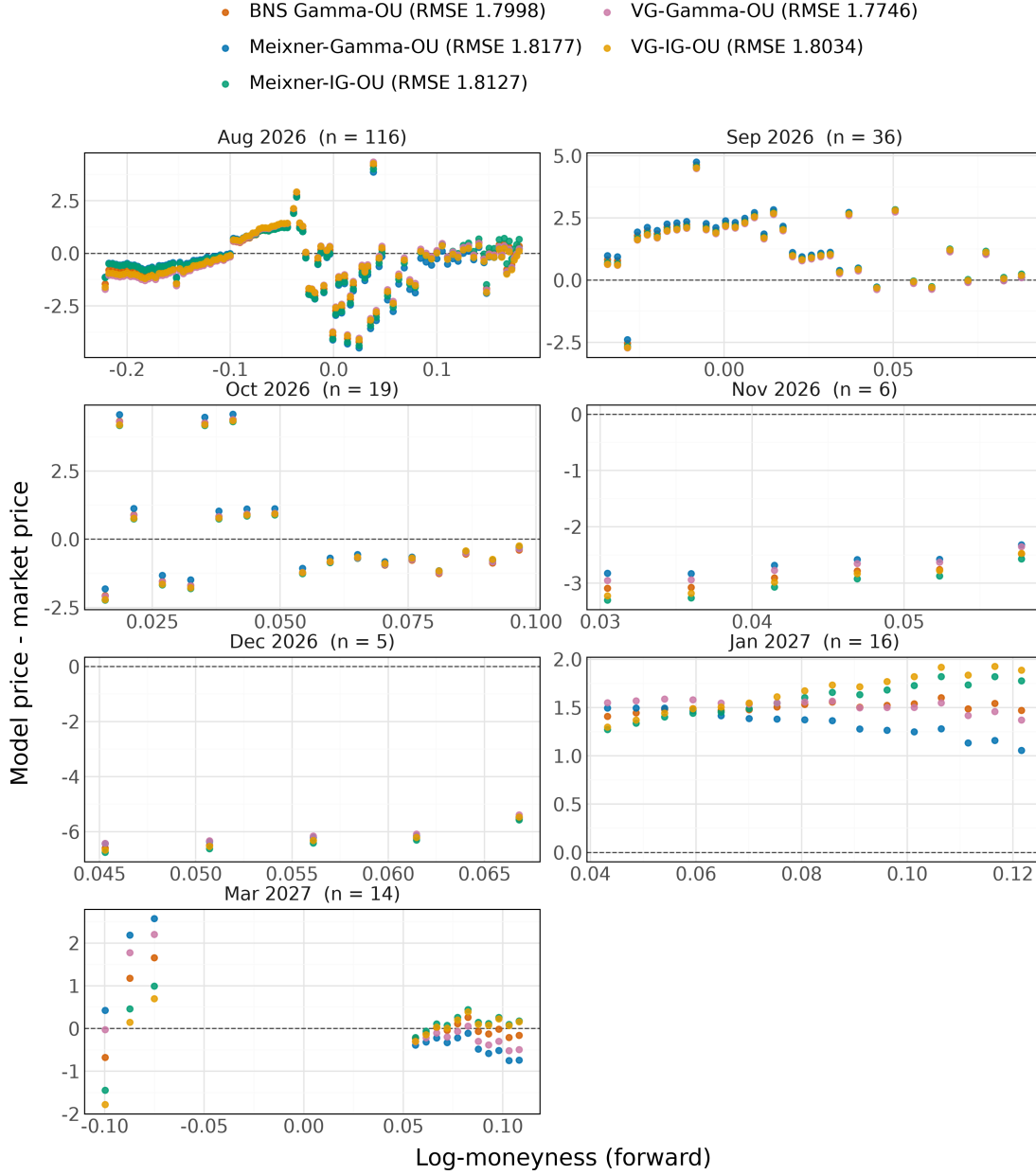


Figure 21: Pricing errors of the five best-fitting models of Table 27, plotted against forward log-moneyness and split by expiry, on the KOSPI 200 (2026) chain. Each panel carries its own vertical scale.

We can also recover each density curve using the process described in Appendix D.7.1*. The resulting density curves for the best models are shown in Figure 22. The five models produce very similar densities to one another, including to the already discussed BNS-

*Except we replace the exponential form $\phi_T(u) = \exp(T\psi_X(u) + iumT)$ with Equation (23). Since it returns the characteristic function of $\log S_T$ rather than of the log return, it is shifted by $e^{-iu \log S_0}$ before inversion so that the densities are drawn on a common scale.

Gamma-OU model, but differ from the Black–Scholes normal in opposite directions at the two maturities. For the August 2026 contracts there is more spread and a lower density at the mean, indicating increased variance relative to the Black–Scholes model; for March 2027 there is greater density at the mean and less spread, indicating reduced variance. Kurtosis is difficult to examine solely from the chart, given it is most noticeable at the tails.

The other stochastic time models had more unusual distributions, shown in Figures 25 and 26 of Appendix C, which have contributed to their higher RMSE relative to the best models. The exception is the NIG-Gamma-OU model, which has a fit similar to the best models. The CIR models had the most noticeably misfitting distributions.

We can computationally check the kurtosis using the characteristic function and the method of Appendix D.7.2; this is shown for the best-fitting models in Table 29. Across the best models the kurtosis differs slightly, ranging from 0.541 for the Meixner-IG-OU model at August 2026 to 0.005 for the VG-Gamma-OU model at the same maturity. The highest excess kurtosis is therefore only fairly modest. The most interesting feature is how the kurtosis changes over time. The VG-Gamma-OU model has increasing kurtosis as T increases; the BNS-Gamma-OU, VG-IG-OU, and Meixner-IG-OU models have decreasing kurtosis; and the Meixner-Gamma-OU model has a U-shaped structure. The kurtoses of the remaining models are given in Table 38 of Appendix C; these are higher than the best models at all maturities.

Table 29: Excess kurtosis of the risk-neutral log-return law for the best-fitting models on the KOSPI 200 (2026) chain, by maturity. Columns are ordered by RMSE.

Expiry	VG– Gamma–OU	BNS Gamma–OU	VG– IG–OU	Meixner– IG–OU	Meixner– Gamma–OU
Aug 26	0.005	0.304	0.134	0.541	0.473
Sep 26	0.033	0.152	0.046	0.267	0.294
Oct 26	0.075	0.125	0.048	0.172	0.318
Nov 26	0.132	0.116	0.028	0.144	0.364
Dec 26	0.168	0.112	0.029	0.130	0.389
Jan 27	0.199	0.107	0.039	0.107	0.406
Mar 27	0.222	0.096	0.017	0.102	0.406
RMSE	1.775	1.800	1.803	1.813	1.818

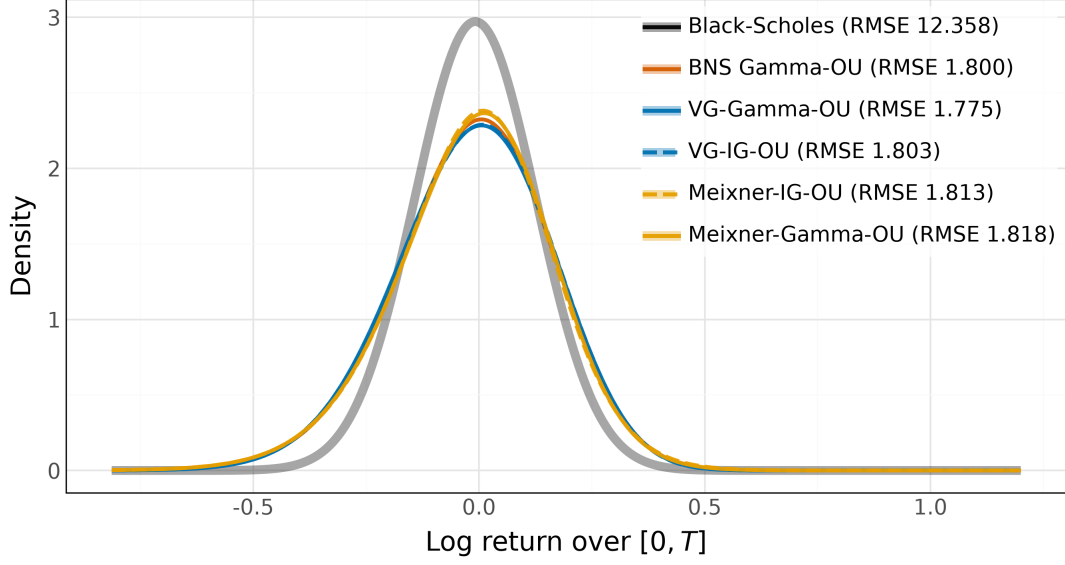
We also examine the structure of implied volatility for the best-fitting models. All models follow the per-maturity BlackScholes model volatilities closely. The implied volatility curves for the remaining models are shown in Figure 27 of Appendix C; these differ markedly from the per-maturity Black–Scholes fit, which suggests why the models fit worse.

6.5 Model Parameterisations

So far we have examined the model fits, their corresponding densities, and their volatility. The next step is to examine what in the parameters caused these results. To do this, we

Log-Return Densities: Best-Fitting Models

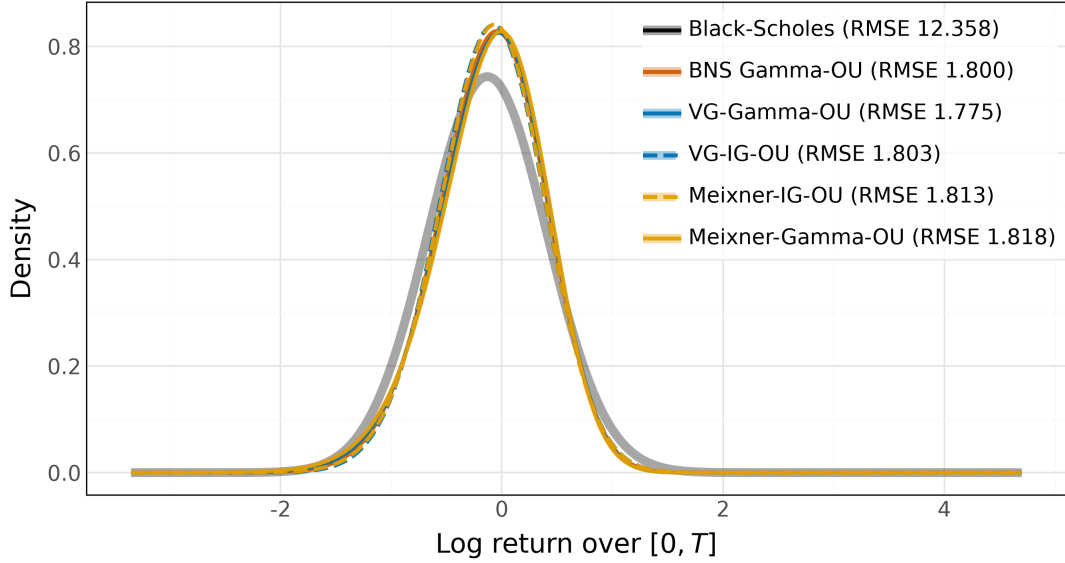
August 2026



(a) Front month August 2026, $T = 0.038$.

Log-Return Densities: Best-Fitting Models

March 2027



(b) Longest maturity March 2027, $T = 0.614$.

Figure 22: Log-return densities of the five best-fitting models of Table 27, against the Black–Scholes model.

need to understand what within the time-change model specifically drives the volatility curve. We therefore work from the closed-form variance $\text{Var}(X_{Y_T})$,[†] which is derived in

[†]The log return of Equation (27) differs from X_{Y_T} only by the deterministic term $\exp((r - q)T)/\mathbb{E}[\exp(X_{Y_T}) \mid y_0]$, which contributes an additive constant to $\log(S_T/S_0)$ and so leaves the variance unchanged. Hence $\text{Var}(\log(S_T/S_0)) = \text{Var}(X_{Y_T})$, and the two are used interchangeably.

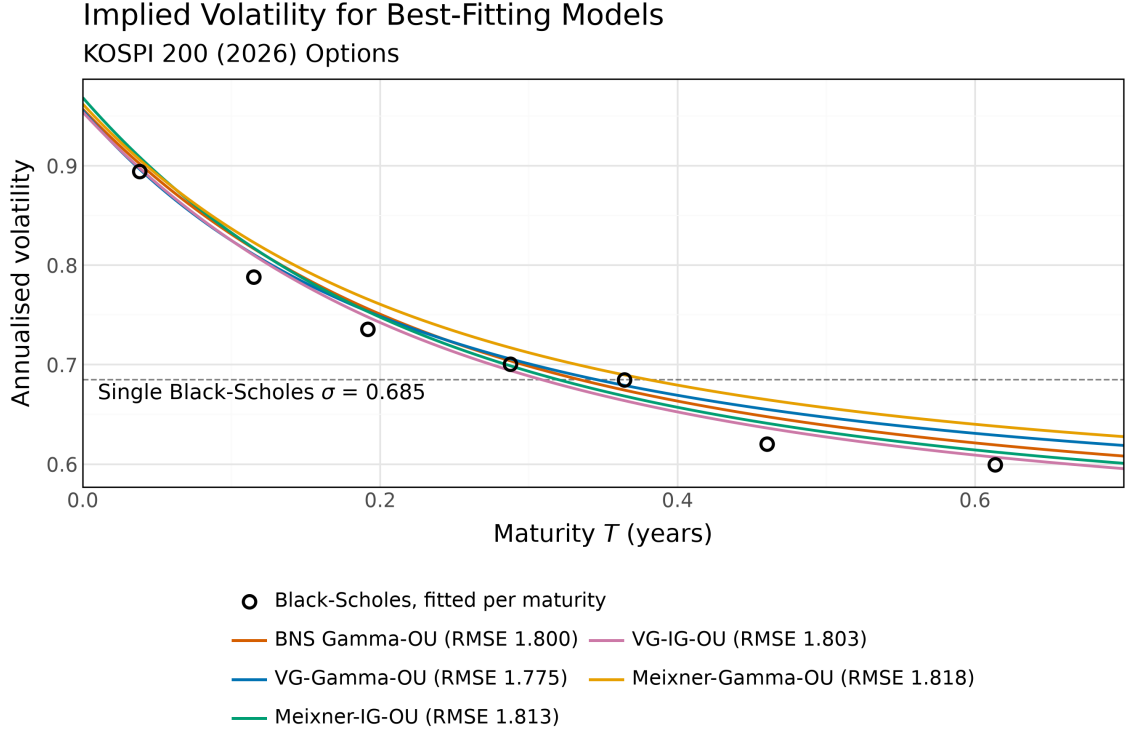


Figure 23: Annualised volatility $\sqrt{\text{Var}(\log(S_T/S_0))/T}$ implied by the five best-fitting models, over the range of maturities quoted on the KOSPI 200 (2026) chain. Rings mark the per-maturity Black–Scholes volatilities of Table 6 and the dashed line the single volatility fitted to the whole chain.

Appendix D.4. For the Gamma-OU case,

$$\text{Var}(X_{Y_T}) = \text{Var}(X_1) \left[\varepsilon(T) + \frac{a}{b} (T - \varepsilon(T)) \right] + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^2} \Psi(T); \quad (29)$$

for the IG-OU case,

$$\text{Var}(X_{Y_T}) = \text{Var}(X_1) \left[\varepsilon(T) + \frac{a}{b} (T - \varepsilon(T)) \right] + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^3} \Psi(T); \quad (30)$$

and for the CIR case,

$$\begin{aligned} \text{Var}(X_{Y_T}) = \text{Var}(X_1) & \left[\eta T + \varepsilon_\kappa(T) (1 - \eta) \right] \\ & + (\mathbb{E}[X_1])^2 \left[\frac{\lambda^2 \eta}{2\kappa^3} (2\kappa T - 3 + 4e^{-\kappa T} - e^{-2\kappa T}) \right. \\ & \quad \left. + \frac{\lambda^2 (1 - \eta)}{\kappa^3} (1 - 2\kappa T e^{-\kappa T} - e^{-2\kappa T}) \right], \quad (31) \end{aligned}$$

where $\varepsilon(T) = \lambda^{-1} (1 - e^{-\lambda T})$, $\Psi(T) = T - \varepsilon(T) - \frac{1}{2} \lambda \varepsilon(T)^2$, $\varepsilon_\kappa(T) = (1 - e^{-\kappa T}) / \kappa$. For the intOU models $\varepsilon(T)$ governs how the variance evolves over T , while $\varepsilon_\kappa(T)$ plays the same role for CIR.

At our fitted parameters the implied volatility curves in Figure 23, and Figure 27 of Appendix C are monotone, so each model is characterised by its behaviour in the two limits $T \rightarrow 0$ and $T \rightarrow \infty$ and its rate of change, determined by either $\varepsilon(T)$ or $\varepsilon_\kappa(T)$.

For all models

$$\sigma(T) \longrightarrow \sqrt{\text{Var}(X_1)} \equiv \sigma(0) \quad \text{as } T \rightarrow 0, \quad (32)$$

so the Lévy parameters alone set the short-maturity level.

As T becomes large

$$\sigma(\infty) = \sqrt{\text{Var}(X_1) \frac{a}{b} + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^2}} \quad (\text{Gamma-OU}), \quad (33)$$

$$\sigma(\infty) = \sqrt{\text{Var}(X_1) \frac{a}{b} + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^3}} \quad (\text{IG-OU}), \quad (34)$$

$$\sigma(\infty) = \sqrt{\text{Var}(X_1) \eta + (\mathbb{E}[X_1])^2 \frac{\lambda^2 \eta}{\kappa^2}} \quad (\text{CIR}). \quad (35)$$

The term structure slopes downward when $\sigma(\infty) < \sigma(0)$, that is, when

$$\frac{a}{b} + \frac{(\mathbb{E}[X_1])^2}{\text{Var}(X_1)} \cdot \frac{2a}{\lambda b^2} < 1 \quad (\text{Gamma-OU}), \quad (36)$$

$$\frac{a}{b} + \frac{(\mathbb{E}[X_1])^2}{\text{Var}(X_1)} \cdot \frac{2a}{\lambda b^3} < 1 \quad (\text{IG-OU}), \quad (37)$$

$$\eta \left[1 + \frac{(\mathbb{E}[X_1])^2}{\text{Var}(X_1)} \cdot \frac{\lambda^2}{\kappa^2} \right] < 1 \quad (\text{CIR}). \quad (38)$$

All limits are worked through in Appendix D.4.

6.5.1 Meixner Parameters

Our parameter estimates for the Meixner stochastic time models are available in Table 30.

In the parameterisation, all the time-change models diverge from the pure Meixner fit. The α and β parameters have moved away from the boundary, and δ is significantly reduced. This reduction in δ is why the kurtosis[‡]

$$\text{kurt}(X_1) = 3 + \frac{2 - \cos \beta}{\delta}, \quad (39)$$

in Table 29 and Table 38 of Appendix C increased relative to the Black–Scholes model, and therefore relative to the pure Meixner fit. It also shows why the intOU models had lower kurtosis than the CIR models, since their δ is higher, while β is similar across the three fits.

[‡]Equation (39) is the kurtosis of the Meixner distribution, and is not the kurtosis of the time-changed model, but their form will be similar.

Table 30: Calibrated Meixner parameters on the KOSPI 200 (2026) chain. Highlighted models were part of the best-model group.

	Lévy			Time change		
	α	β	δ			
Meixner	0.001	−3.043	1350.910	—	—	—
Meixner-CIR	1.066	−0.770	2.416	κ	η	λ
				11.905	0.196	1.134
Meixner-Gamma-OU	0.140	−0.732	82.520	λ	a	b
				8.412	1.573	7.984
Meixner-IG-OU	0.162	−0.606	64.966	9.684	2.169	8.047

The starting level of volatility is set by the underlying Meixner process, since

$$\sigma(T) \rightarrow \sqrt{\text{Var}(X_1)} = \sqrt{\frac{\alpha^2 \delta}{2 \cos^2(\beta/2)}} \quad \text{as } T \rightarrow 0,$$

giving 0.962, 0.969, and 1.264 for the Gamma-OU, IG-OU, and CIR fits respectively. In the Lévy parameters, the CIR fit is misparameterised, as its initial volatility is too high. The terminal volatility is 0.545, 0.512, and 0.561 for the Gamma-OU, IG-OU, and CIR fits respectively. The values of λ for the Gamma-OU and IG-OU models are very similar, at 8.412 and 9.684, while κ , which acts in much the same way as λ in the intOU models, is slightly higher at 11.905. This suggests that the CIR model’s rate of decay is steeper, so it converges faster to a similar terminal volatility. However, its high starting value means it overestimates volatility at short-term maturities.

6.5.2 VG Parameters

The variance gamma stochastic time parameters are in Table 31.

Table 31: Calibrated Variance Gamma parameters on the KOSPI 200 (2026) chain. Highlighted models were part of best-model group.

	Lévy			Time change		
	C	G	M			
Variance Gamma	1.05×10^8	9.29×10^7	14970.913	—	—	—
VG-CIR	20.049	5.680	7.209	κ	η	λ
				3.627	0.211	3.514
VG-Gamma-OU	2273.296	50.103	542.433	λ	a	b
				7.504	2.024	59.556
VG-IG-OU	1289.974	39.722	118.125	9.322	28.730	103.776

In the parameterisation, all the time-change models diverge from the pure variance gamma

fit. Under a time-change model, C falls to 2273, 1290, and 20.0 for the Gamma-OU, IG-OU, and CIR models respectively. This is why the kurtosis[§]

$$\text{kurt}(X_1) = 3 \left(1 + \frac{2}{C} \cdot \frac{G^4 + M^4}{(G^2 + M^2)^2} \right), \quad (40)$$

reported in Table 29 and Table 38 of Appendix C, rises relative to the pure fit, and why the CIR model, with the smallest C , is far more heavy-tailed than the two intOU models.

The starting volatility level is of the form

$$\sigma(T) \rightarrow \sqrt{\text{Var}(X_1)} = \sqrt{\frac{C(G^2 + M^2)}{(MG)^2}} \quad \text{as } T \rightarrow 0,$$

giving 0.956, 0.954, and 1.004 for the Gamma-OU, IG-OU, and CIR fits respectively. The terminal level $\sigma(\infty)$ is 0.538, 0.504, and 0.569 respectively, so all three converge to similar values. The rate at which they fall, however, differs markedly. For the CIR model, with $\kappa = 3.627$, volatility falls far more slowly, which contributes significantly to its misfit. The intOU models have a higher λ , and together with their lower starting values this allows them to approximate the volatility term structure much better.

6.5.3 CGMY Parameters

The CGMY stochastic time parameters are given in Table 32.

Table 32: Calibrated CGMY parameters on the KOSPI 200 (2026) chain. No models were part of best-model group.

	Lévy				Time change		
	C	G	M	Y			
CGMY	2.939	60.043	130.260	1.356	—	—	—
CGMY-CIR	0.000	0.088	1.191	2.000	κ	η	λ
					0.414	0.599	5.279
CGMY-Gamma-OU	0.013	1.599	2.656	1.964	λ	a	b
					1.739	0.292	0.015
CGMY-IG-OU	0.053	0.000	18.765	1.901	8.996	1.035	5.353

No model was part of the best group and only the CIR model outperformed its VG counterpart, suggesting these are not the globally optimal parameters, but a local optimum. In the parameterisation, the CGMY time-change models diverge from the pure fit, with all parameters, except Y , falling considerably. This is why the kurtosis[¶]

$$\text{kurt}(X_1) = 3 + \frac{C(M^{Y-4} + G^{Y-4})\Gamma(4-Y)}{(C(M^{Y-2} + G^{Y-2})\Gamma(2-Y))^2} \quad (41)$$

[§]Equation (40) is the kurtosis of the variance gamma distribution, not that of the time-changed model, though their form is similar.

[¶]Equation (41) is the kurtosis of the CGMY distribution, not that of the time-changed model, though their form is similar.

reported in Table 38 of Appendix C rises considerably. The IG-OU fit reaches $G = 2.4 \times 10^{-7}$, which makes G^{Y-4} diverge, so its kurtosis is not calculated.

The starting level of implied volatility is determined by the Lévy parameters through

$$\sigma(T) \rightarrow \sqrt{\text{Var}(X_1)} = \sqrt{C(M^{Y-2} + G^{Y-2})\Gamma(2-Y)} \quad \text{as } T \rightarrow 0,$$

giving 0.813, 1.644, and 0.908 for the Gamma-OU, IG-OU, and CIR fits respectively. The terminal level $\sigma(\infty)$ is then 14.06, 0.787, and 4.539 respectively. For the Gamma-OU and CIR models, implied volatility therefore *increases* as T increases, a complete misfit that does not reflect the data. For the CIR model, κ is far smaller than λ , so the ratio λ^2/κ^2 in Equation (35) for $\sigma(\infty)$ is very large, which pushes the terminal volatility level higher. In the Gamma-OU model, $a > b$, which drives $\sigma(\infty)$ above $\sigma(0)$. These misparameterisations are the main cause of the poor fit. In the IG-OU case, the model itself fits reasonably well, but its volatility curve fits poorly. The curve is the right shape, decreasing with a reasonable decay rate λ , but the Lévy parameters are misfitted, as the starting volatility is too high.

6.5.4 NIG Parameters

The parameter estimates for each model are given in Table 33.

Table 33: Calibrated NIG parameters on the KOSPI 200 (2026) chain. No models were part of best model group.

	Lévy			Time change		
	α	β	δ			
NIG	1.423×10^5	3.601×10^4	6.047×10^4	—	—	—
NIG-CIR	3.563	−0.597	3.322	κ	η	λ
				0.000	1.246	4.962
NIG-Gamma-OU	11.573	−1.692	10.502	λ	a	b
				9.167	0.497	1.773
NIG-IG-OU	4.974	−0.556	4.276	3.507	0.000	0.000

The NIG models fit poorly overall. In the parameters, the NIG time-change models diverge from the pure fit, with all variables falling considerably. This is why the kurtosis^l

$$\text{kurt}(X_1) = 3 \left(1 + \frac{\alpha^2 + 4\beta^2}{\delta\alpha^2\sqrt{\alpha^2 - \beta^2}} \right) \quad (42)$$

reported in Table 38 of Appendix C rises relative to the pure fit. It also explains the ordering, since the CIR fit, with the smallest δ , has the largest kurtosis.

^lEquation (42) is the kurtosis of the NIG distribution, not that of the time-changed model, though their form is similar.

The implied volatility measure fits poorly for two of the three models. The starting values are given by

$$\sigma(T) \rightarrow \sqrt{\text{Var}(X_1)} = \sqrt{\frac{\alpha^2 \delta}{(\alpha^2 - \beta^2)^{3/2}}} \quad \text{as } T \rightarrow 0,$$

which gives 0.968, 0.936, and 0.986 for the Gamma-OU, IG-OU, and CIR fits respectively, driven by the Lévy parameters. The terminal values $\sigma(\infty)$ are 0.588, 3.1×10^8 , and undefined respectively. The Gamma-OU model is the only one with sensible parameters, evident in $\lambda \approx 9$, in line with some of the other best models, and $a < b$. In the CIR model $\kappa \approx 0$, which explains why the terminal value is so large, since $\lambda^2/\kappa^2 \rightarrow \infty$, although this divergence also occurs slowly because κ is so small. For the IG-OU model the fit is clearly degenerate, as a/b is undefined.

7 Concluding Remarks

This dissertation set out with the task of pricing options on a highly volatile stock market index, the KOSPI. The Black–Scholes model had a significant error attached to it, and clearly exhibited distinct implied volatilities across maturity dates, breaking the assumption of constant volatility. On examining the log-returns data, we found that the KOSPI exhibits non-normality; specifically, it has an empirical kurtosis exceeding that of the normal distribution. For this reason we attempted models with alternative distributions, namely the Meixner, variance gamma, CGMY, NIG, and generalised hyperbolic models.

They did not improve significantly upon the Black–Scholes model, with many converging to the normal distribution, and hence to the Black–Scholes model itself. The generalised hyperbolic model did not converge to the normal distribution, but nor did it exhibit much improvement. The key issue appeared to lie in volatility. On examining the Black–Scholes model, we discovered that each maturity has a distinct σ attached to it, violating the assumption in both the Black–Scholes and Lévy models that volatility is constant. We therefore required a model in which volatility changed with the maturity date, specifically one in which an increasing T led to a decreasing σ . This led us to stochastic time and stochastic volatility models.

We fit a model in which the stock price followed a specific process and volatility itself also followed a random process. These models were significantly better than any considered previously; all of them produced considerably lower error than their constant-volatility counterparts. The best fit proved to be a group of models with very similar error, highlighted in Table 27. Figure 23 shows that the implied volatility of these models moves across maturity and matches that of a per-maturity Black–Scholes model. Among these best models, the BNS-Gamma-OU is the most parsimonious with one fewer parameter.

A Schoutens Data

Table 34: 77 S&P 500 call option prices at the close of 18 April 2002, as tabulated in Appendix C of Schoutens (2003). On that day the index closed at $S_0 = 1124.47$, with $r = 1.9\%$ and $q = 1.2\%$ per year. Blank entries indicate strikes not quoted at that expiry.

Strike	May 2002	June 2002	Sep. 2002	Dec. 2002	March 2003	June 2003	Dec. 2003
975			161.60	173.30			
995			144.80	157.00		182.10	
1025			120.10	133.10	146.50		
1050		84.50	100.70	114.80		143.00	171.40
1075		64.30	82.50	97.60			
1090	43.10						
1100	35.60		65.50	81.20	96.20	111.30	140.40
1110		39.50					
1120	22.90	33.50					
1125	20.20	30.70	51.00	66.90	81.70	97.00	
1130		28.00					
1135		25.60	45.50				
1140	13.30	23.20		58.90			
1150		19.10	38.10	53.90	68.30	83.30	112.80
1160		15.30					
1170		12.10					
1175		10.90	27.70	42.50	56.60		99.80
1200			19.60	33.00	46.10	60.90	
1225			13.20	24.90	36.90	49.80	
1250				18.30	29.30	41.20	66.90
1275				13.20	22.50		
1300					17.20	27.10	49.50
1325					12.80		
1350						17.10	35.70
1400						10.10	25.20
1450							17.00
1500							12.20

B My Data

Table 35: KOSPI 200 call option prices, bid–ask midpoints from the market close of 30 July 2026, in the layout of Appendix C of Schoutens (2003). $S_0 = 872.49$, $r = 2.91\%$ and $q = 0.79\%$ per year. For legibility only strikes divisible by ten are shown, which retains 76 of the 212 contracts used; the full chain quotes strikes every 2.5 points in the August expiry and is available upon request. Blank entries indicate strikes not quoted at that expiry. Prices were obtained via Interactive Brokers API.

Strike	Aug. 2026	Sep. 2026	Oct. 2026	Nov. 2026	Dec. 2026	Jan. 2027	March 2027
700	182.40						
710	173.12						
720	164.57						
730	156.35						
740	148.02						
750	140.55						
760	132.07						
770	124.30						
780	116.78						
790	109.50						
800	101.85						200.60
810	94.97						193.85
820	88.43						188.57
830	82.12						
840	75.60						
850	72.12	108.58					
860	67.10	99.00					
870	61.55	94.22					
880	59.55	89.42					
890	53.98	85.32	108.22				
900	49.23	82.15	103.50				
910	47.12	78.65	97.05	117.05			
920	42.40	72.38	93.00	112.82	126.93	130.18	
930	38.15	71.72	90.97	108.70	122.82	126.30	
940	35.15	67.85	87.40		118.32	122.60	140.20
950	30.65	64.33	84.15			118.95	136.53
960	28.17		80.33			115.38	132.70
970	24.93					111.98	129.65
980	22.90					108.55	126.38
990	20.45					105.35	
1000	18.75						
1010	17.05						
1020	15.12						
1030	13.12						
1040	12.25						

Table 36: Coverage of the KOSPI 200 (2026) option chain by expiry. Spreads are quoted bid–ask spreads as a percentage of the mid price.

Expiry	TTM	Contracts	$K_{\min}$	$K_{\max}$	$C_{\min}$	$C_{\max}$	Median spread (%)
Aug 26	0.038	116	700	1045	11.15	182.40	30.3
Sep 26	0.115	36	845	955	62.53	108.58	37.4
Oct 26	0.192	19	890	965	78.18	108.23	46.0
Nov 26	0.288	6	905	930	108.70	119.08	48.2
Dec 26	0.364	5	920	940	118.33	126.93	47.8
Jan 27	0.460	16	920	995	103.83	130.18	45.7
Mar 27	0.614	14	800	985	124.68	200.60	47.6
Total		212	700	1045	11.15	200.60	38.5

B.1 Filtering of the KOSPI 200 Option Chain

Our original pull returned 2,284 contracts. Table 37 records the process applied to reach the 212 contracts we used.

Table 37: Filtering of the KOSPI 200 chain.

Filter	Remaining	Dropped
Original data	2,284	—
Remove weekly contracts	1,642	642
Calls only	821	821
Bid and ask both present	572	249
Spread $\leq 50\%$ of mid price	212	360

We choose to remove weekly contracts to prune the dataset. We also remove puts. We then remove options that do not have both a bid and an ask price present, because we choose the mid-price as the option’s price.

Quoted spreads,

$$\text{spread\%} = 100 \times \frac{\text{ask} - \text{bid}}{\text{mid}}, \quad \text{mid} = \frac{\text{bid} + \text{ask}}{2},$$

on this chain are wide. The median before filtering is 59% of the mid price, the upper quartile 134%, and the widest quotes reach 200%, with a bid of 0.03 and an ask of 400, giving a nominal mid of 200 at which no trade would occur. Including these contracts would be unusual given they are not actually traded.

We choose a threshold of 50%. We chose this somewhat arbitrarily, but a lower threshold would reduce the number of expiries excessively, and a higher threshold would keep too many illiquid options. At 50% the chain keeps 212 contracts across seven expiries while removing the worst half of the quotation noise.

C Supplementary Figures and Tables

This appendix collects figures and tables that support the results of the main text but are too large to sit within it.

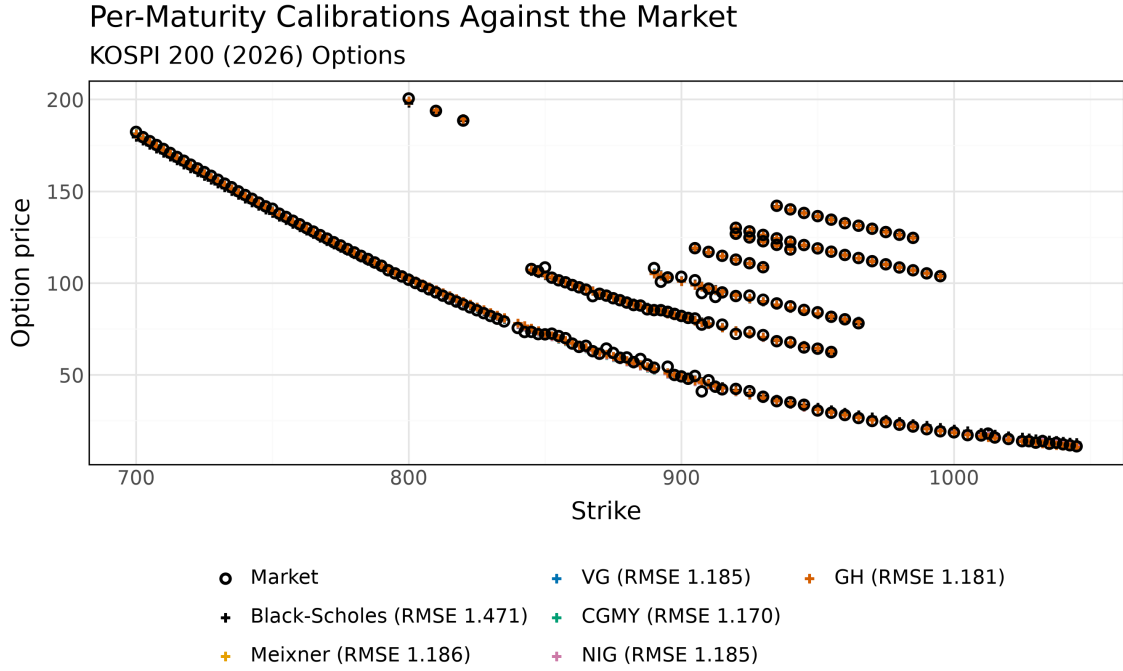


Figure 24: Market prices against the six per-maturity calibrations of Table 18 on the KOSPI 200 (2026) chain.

Table 38: Excess kurtosis for the models outside the best-fitting group, on the KOSPI 200 (2026) chain, by maturity. Models ordered by RMSE. The CGMY–IG–OU entries cannot be computed, its fourth cumulant is not finite.

Expiry	CGMY– IG–OU	NIG– Gamma–OU	CGMY– CIR	CGMY– Gamma–OU	VG– CIR	Meixner– CIR	NIG– CIR	NIG– IG–OU
Aug 26	—	0.939	1.049	150.166	4.846	16.478	8.445	4.220
Sep 26	—	0.790	3.287	140.233	3.041	7.413	5.599	1.582
Oct 26	—	0.977	5.765	98.756	3.432	5.569	7.032	1.075
Nov 26	—	1.151	9.328	68.580	4.320	4.544	10.075	0.831
Dec 26	—	1.224	12.493	54.193	5.099	4.024	13.012	0.733
Jan 27	—	1.252	16.683	42.432	6.049	3.539	16.976	0.662
Mar 27	—	1.216	23.530	30.998	7.382	2.976	23.435	0.603
RMSE	1.804	1.878	2.049	2.637	3.322	3.394	3.748	3.843

Table 39: Our calibration of each stochastic time-change model to the S&P 500 (2002) chain, against the values published in Schoutens (2003, Tables 7.3 and 7.4). Each cell is shaded by how far our value departs from Schoutens’: green where the two agree to within 30% in absolute terms, and red where they do not.

	Lévy process			Time change			RMSE
<i>Meixner</i>							
	α	β	δ	κ	η	λ	
CIR, ours	0.1159	−0.5432	3.8954	0.5510	1.6258	1.9981	0.6197
CIR, Schoutens	0.1231	−0.5875	3.3588	0.5705	1.5863	1.9592	0.4896
	α	β	δ	λ	a	b	
Gamma–OU, ours	0.0962	−1.1439	4.5175	1.0847	0.5539	0.7596	0.4948
Gamma–OU, Schoutens	0.1108	−0.9858	3.6288	1.1729	0.5914	0.6558	0.4180
IG–OU, ours	0.0715	−1.3732	6.7216	1.1473	0.5779	0.9443	0.4899
IG–OU, Schoutens	0.0890	−1.1323	5.0262	1.2190	0.6564	0.8266	0.4140
<i>Variance gamma</i>							
	C	G	M	κ	η	λ	
CIR, ours	14.4078	28.5137	37.8427	0.5946	1.6184	2.0852	0.6297
CIR, Schoutens	11.9896	25.8523	35.5344	0.6020	1.5560	1.9992	0.5003
	C	G	M	λ	a	b	
Gamma–OU, ours	13.6273	24.2562	47.2151	1.1448	0.5552	0.7491	0.5154
Gamma–OU, Schoutens	11.4838	23.2880	40.1291	1.2517	0.5841	0.6282	0.4393
IG–OU, ours	18.5576	27.7163	64.8756	1.1616	0.5826	0.9295	0.5010
IG–OU, Schoutens	14.9248	26.1529	50.4425	1.2801	0.6615	0.8104	0.4306
<i>CGMY</i>							
	C	G	M	Y	κ	η	λ
CIR, ours	0.0082	0.1614	11.5171	1.6615	0.2930	1.4501	1.2843
CIR, Schoutens	0.0074	0.1025	11.3940	1.6765	0.3881	1.4012	1.3612
	C	G	M	Y	λ	a	b
Gamma–OU, ours	0.0319	3.3120	68.7920	1.4605	0.8833	0.5473	1.5424
Gamma–OU, Schoutens	0.0415	3.9134	30.6322	1.3664	0.8826	0.5945	0.8524
IG–OU, ours	0.0165	1.8821	185.5035	1.6101	1.0435	0.3614	1.9177
IG–OU, Schoutens	0.0672	6.1316	44.7448	1.2911	1.0622	0.6092	0.9999
<i>NIG</i>							
	α	β	δ	κ	η	λ	
CIR, ours	19.5470	−4.7230	0.5098	0.5211	1.6258	1.9289	0.6127
CIR, Schoutens	18.4815	−4.8412	0.4685	0.5391	1.5746	1.8772	0.4814
	α	β	δ	λ	a	b	
Gamma–OU, ours	27.1041	−13.1882	0.5417	1.0200	0.5536	0.8086	0.4749
Gamma–OU, Schoutens	29.4722	−15.9048	0.5071	0.6252	0.4239	0.5962	0.4510
IG–OU, ours	39.2267	−22.3726	0.6332	1.1008	0.5571	0.9990	0.4742
IG–OU, Schoutens	29.1553	−13.9331	0.5600	1.1559	0.6496	0.8572	0.4156
<i>Generalised hyperbolic</i>							
CIR, Gamma–OU, IG–OU Schoutens’ published fit could not be reproduced; see text.							

RMSE is defined in Equation (6).

Log-Return Densities: Remaining Stochastic Time Models August 2026

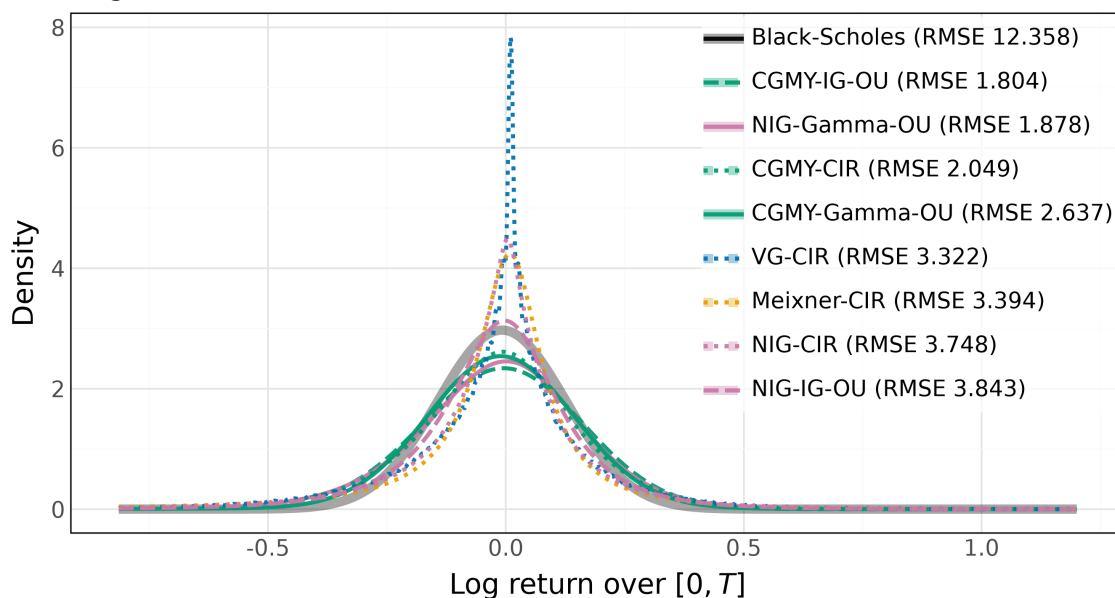


Figure 25: As Figure 22, for the remaining stochastic time models, against the Black-Scholes model. Front month August 2026, $T = 0.038$.

Log-Return Densities: Remaining Stochastic Time Models March 2027

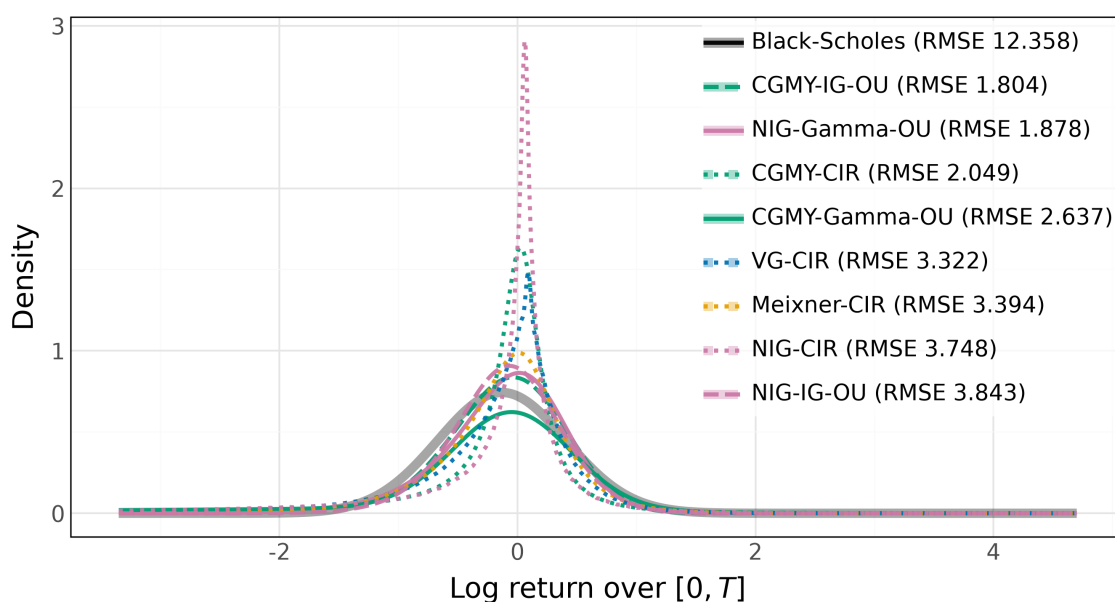


Figure 26: As Figure 22, for the remaining stochastic time models, against the Black-Scholes model. Longest maturity March 2027, $T = 0.614$.

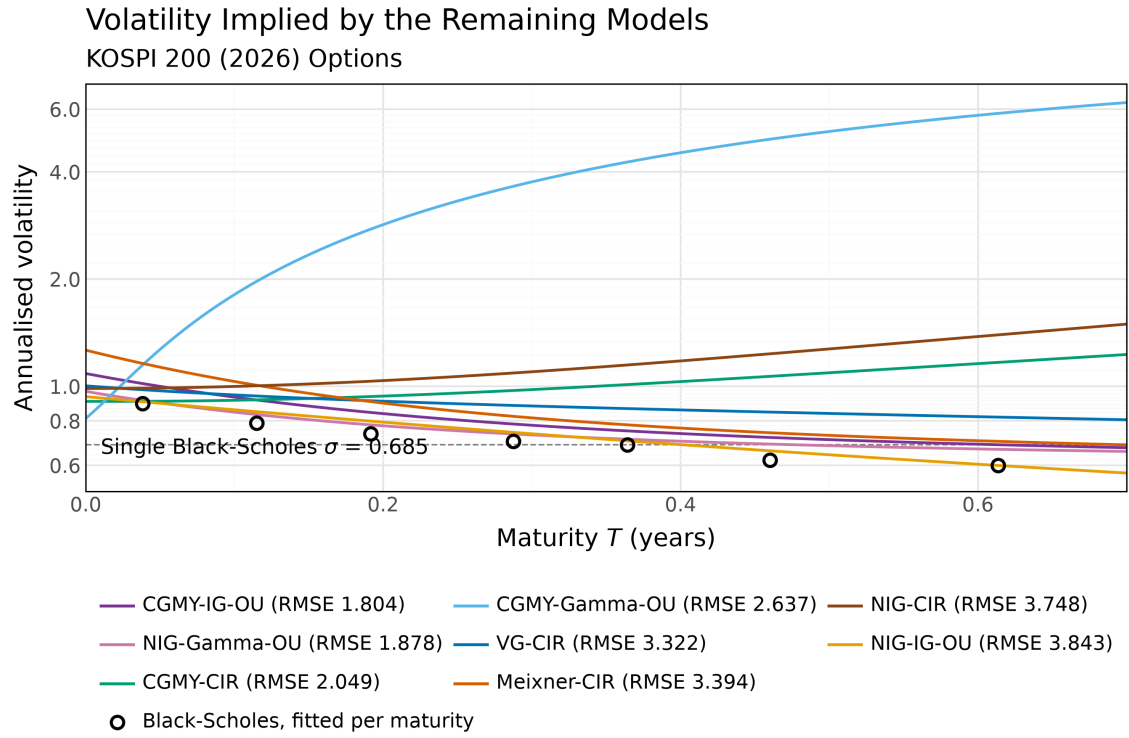


Figure 27: Annualised volatility $\sqrt{\text{Var}(\log(S_T/S_0))/T}$ implied by the eight models outside the best-fitting group, over the range of maturities quoted on the KOSPI 200 (2026) chain. Rings mark the per-maturity Black–Scholes volatilities of Table 6 and the dashed line the single volatility fitted to the whole chain. The vertical axis is logarithmic.

D Mathematical Results

This appendix collects the mathematical material supporting the main text: the characteristic functions of the stochastic time-change models, the moments of the driving Lévy processes, the derivations of the BNS variance decomposition, variance under a stochastic time change and the integrated CIR variance, the sample moment estimators, and the numerical routines used throughout.

D.1 Characteristic Functions of the Stochastic Time-Change Models

This section collects the characteristic functions of the twelve stochastic time-change models fitted in Section 6. Each is built from the general form of Equation (28), in which ψ_X is the characteristic exponent of the Lévy process.

For the Meixner process*,

$$\psi_X(u) = 2\delta \left(\log \cos(\beta/2) - \log \cosh\left(\frac{\alpha u - i\beta}{2}\right) \right);$$

for the Variance Gamma process,

$$\psi_X(u) = C \left(\log(GM) - \log(GM + (M - G)iu + u^2) \right);$$

for the CGMY process,

$$\psi_X(u) = C\Gamma(-Y) \left((M - iu)^Y - M^Y + (G + iu)^Y - G^Y \right);$$

and for the NIG process,

$$\psi_X(u) = -\delta \left(\sqrt{\alpha^2 - (\beta + iu)^2} - \sqrt{\alpha^2 - \beta^2} \right).$$

In each case ϕ_Y is the characteristic function unique to the stochastic time volatility method used, for example $\phi_{\text{CIR}}(u; t, \kappa, \eta, \lambda, y_0)$, $\phi_{\text{Gamma-OU}}(u; t, \lambda, a, b, y_0)$, or $\phi_{\text{IG-OU}}(u; t, \lambda, a, b, y_0)$. In turn this is solved by RMSE minimisation to find the optimal parameters.

Schoutens (2003, pp. 95–96) shows that the characteristic function has the following

*In prior sections we fit the Meixner model using the density function directly. Under a stochastic time change that density is unavailable, because X_{Y_T} is a Meixner process evaluated at a random time, and the density of that mixture has no closed form.

equalities

$$\begin{aligned}
\phi_{\text{Meixner-CIR}}(u; \alpha, \beta, \delta, \kappa, \eta, \lambda, y_0) &= \phi_{\text{Meixner-CIR}}(u; \alpha, \beta, \delta y_0, \kappa, \eta/y_0, \lambda/\sqrt{y_0}, 1), \\
\phi_{\text{Meixner-Gamma-OU}}(u; \alpha, \beta, \delta, \lambda, a, b, y_0) &= \phi_{\text{Meixner-Gamma-OU}}(u; \alpha, \beta, \delta y_0, \lambda, a, b y_0, 1), \\
\phi_{\text{Meixner-IG-OU}}(u; \alpha, \beta, \delta, \lambda, a, b, y_0) &= \phi_{\text{Meixner-IG-OU}}(u; \alpha, \beta, \delta y_0, \lambda, a/\sqrt{y_0}, b\sqrt{y_0}, 1), \\
\phi_{\text{VG-CIR}}(u; C, G, M, \kappa, \eta, \lambda, y_0) &= \phi_{\text{VG-CIR}}(u; C y_0, G, M, \kappa, \eta/y_0, \lambda/\sqrt{y_0}, 1), \\
\phi_{\text{VG-Gamma-OU}}(u; C, G, M, \lambda, a, b, y_0) &= \phi_{\text{VG-Gamma-OU}}(u; C y_0, G, M, \lambda, a, b y_0, 1), \\
\phi_{\text{VG-IG-OU}}(u; C, G, M, \lambda, a, b, y_0) &= \phi_{\text{VG-IG-OU}}(u; C y_0, G, M, \lambda, a/\sqrt{y_0}, b\sqrt{y_0}, 1), \\
\phi_{\text{CGMY-CIR}}(u; C, G, M, Y, \kappa, \eta, \lambda, y_0) &= \phi_{\text{CGMY-CIR}}(u; C y_0, G, M, Y, \kappa, \eta/y_0, \lambda/\sqrt{y_0}, 1), \\
\phi_{\text{CGMY-Gamma-OU}}(u; C, G, M, Y, \lambda, a, b, y_0) &= \phi_{\text{CGMY-Gamma-OU}}(u; C y_0, G, M, Y, \lambda, a, b y_0, 1), \\
\phi_{\text{CGMY-IG-OU}}(u; C, G, M, Y, \lambda, a, b, y_0) &= \phi_{\text{CGMY-IG-OU}}(u; C y_0, G, M, Y, \lambda, a/\sqrt{y_0}, b\sqrt{y_0}, 1), \\
\phi_{\text{NIG-CIR}}(u; \alpha, \beta, \delta, \kappa, \eta, \lambda, y_0) &= \phi_{\text{NIG-CIR}}(u; \alpha, \beta, \delta y_0, \kappa, \eta/y_0, \lambda/\sqrt{y_0}, 1), \\
\phi_{\text{NIG-Gamma-OU}}(u; \alpha, \beta, \delta, \lambda, a, b, y_0) &= \phi_{\text{NIG-Gamma-OU}}(u; \alpha, \beta, \delta y_0, \lambda, a, b y_0, 1), \\
\phi_{\text{NIG-IG-OU}}(u; \alpha, \beta, \delta, \lambda, a, b, y_0) &= \phi_{\text{NIG-IG-OU}}(u; \alpha, \beta, \delta y_0, \lambda, a/\sqrt{y_0}, b\sqrt{y_0}, 1),
\end{aligned}$$

so a model with free y_0 is equivalent to one with $y_0 = 1$ and rescaled parameters. Hence, we set $y_0 = 1$, giving the following characteristic functions.

$$\begin{aligned}
\phi_{\text{Meixner-CIR}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{CIR}}(-i\psi_X(u); t, y_0)}{\phi_{\text{CIR}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{Meixner-Gamma-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{Gamma-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{Gamma-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{Meixner-IG-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{IG-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{IG-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{VG-CIR}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{CIR}}(-i\psi_X(u); t, y_0)}{\phi_{\text{CIR}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{VG-Gamma-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{Gamma-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{Gamma-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{VG-IG-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{IG-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{IG-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{CGMY-CIR}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{CIR}}(-i\psi_X(u); t, y_0)}{\phi_{\text{CIR}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{CGMY-Gamma-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{Gamma-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{Gamma-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{CGMY-IG-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{IG-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{IG-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{NIG-CIR}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{CIR}}(-i\psi_X(u); t, y_0)}{\phi_{\text{CIR}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{NIG-Gamma-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{Gamma-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{Gamma-OU}}(-i\psi_X(-i); t, y_0)^{iu}} \\
\phi_{\text{NIG-IG-OU}}(u) &= \exp(iu((r-q)t + \log S_0)) \frac{\phi_{\text{IG-OU}}(-i\psi_X(u); t, y_0)}{\phi_{\text{IG-OU}}(-i\psi_X(-i); t, y_0)^{iu}}
\end{aligned}$$

D.2 Moments of the Driving Lévy Processes

The mean and variance of each Lévy process over one unit of time, used in the variance expressions of Appendix D.4, are collected in Table 40.

Table 40: Mean and variance of each Lévy process over one unit of time, as used in Equations (29), (30) and (31). Taken from Schoutens (2003, §5.3).

Process	$\mathbb{E}[X_1]$	$\text{Var}(X_1)$
Meixner	$\alpha\delta \tan(\beta/2)$	$\frac{\alpha^2\delta}{2\cos^2(\beta/2)}$
Variance gamma	$\frac{C(G-M)}{MG}$	$\frac{C(G^2+M^2)}{(MG)^2}$
CGMY	$C(M^{Y-1} - G^{Y-1})\Gamma(1-Y)$	$C(M^{Y-2} + G^{Y-2})\Gamma(2-Y)$
Normal inverse Gaussian	$\frac{\delta\beta}{\sqrt{\alpha^2 - \beta^2}}$	$\frac{\alpha^2\delta}{(\alpha^2 - \beta^2)^{3/2}}$

D.3 Derivation of the BNS Variance Decomposition

This section derives Equation (25), the three-way split of $\text{Var}(\log(S_T/S_0))$ reported in Section 5.7.1.

Integrating Equation (22) over $[0, T]$, the fundamental theorem of calculus gives $\int_0^T d\log S_t = \log S_T - \log S_0 = \log(S_T/S_0)$ on the left, so that

$$\log(S_T/S_0) = \underbrace{\int_0^T \left(\mu - \frac{1}{2}\sigma_s^2\right) ds}_{\text{drift}} + \underbrace{\int_0^T \sigma_s dW_s}_{M_T} + \rho z_{\lambda T}.$$

To determine the variance of $\log(S_T/S_0)$, we note three facts. First, from Schoutens (2003, p. 86), W and the BDLP z are independent. Second, by the Itô isometry (Corollary 3.1.7 of Øksendal (2003, p. 29)),

$$\mathbb{E}\left[\left(\int_0^T \sigma_s dW_s\right)^2\right] = \mathbb{E}\left[\int_0^T \sigma_s^2 ds\right],$$

so that $\mathbb{E}[M_T^2] = \mathbb{E}[I_T]$, the expected integrated variance; and since M is a martingale with $M_0 = 0$, this second moment is its variance. Third, the integrated variance is

$$I_T = \int_0^T \sigma_s^2 ds = \underbrace{\lambda^{-1}(1 - e^{-\lambda T})\sigma_0^2}_{\text{contribution of today's variance}} + \underbrace{\lambda^{-1} \int_0^T (1 - e^{-\lambda(T-s)}) dz_{\lambda s}}_{\text{contribution of future jumps}},$$

a decomposition that follows Schoutens (2003, eq. (5.6), p. 49). Combining these points,

and dropping the drift term, gives

$$\begin{aligned}\text{Var}(\log(S_T/S_0)) &\approx \mathbb{E}[I_T] + \rho^2 \text{Var}(z_{\lambda T}) \\ &= \lambda^{-1}(1 - e^{-\lambda T}) \sigma_0^2 + \lambda^{-1} \int_0^T (1 - e^{-\lambda(T-s)}) \mathbb{E}[dz_{\lambda s}] + \rho^2 \text{Var}(z_{\lambda T}).\end{aligned}\tag{43}$$

Since z is a Lévy process, $\mathbb{E}[z_{\lambda T}] = \lambda T \mathbb{E}[z_1]$ and $\text{Var}(z_{\lambda T}) = \lambda T \text{Var}(z_1)$, where the mean and variance of z_1 follow from the derivatives of the cumulant function k at the origin. Differentiating Equation (20) gives

$$\mathbb{E}[z_1] = a/b \quad \text{and} \quad \text{Var}(z_1) = 2a/b^2$$

for the Gamma-OU specification, while Equation (21) gives

$$\mathbb{E}[z_1] = a/b \quad \text{and} \quad \text{Var}(z_1) = 2a/b^3$$

for IG-OU. Substituting $\mathbb{E}[dz_{\lambda s}] = \lambda(a/b) ds$ into the middle term of Equation (43) and evaluating the integral[†] gives

$$\text{Var}(\log(S_T/S_0)) \approx \underbrace{\lambda^{-1}(1 - e^{-\lambda T}) \sigma_0^2}_{(i)} + \underbrace{\frac{\lambda a}{b} (T - \lambda^{-1}(1 - e^{-\lambda T}))}_{(ii)} + \underbrace{\rho^2 \lambda T \text{Var}(z_1)}_{(iii)},$$

which is Equation (25).

D.4 Variance of the Log Return under a Time Change

This section derives the closed-form variance $\text{Var}(X_{Y_T})$ of the time-change models, the three variance formulas of Equations (29), (30) and (31), and the two limiting cases reported in Section 6.

D.4.1 Deriving Closed Form Variance

Applying the law of total variance,

$$\text{Var}(X_{Y_T}) = \mathbb{E}[\text{Var}(X_{Y_T} \mid Y_T)] + \text{Var}(\mathbb{E}[X_{Y_T} \mid Y_T]).\tag{44}$$

Because a Lévy process has stationary and independent increments, $\mathbb{E}[X_t] = t \mathbb{E}[X_1]$ and $\text{Var}(X_t) = t \text{Var}(X_1)$. This implies $\mathbb{E}[X_{Y_T} \mid Y_T] = \mathbb{E}[X_1] Y_T$ and $\text{Var}(X_{Y_T} \mid Y_T) = \text{Var}(X_1) Y_T$. Substituting these into Equation (44) gives

$$\text{Var}(X_{Y_T}) = \text{Var}(X_1) \mathbb{E}[Y_T] + (\mathbb{E}[X_1])^2 \text{Var}(Y_T),\tag{45}$$

where $\mathbb{E}[X_1]$ and $\text{Var}(X_1)$ are the mean and variance of the driving Lévy process over one unit of time, given for each family in Table 40 of Appendix D.2. Here the CIR and

[†]The surviving factor of λ in the middle term arises because the subordinator is evaluated on the time scale λs rather than s .

intOU models diverge. The two moments of Y_T for the OU distributions, with $\varepsilon(T) = \lambda^{-1} (1 - e^{-\lambda T})$, are

$$\mathbb{E}[Y_T | y_0] = \varepsilon(T) y_0 + \mathbb{E}[z_1] (T - \varepsilon(T)), \quad (46)$$

$$\text{Var}(Y_T | y_0) = \frac{\text{Var}(z_1)}{\lambda} \left(T - 2\varepsilon(T) + \frac{1 - e^{-2\lambda T}}{2\lambda} \right), \quad (47)$$

where z is the background driving Lévy process. Since the initial rate of time change is fixed at $y_0 = 1$, the conditioning is redundant. Substituting Equations (46) and (47) into Equation (45) gives

$$\begin{aligned} \text{Var}(X_{Y_T}) &= \text{Var}(X_1) \left[\varepsilon(T) + \mathbb{E}[z_1] (T - \varepsilon(T)) \right] \\ &\quad + (\mathbb{E}[X_1])^2 \frac{\text{Var}(z_1)}{\lambda} \left(T - 2\varepsilon(T) + \frac{1 - e^{-2\lambda T}}{2\lambda} \right). \end{aligned} \quad (48)$$

The CIR models have no background driving Lévy process; their expectation is given by Schoutens (2003, p. 89),

$$\mathbb{E}[Y_T] = \eta T + \varepsilon_\kappa(T) (y_0 - \eta), \quad \varepsilon_\kappa(T) = \frac{1 - e^{-\kappa T}}{\kappa}, \quad (49)$$

and their variance, which Schoutens (2003) does not state, is

$$\text{Var}(Y_T) = \frac{\lambda^2 \eta}{2\kappa^3} (2\kappa T - 3 + 4e^{-\kappa T} - e^{-2\kappa T}) + \frac{\lambda^2 (y_0 - \eta)}{\kappa^3} (1 - 2\kappa T e^{-\kappa T} - e^{-2\kappa T}), \quad (50)$$

which follows as $\mathbb{E}[Y_T^2] - (\mathbb{E}[Y_T])^2$ from the first two moments of the integrated square-root process given by Dufresne (2001, p. 16), with the algebra set out in full in Appendix D.5. This leaves three variance formulas, one for each time-change specification. Substituting $\mathbb{E}[z_1] = a/b$ with $\text{Var}(z_1) = 2a/b^2$ into Equation (48) gives the Gamma-OU case,

$$\text{Var}(X_{Y_T}) = \underbrace{\text{Var}(X_1) \left[\varepsilon(T) + \frac{a}{b} (T - \varepsilon(T)) \right]}_{(i)} + \underbrace{(\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^2} \Psi(T)}_{(ii)};$$

substituting $\text{Var}(z_1) = 2a/b^3$ instead gives the IG-OU case,

$$\text{Var}(X_{Y_T}) = \underbrace{\text{Var}(X_1) \left[\varepsilon(T) + \frac{a}{b} (T - \varepsilon(T)) \right]}_{(i)} + \underbrace{(\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^3} \Psi(T)}_{(ii)};$$

and substituting Equations (49) and (50) into Equation (45) gives the CIR case,

$$\begin{aligned} \text{Var}(X_{Y_T}) = & \underbrace{\text{Var}(X_1) \left[\eta T + \varepsilon_\kappa(T) (1 - \eta) \right]}_{(i)} \\ & + (\mathbb{E}[X_1])^2 \left[\underbrace{\frac{\lambda^2 \eta}{2\kappa^3} (2\kappa T - 3 + 4e^{-\kappa T} - e^{-2\kappa T})}_{(ii)} \right. \\ & \left. + \underbrace{\frac{\lambda^2 (1 - \eta)}{\kappa^3} (1 - 2\kappa T e^{-\kappa T} - e^{-2\kappa T})}_{(iii)} \right], \end{aligned}$$

where $\Psi(T) = T - \varepsilon(T) - \frac{1}{2}\lambda\varepsilon(T)^2$ and $y_0 = 1$ throughout. These are Equations (29), (30) and (31) of the main text. Each was checked against the corresponding characteristic function of Appendix D.1 and agrees to four significant figures at every maturity in the chain.

D.4.2 Influence of the Time-Change Parameters on Volatility

At our fitted parameters the term structures of Figures 23 and 27 of Appendix C are monotone, so each model is characterised by its behaviour in the two limits $T \rightarrow 0$ and $T \rightarrow \infty$ and its rate of change, determined by either $\varepsilon(T)$ or $\varepsilon_\kappa(T)$. In understanding the limiting cases we take the intOU models first and the CIR model second.

Integrated OU models. At near-term maturities $\varepsilon(T) \approx T$, so the factor $T - \varepsilon(T)$ multiplying a/b vanishes and term (i) reduces to $\text{Var}(X_1) T$, while term (ii) is $O(T^3)$ by Taylor expansion[‡] and therefore negligible. Dividing by T ,

$$\sigma(T) = \sqrt{\frac{\text{Var}(X_{Y_T})}{T}} \longrightarrow \sqrt{\text{Var}(X_1)} \equiv \sigma(0) \quad \text{as } T \rightarrow 0,$$

which is Equation (32), so the Lévy parameters alone set the short-maturity level.

As T becomes large, $e^{-\lambda T}$ approaches zero, leaving

$$\varepsilon(T) \approx \frac{1}{\lambda}, \quad \Psi(T) \approx T - \frac{2}{\lambda} + \frac{1}{2\lambda} = T - \frac{3}{2\lambda}.$$

Substituting into Equation (29) and dividing by T ,

$$\sigma^2(T) = \text{Var}(X_1) \left[\frac{1}{\lambda T} + \frac{a}{b} \left(1 - \frac{1}{\lambda T} \right) \right] + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^2} \left(1 - \frac{3}{2\lambda T} \right),$$

[‡]Expanding $\varepsilon(T) = (1 - e^{-\lambda T})/\lambda = T - \frac{1}{2}\lambda T^2 + O(T^3)$ and substituting into $\Psi(T) = T - \varepsilon(T) - \frac{1}{2}\lambda\varepsilon(T)^2$ gives $T - \varepsilon(T) = \frac{1}{2}\lambda T^2 + O(T^3)$ and $\frac{1}{2}\lambda\varepsilon(T)^2 = \frac{1}{2}\lambda T^2 + O(T^3)$. The two quadratic terms cancel, leaving $\Psi(T) = O(T^3)$.

in which every term in $1/(\lambda T)$ vanishes as T grows, giving

$$\sigma(\infty) = \sqrt{\text{Var}(X_1) \frac{a}{b} + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^2}} \quad (\text{Gamma-OU}),$$

$$\sigma(\infty) = \sqrt{\text{Var}(X_1) \frac{a}{b} + (\mathbb{E}[X_1])^2 \frac{2a}{\lambda b^3}} \quad (\text{IG-OU}),$$

which are Equations (33) and (34). Imposing $\sigma(\infty) < \sigma(0)$ then gives

$$\frac{a}{b} + \frac{(\mathbb{E}[X_1])^2}{\text{Var}(X_1)} \cdot \frac{2a}{\lambda b^2} < 1 \quad (\text{Gamma-OU}),$$

$$\frac{a}{b} + \frac{(\mathbb{E}[X_1])^2}{\text{Var}(X_1)} \cdot \frac{2a}{\lambda b^3} < 1 \quad (\text{IG-OU}),$$

the downward-sloping conditions of Equations (36) and (37).

Integrated CIR model. The short-maturity behaviour is the same. As $T \rightarrow 0$ we have $\varepsilon_\kappa(T) = (1 - e^{-\kappa T}) / \kappa \approx T$, so term (i) of Equation (31) becomes

$$\eta T + \varepsilon_\kappa(T) (1 - \eta) \approx \eta T + T (1 - \eta) = T,$$

while Taylor expansion of $e^{-\kappa T}$ and $e^{-2\kappa T}$ in terms (ii) and (iii) cancels the low-order terms, leaving both $O(T^3)$ and hence negligible. Dividing by T returns the same short-maturity limit as Equation (32), $\sigma(T) \rightarrow \sigma(0) = \sqrt{\text{Var}(X_1)}$.

As T becomes large, $e^{-\kappa T}$ again approaches zero, so $\varepsilon_\kappa(T)$ settles at the constant $1/\kappa$. Dividing Equation (31) by T , term (i) gives

$$\eta + \frac{\varepsilon_\kappa(T) (1 - \eta)}{T} \rightarrow \eta,$$

term (ii) grows linearly, since $2\kappa T - 3 + 4e^{-\kappa T} - e^{-2\kappa T} \approx 2\kappa T$, and term (iii) tends to the constant 1 and so vanishes after division by T . Hence

$$\sigma(\infty) = \sqrt{\text{Var}(X_1) \eta + (\mathbb{E}[X_1])^2 \frac{\lambda^2 \eta}{\kappa^2}},$$

which is Equation (35). Its structure matches the OU-driven case, with the long-run rate of time change η in place of a/b . Imposing $\sigma(\infty) < \sigma(0)$ gives

$$\eta \left[1 + \frac{(\mathbb{E}[X_1])^2}{\text{Var}(X_1)} \cdot \frac{\lambda^2}{\kappa^2} \right] < 1,$$

the condition of Equation (38).

D.5 Derivation of the Integrated CIR Variance

This section derives the variance of the integrated CIR process from the moments of the integrated square-root process given by Dufresne (2001, p. 16), who writes the square-root process as

$$dX = (\alpha X + \beta) dt + \gamma \sqrt{X} dW, \quad X_0 = \bar{x}, \quad (51)$$

whereas the CIR time change of Section 6 is

$$dy_t = \kappa (\eta - y_t) dt + \lambda \sqrt{y_t} dW_t, \quad y_0 = 1.$$

Comparing with Equation (51), the two are the same equation under

$$\alpha = -\kappa, \quad \beta = \kappa\eta, \quad \gamma = \lambda, \quad \bar{x} = y_0. \quad (52)$$

Dufresne's expectation is

$$\mathbb{E}[Y_T] = -\frac{\bar{x}}{\alpha} - \frac{\beta}{\alpha^2} - \frac{T\beta}{\alpha} + e^{T\alpha} \left(\frac{\bar{x}}{\alpha} + \frac{\beta}{\alpha^2} \right).$$

Substituting Equation (52),

$$\begin{aligned} \mathbb{E}[Y_T] &= \frac{y_0}{\kappa} - \frac{\eta}{\kappa} + \eta T + e^{-\kappa T} \left(\frac{\eta}{\kappa} - \frac{y_0}{\kappa} \right) \\ &= \eta T + \frac{y_0 - \eta}{\kappa} (1 - e^{-\kappa T}), \end{aligned}$$

which is the expression Schoutens (2003, p. 89) states independently, the two agree. Letting $d = y_0 - \eta$,

$$\mathbb{E}[Y_T] = \eta T + \frac{d}{\kappa} (1 - e^{-\kappa T}). \quad (53)$$

Dufresne gives $\mathbb{E}[Y_T^2]$

$$\begin{aligned} \mathbb{E}[Y_t^2] &= \frac{\bar{x}^2}{\alpha^2} + \frac{2\bar{x}\beta}{\alpha^3} + \frac{\beta^2}{\alpha^4} - \frac{\bar{x}\gamma^2}{\alpha^3} - \frac{5\beta\gamma^2}{2\alpha^4} + t \left(\frac{2\bar{x}\beta}{\alpha^2} + \frac{2\beta^2}{\alpha^3} - \frac{\beta\gamma^2}{\alpha^3} \right) + \frac{t^2\beta^2}{\alpha^2} \\ &+ e^{t\alpha} \left[-\frac{2\bar{x}^2}{\alpha^2} - \frac{4\bar{x}\beta}{\alpha^3} - \frac{2\beta^2}{\alpha^4} + \frac{2\beta\gamma^2}{\alpha^4} + t \left(-\frac{2\bar{x}\beta}{\alpha^2} - \frac{2\beta^2}{\alpha^3} - \frac{2\bar{x}\gamma^2}{\alpha^2} - \frac{2\beta\gamma^2}{\alpha^3} \right) \right] \\ &+ e^{2t\alpha} \left(\frac{\bar{x}^2}{\alpha^2} + \frac{2\bar{x}\beta}{\alpha^3} + \frac{\beta^2}{\alpha^4} + \frac{\bar{x}\gamma^2}{\alpha^3} + \frac{\beta\gamma^2}{2\alpha^4} \right) \end{aligned}$$

Substituting Equation (52) and using $\alpha^2 = \kappa^2$, $\alpha^3 = -\kappa^3$ and $\alpha^4 = \kappa^4$, and rearranging,

$$\begin{aligned} \mathbb{E}[Y_T^2] &= \eta^2 T^2 + \frac{2\eta T d}{\kappa} + \frac{d^2}{\kappa^2} + \frac{\lambda^2 \eta T}{\kappa^2} + \frac{\lambda^2}{\kappa^3} \left(d - \frac{3}{2}\eta \right) \\ &+ e^{-\kappa T} \left[-\frac{2\eta T d}{\kappa} - \frac{2d^2}{\kappa^2} - \frac{2\lambda^2 T d}{\kappa^2} + \frac{2\lambda^2 \eta}{\kappa^3} \right] \\ &+ e^{-2\kappa T} \left[\frac{d^2}{\kappa^2} - \frac{\lambda^2}{\kappa^3} \left(d + \frac{1}{2}\eta \right) \right]. \quad (54) \end{aligned}$$

Squaring Equation (53),

$$(\mathbb{E}[Y_T])^2 = \eta^2 T^2 + \frac{2\eta T d}{\kappa} + \frac{d^2}{\kappa^2} - e^{-\kappa T} \left[\frac{2\eta T d}{\kappa} + \frac{2d^2}{\kappa^2} \right] + e^{-2\kappa T} \frac{d^2}{\kappa^2}. \quad (55)$$

To find the variance we take

$$\text{Var}(Y_T) = \mathbb{E}[Y_T^2] - (\mathbb{E}[Y_T])^2$$

Every term of Equation (54) not carrying a factor λ^2 appears identically in Equation (55) and cancels giving,

$$\text{Var}(Y_T) = \lambda^2 \left[\frac{\eta T}{\kappa^2} + \frac{d}{\kappa^3} - \frac{3\eta}{2\kappa^3} + e^{-\kappa T} \left(\frac{2\eta}{\kappa^3} - \frac{2Td}{\kappa^2} \right) - e^{-2\kappa T} \left(\frac{d}{\kappa^3} + \frac{\eta}{2\kappa^3} \right) \right].$$

Grouping the η and d terms separately and restoring $d = y_0 - \eta$,

$$\text{Var}(Y_T) = \frac{\lambda^2 \eta}{2\kappa^3} (2\kappa T - 3 + 4e^{-\kappa T} - e^{-2\kappa T}) + \frac{\lambda^2 (y_0 - \eta)}{\kappa^3} (1 - 2\kappa T e^{-\kappa T} - e^{-2\kappa T}),$$

which is Equation (50). Setting $\lambda = 0$ gives $\text{Var}(Y_T) = 0$ as required, and both bracketed factors vanish as $T \rightarrow 0$.

D.6 Sample Moment Estimators

This section sets out the empirical moment estimators reported in Table 7 of Section 3.6. They are applied to the same daily log returns $r_t = \ln(S_t/S_{t-1})$ used for the annualised volatility of Equation (8). Writing

$$m_k = \frac{1}{n} \sum_{t=1}^n (r_t - \bar{r})^k$$

for the k th central moment of the sample, the reported quantities are

$$\begin{aligned} \bar{r} &= \frac{1}{n} \sum_{t=1}^n r_t, & \hat{\sigma} &= \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}, \\ \text{skewness} &= \frac{m_3}{m_2^{3/2}}, & \text{kurtosis} &= \frac{m_4}{m_2^2}, \end{aligned} \quad (56)$$

where $\bar{r}$ is the mean and $\hat{\sigma}$ the standard deviation of the daily log returns. Note that the central moments m_k divide by n , whereas the standard deviation and the annualised volatility of Equation (8) use the unbiased $1/(n-1)$ estimator, so m_2 and $\hat{\sigma}^2$ differ by a factor of $(n-1)/n$. On samples of the size used here the distinction is immaterial.

D.7 Computational Methods

This section sets out the two numerical routines used throughout the results chapters: recovering a density from a characteristic function, and recovering the variance and kurtosis from one.

D.7.1 Recovering Densities from the Characteristic Function

Of the five processes only the Meixner has a density in closed form, therefore recovering the distribution is straightforward. For the variance gamma, CGMY, and NIG we recover the density by Fourier inversion of the same characteristic function used to price. For a random variable with characteristic function ϕ ,

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} \phi(u) du = \frac{1}{\pi} \int_0^{\infty} \operatorname{Re}[e^{-iux} \phi(u)] du,$$

the second equality following from $\phi(-u) = \overline{\phi(u)}$, which holds because f is real valued. We evaluate the second form numerically, with the characteristic function of the log return over $[0, T]$ assembled as in the pricing routine,

$$\phi_T(u) = \exp(T \psi_X(u) + iu mT),$$

where ψ_X is the characteristic exponent of the process and m the mean-correcting drift of Equation (10). Both are taken from the pricing code, so the densities plotted are those of the laws the calibrations were run on.

As a check, the mean, standard deviation, skewness and excess kurtosis were computed by numerical integration against each recovered density and compared with the analytic moments; the two agree throughout.

D.7.2 Moments from the Characteristic Function

The variance and kurtosis for each model can be determined by examining its characteristic function. Let $\phi(u)$ denote the characteristic function of the log return $\log(S_T/S_0)$. Section 6.4 gives the characteristic function of $\log S_T$ rather than of the log return, so it is shifted by $e^{-iu \log S_0}$ to obtain ϕ . Finally, let $K(u) = \log \phi(u)$ be the corresponding cumulant generating function. The variance and excess kurtosis of the log return are

$$\operatorname{Var} = \kappa_2, \quad \operatorname{exkurt} = \frac{\kappa_4}{\kappa_2^2},$$

where κ_2 and κ_4 are its second and fourth cumulants,

$$\kappa_2 = -K''(0), \quad \kappa_4 = K''''(0).$$

The derivatives are evaluated by using the following approximations for $h \rightarrow 0$,

$$K''(0) \approx \frac{K(h) - 2K(0) + K(-h)}{h^2},$$

$$K''''(0) \approx \frac{K(2h) - 4K(h) + 6K(0) - 4K(-h) + K(-2h)}{h^4},$$

with $K(0) = \log \phi(0) = 0$. In our computation h cannot simply be made small. Each entry of h is computed across a ladder of steps from 3×10^{-3} to 1.2×10^{-1} , and the value accepted is the first at which consecutive steps agree to within 2%. The variance, needed only through K'' , is far better conditioned than the kurtosis and converges at every step in the ladder[§].

[§]The ladder is retained for it regardless, so that both quantities are produced by one routine.

E Implied Volatility by Newton–Raphson

This appendix sets out the Newton–Raphson iteration used to invert Equation (7) for the implied volatility of a single contract in Section 3.5. The goal is to solve

$$g(\sigma) := C(\sigma) - C_{\text{mkt}} = 0$$

for σ , where $C(\sigma)$ denotes the Black–Scholes price (C_{BS}) viewed as a function of volatility alone, all other inputs being fixed and observable. No closed-form inverse exists, so the root must be located iteratively. Newton–Raphson does this by *repeatedly replacing the curve g with a straight line and solving the straight line exactly*. Given a current estimate σ_n , take the first-order Taylor expansion of g about that point,

$$g(\sigma) \approx g(\sigma_n) + g'(\sigma_n)(\sigma - \sigma_n), \quad (57)$$

and ask where this linear approximation, the tangent to g at σ_n , crosses zero. Setting the right-hand side of Equation (57) to zero and rearranging gives the next iterate,

$$\sigma_{n+1} = \sigma_n - \frac{g(\sigma_n)}{g'(\sigma_n)} = \sigma_n - \frac{C(\sigma_n) - C_{\text{mkt}}}{C'(\sigma_n)}, \quad (58)$$

and the procedure repeats until $|g(\sigma_n)|$ falls below a tolerance, which we choose to be 10^{-9} . The derivative in the denominator of Equation (58) is the option’s *vega*. Differentiating the pricing formula, Equation (4), with respect to σ gives

$$\mathcal{V}(\sigma_n) = C'(\sigma_n) = S_0 e^{-qT} \sqrt{T} \varphi(d_1) = S_0 e^{-qT} \sqrt{T} \varphi\left(\frac{\log(S_0/K) + (r - q + \frac{1}{2}\sigma_n^2)T}{\sigma_n \sqrt{T}}\right),$$

where φ denotes the standard normal *density*. *

*Schoutens (2003, p. 41) states this derivative as $S_0 \sqrt{T} N(d_1)$ with N the cumulative distribution function. This appears to be in error, since differentiation of (4) yields the density.

F The Nelder–Mead Simplex Method

This explanation leans on material from Virtanen et al. (2020). Writing θ for the parameter vector of whichever model is being fitted, the objective is the RMSE of Equation (6),

$$\text{RMSE}(\theta) = \sqrt{\frac{1}{n} \sum_{i=1}^n (C_{\text{mkt},i} - C_{\text{model},i}(\theta))^2}, \quad \hat{\theta} = \arg \min_{\theta} \text{RMSE}(\theta), \quad (59)$$

where $C_{\text{model},i}(\theta)$ is the price the model assigns to contract i at parameters θ , and the minimisation runs over all n contracts of the chain. The dimension of θ is different depending on the model used.

Each evaluation of (59) requires a Carr–Madan inversion at every maturity, and there is no analytic derivative of the price with respect to θ . A derivative-free method is needed, and we use the simplex method of Nelder and Mead (1965).

The algorithm is as follows. Let p denote the number of parameters. For a p -dimensional parameter vector the method needs $p + 1$ candidate parameter vectors, forming a simplex. The initial simplex is built from the starting point $\theta^{(0)}$ of Appendix G, which forms the first vertex; the remaining p vertices are generated from it. Each is a copy of the starting point with a single parameter increased by 5%: the second vertex bumps the first parameter, the third bumps a different parameter, and so on until $p + 1$ vectors are obtained. The result is $p + 1$ vectors that are identical except that p of them each have one parameter increased by 5%, with the last being the original starting point. We then order the vertices so that $\text{RMSE}(\theta_1) \leq \dots \leq \text{RMSE}(\theta_{p+1})$ and, writing $\bar{\theta}$ for the centroid of all but the worst vertex, each iteration attempts to replace that worst vertex. This gives a decision tree that determines the next step.

Each iteration first computes the reflection of the worst vertex through the centroid of the others,

$$\theta_r = \bar{\theta} + (\bar{\theta} - \theta_{p+1}),$$

and the RMSE it attains then decides which of four moves is taken. Recall that the vertices are ordered so that θ_1 is the best-fitting and θ_{p+1} the worst.

1. If $\text{RMSE}(\theta_r) < \text{RMSE}(\theta_1)$, the reflected vector fits better than every current vertex, so that direction in parameter space is promising. We try twice as far, $\theta_e = \bar{\theta} + 2(\bar{\theta} - \theta_{p+1})$, and replace θ_{p+1} with whichever of θ_e and θ_r fits better.
2. If $\text{RMSE}(\theta_r)$ falls between $\text{RMSE}(\theta_1)$ and $\text{RMSE}(\theta_p)$, the reflected vector improves on the second-worst vertex without being a new best, so we accept it in place of θ_{p+1} .
3. If $\text{RMSE}(\theta_r) \geq \text{RMSE}(\theta_p)$, the simplex has overshoot, so we try a shorter step, a parameter vector halfway between $\bar{\theta}$ and the better of θ_{p+1} and θ_r , accepted if it improves on both.
4. If that also fails, we move every vertex halfway towards the best-fitting one, $\theta_i \mapsto \theta_1 + \frac{1}{2}(\theta_i - \theta_1)$.

The simplex thus walks through parameter space towards lower RMSE, stretching when it finds a productive direction and collapsing towards its best vertex once it has bracketed a minimum.

Termination is judged by whether the simplex itself has stopped moving. The search continues until *both*

$$\max_{i=2,\dots,p+1} \|\theta_i - \theta_1\|_\infty \leq \text{xatol} \quad \text{and} \quad \max_{i=2,\dots,p+1} |\text{RMSE}(\theta_i) - \text{RMSE}(\theta_1)| \leq \text{fatol} \quad (60)$$

hold at the same iteration. The second condition asks that all $p + 1$ candidate parameter vectors price the chain to within `fatol` of one another in RMSE; the first asks that they have also converged on the same parameter values, every vertex lying within `xatol` of the best in every coordinate of θ .

The tolerances are `xatol` = `fatol` = 10^{-6} with a limit of 4000 evaluations for the variance gamma, CGMY and BNS calibrations, 10^{-5} with 1500 for the Meixner fit, 10^{-5} with 3000 for the time-changed models, and 10^{-8} with 1500 for the per-maturity Lévy fits, which are cheaper to evaluate. To incorporate boundaries into the search, we imposed that whenever a parameter would exceed a boundary, the RMSE is set to 10^6 . The downside of this is that it can force parameters to become stuck at the boundary. This happened for the CGMY intOU models, which failed to improve on their VG counterparts, suggesting a failed optimisation.

G Optimisation Starting Points

The starting points for each calibration are listed here, one table per model.

Table 41: Meixner starting points.

Start	α	β	δ
1	0.10	-0.10	1.00
2	0.50	-1.00	0.50
3	0.20	0.50	2.00
4	1.00	-2.00	0.20
5	0.05	-0.50	3.00
6	0.30	-1.50	0.35
7	0.80	0.00	1.00
8	1.50	-1.00	0.50
9	2.00	-2.50	0.30
10	1.00	-0.50	2.00

Table 42: Variance gamma starting points.

Start	C	G	M
1	1.50	6.00	15.00
2	5.00	20.00	30.00
3	0.50	2.00	5.00
4	20.00	40.00	60.00
5	1.00	0.30	1.60
6	50.00	90.00	150.00

Table 43: CGMY starting points.

Start	C	G	M	Y
1	0.02	0.08	7.60	1.29
2	1.00	5.00	15.00	0.50
3	3.00	60.00	130.00	1.36
4	0.50	2.00	8.00	1.00
5	0.10	1.00	20.00	1.50
6	10.00	30.00	60.00	0.20

Table 44: Normal inverse Gaussian starting points.

Start	α	β	δ
1	6.20	−3.90	0.16
2	20.00	−5.00	0.50
3	2.00	−0.50	0.30
4	50.00	−20.00	1.00
5	100.00	10.00	50.00
6	10.00	0.00	0.20

Table 45: Generalised hyperbolic starting points.

Start	α	β	δ	ν
1	3.80	−3.80	0.24	−1.76
2	10.00	−3.00	0.50	−2.00
3	36.50	3.10	18.80	−83.70
4	5.00	0.00	1.00	−1.00
5	20.00	−10.00	2.00	−5.00
6	2.00	−1.00	0.30	1.00

Table 46: BNS Gamma–OU starting points. The BNS starting points listed here are those of a re-calibration run, seeded on Schoutens (2003, Table 7.1)’s estimate; the original starting values were not preserved. The re-calibration recovers the same Gamma–OU optimum of 1.7998.

Start	ρ	λ	a	b	σ_0^2
1	−1.2606	0.5783	1.4338	11.6641	0.0145
2	−1.0000	1.0000	1.5000	10.0000	0.0500
3	−2.0000	2.0000	1.0000	5.0000	0.2000
4	−0.5000	5.0000	2.0000	20.0000	0.4000
5	−5.0000	8.0000	3.0000	30.0000	0.6500
6	−0.2000	0.5000	0.5000	2.0000	0.7000
7	−10.0000	10.0000	5.0000	50.0000	0.8000
8	−1.0000	3.0000	0.3000	1.0000	0.3000

Table 47: BNS IG–OU starting points.

Start	ρ	λ	a	b	σ_0^2
1	−0.1926	0.0636	6.2410	0.7995	0.0156
2	−0.5000	1.0000	3.0000	1.0000	0.0500
3	−1.0000	3.0000	1.0000	2.0000	0.3000
4	−0.3000	8.0000	0.1000	2.0000	0.8500
5	−2.0000	5.0000	0.5000	1.5000	0.5000
6	−0.1000	0.5000	5.0000	0.8000	0.2000
7	−3.0000	10.0000	0.0500	3.0000	0.7500
8	−0.7000	2.0000	2.0000	0.5000	0.4000

Table 48: Meixner–CIR starting points.

Start	α	β	δ	κ	η	λ
1	0.1231	−0.5875	3.3588	0.5391	1.5746	1.8772
2	0.1108	−0.9858	3.6288	0.5145	0.9029	1.3750
3	0.0890	−1.1323	5.0262	0.6020	1.5560	1.9992
4	0.3000	−0.5000	1.5000	0.3881	1.4012	1.3612
5	0.0500	−1.5000	8.0000	0.6000	1.5000	1.0000
6	0.2000	−0.8000	2.5000	1.0000	1.0000	1.0000
7	0.1500	−2.0000	4.0000	0.4000	1.4000	1.3600
8	0.5000	−0.3000	1.0000	1.5000	0.8000	2.5000

Table 49: Meixner–Gamma–OU starting points.

Start	α	β	δ	λ	a	b
1	0.1231	−0.5875	3.3588	0.6252	0.4239	0.5962
2	0.1108	−0.9858	3.6288	1.2517	0.5841	0.6282
3	0.0890	−1.1323	5.0262	0.8826	0.5945	0.8524
4	0.3000	−0.5000	1.5000	1.0740	0.3573	0.6143
5	0.0500	−1.5000	8.0000	1.1729	0.5914	0.6558
6	0.2000	−0.8000	2.5000	1.0000	1.0000	1.0000
7	0.1500	−2.0000	4.0000	0.5000	0.3000	0.4000
8	0.5000	−0.3000	1.0000	2.0000	0.8000	1.2000

Table 50: Meixner–IG–OU starting points.

Start	α	β	δ	λ	a	b
1	0.1231	−0.5875	3.3588	1.1559	0.6496	0.8572
2	0.1108	−0.9858	3.6288	1.2801	0.6615	0.8104
3	0.0890	−1.1323	5.0262	1.0622	0.6092	0.9999
4	0.3000	−0.5000	1.5000	1.2190	0.6564	0.8266
5	0.0500	−1.5000	8.0000	1.1315	0.8993	0.6168
6	0.2000	−0.8000	2.5000	1.0000	1.0000	1.0000
7	0.1500	−2.0000	4.0000	0.6000	0.4000	0.5000
8	0.5000	−0.3000	1.0000	2.0000	1.0000	1.5000

Table 51: VG–CIR starting points.

Start	C	G	M	κ	η	λ
1	11.9896	25.8523	35.5344	0.5391	1.5746	1.8772
2	11.4838	23.2880	40.1291	0.5145	0.9029	1.3750
3	14.9248	26.1529	50.4425	0.6020	1.5560	1.9992
4	5.0000	10.0000	20.0000	0.3881	1.4012	1.3612
5	20.0000	40.0000	60.0000	0.6000	1.5000	1.0000
6	1.0000	5.0000	10.0000	1.0000	1.0000	1.0000
7	8.0000	15.0000	30.0000	0.4000	1.4000	1.3600
8	30.0000	50.0000	80.0000	1.5000	0.8000	2.5000

Table 52: VG–Gamma–OU starting points.

Start	C	G	M	λ	a	b
1	11.9896	25.8523	35.5344	0.6252	0.4239	0.5962
2	11.4838	23.2880	40.1291	1.2517	0.5841	0.6282
3	14.9248	26.1529	50.4425	0.8826	0.5945	0.8524
4	5.0000	10.0000	20.0000	1.0740	0.3573	0.6143
5	20.0000	40.0000	60.0000	1.1729	0.5914	0.6558
6	1.0000	5.0000	10.0000	1.0000	1.0000	1.0000
7	8.0000	15.0000	30.0000	0.5000	0.3000	0.4000
8	30.0000	50.0000	80.0000	2.0000	0.8000	1.2000

Table 53: VG–IG–OU starting points.

Start	C	G	M	λ	a	b
1	11.9896	25.8523	35.5344	1.1559	0.6496	0.8572
2	11.4838	23.2880	40.1291	1.2801	0.6615	0.8104
3	14.9248	26.1529	50.4425	1.0622	0.6092	0.9999
4	5.0000	10.0000	20.0000	1.2190	0.6564	0.8266
5	20.0000	40.0000	60.0000	1.1315	0.8993	0.6168
6	1.0000	5.0000	10.0000	1.0000	1.0000	1.0000
7	8.0000	15.0000	30.0000	0.6000	0.4000	0.5000
8	30.0000	50.0000	80.0000	2.0000	1.0000	1.5000

Table 54: CGMY–CIR starting points.

Start	C	G	M	Y	κ	η	λ
1	0.0074	0.1025	11.3940	1.6765	0.5391	1.5746	1.8772
2	0.0415	3.9134	30.6322	1.3664	0.5145	0.9029	1.3750
3	0.0672	6.1316	44.7448	1.2911	0.6020	1.5560	1.9992
4	0.5000	5.0000	15.0000	0.5000	0.3881	1.4012	1.3612
5	0.0100	1.0000	5.0000	1.0000	0.6000	1.5000	1.0000
6	0.1000	2.0000	20.0000	1.5000	1.0000	1.0000	1.0000
7	0.0050	0.5000	8.0000	1.8000	0.4000	1.4000	1.3600
8	0.2000	8.0000	35.0000	0.8000	1.5000	0.8000	2.5000

Table 55: CGMY–Gamma–OU starting points.

Start	C	G	M	Y	λ	a	b
1	0.0074	0.1025	11.3940	1.6765	0.6252	0.4239	0.5962
2	0.0415	3.9134	30.6322	1.3664	1.2517	0.5841	0.6282
3	0.0672	6.1316	44.7448	1.2911	0.8826	0.5945	0.8524
4	0.5000	5.0000	15.0000	0.5000	1.0740	0.3573	0.6143
5	0.0100	1.0000	5.0000	1.0000	1.1729	0.5914	0.6558
6	0.1000	2.0000	20.0000	1.5000	1.0000	1.0000	1.0000
7	0.0050	0.5000	8.0000	1.8000	0.5000	0.3000	0.4000
8	0.2000	8.0000	35.0000	0.8000	2.0000	0.8000	1.2000

Table 56: CGMY–IG–OU starting points.

Start	C	G	M	Y	λ	a	b
1	0.0074	0.1025	11.3940	1.6765	1.1559	0.6496	0.8572
2	0.0415	3.9134	30.6322	1.3664	1.2801	0.6615	0.8104
3	0.0672	6.1316	44.7448	1.2911	1.0622	0.6092	0.9999
4	0.5000	5.0000	15.0000	0.5000	1.2190	0.6564	0.8266
5	0.0100	1.0000	5.0000	1.0000	1.1315	0.8993	0.6168
6	0.1000	2.0000	20.0000	1.5000	1.0000	1.0000	1.0000
7	0.0050	0.5000	8.0000	1.8000	0.6000	0.4000	0.5000
8	0.2000	8.0000	35.0000	0.8000	2.0000	1.0000	1.5000

Table 57: NIG–CIR starting points.

Start	α	β	δ	κ	η	λ
1	18.4815	−4.8412	0.4685	0.5391	1.5746	1.8772
2	29.4722	−15.9048	0.5071	0.5145	0.9029	1.3750
3	29.1553	−13.9331	0.5600	0.6020	1.5560	1.9992
4	10.0000	−2.0000	1.0000	0.3881	1.4012	1.3612
5	40.0000	−20.0000	0.3000	0.6000	1.5000	1.0000
6	15.0000	−8.0000	0.7000	1.0000	1.0000	1.0000
7	50.0000	−25.0000	0.4000	0.4000	1.4000	1.3600
8	8.0000	−1.0000	1.5000	1.5000	0.8000	2.5000

Table 58: NIG–Gamma–OU starting points.

Start	α	β	δ	λ	a	b
1	18.4815	−4.8412	0.4685	0.6252	0.4239	0.5962
2	29.4722	−15.9048	0.5071	1.2517	0.5841	0.6282
3	29.1553	−13.9331	0.5600	0.8826	0.5945	0.8524
4	10.0000	−2.0000	1.0000	1.0740	0.3573	0.6143
5	40.0000	−20.0000	0.3000	1.1729	0.5914	0.6558
6	15.0000	−8.0000	0.7000	1.0000	1.0000	1.0000
7	50.0000	−25.0000	0.4000	0.5000	0.3000	0.4000
8	8.0000	−1.0000	1.5000	2.0000	0.8000	1.2000

Table 59: NIG–IG–OU starting points.

Start	α	β	δ	λ	a	b
1	18.4815	−4.8412	0.4685	1.1559	0.6496	0.8572
2	29.4722	−15.9048	0.5071	1.2801	0.6615	0.8104
3	29.1553	−13.9331	0.5600	1.0622	0.6092	0.9999
4	10.0000	−2.0000	1.0000	1.2190	0.6564	0.8266
5	40.0000	−20.0000	0.3000	1.1315	0.8993	0.6168
6	15.0000	−8.0000	0.7000	1.0000	1.0000	1.0000
7	50.0000	−25.0000	0.4000	0.6000	0.4000	0.5000
8	8.0000	−1.0000	1.5000	2.0000	1.0000	1.5000

H Python Code

All analysis was carried out in Python 3.14. This appendix lists a single script that reproduces every numerical result reported in the dissertation, in close to the order in which the results appear. It depends on numpy, pandas and scipy, and on two data files: the pruned KOSPI 200 option chain and Data.xlsx, which holds the S&P 500 chain of Schoutens (2003, Appendix C) used for the validation sections, Sections 4.5, 5.4 and 6.3. The script is organised into ten sections.

The Interactive Brokers pull that produced the raw quotes is given as a comment in Section 1 rather than as running code, since it requires a live TWS session and would return a different snapshot on each call; the chain it wrote is preserved and read from disk instead, so the results below are reproducible without a broker connection. The filtering that reduces the raw pull to the 212 contracts used throughout is also described rather than reproduced. Code for generating the figures is omitted as well. Running the file end to end will take several hours.

```
1 # %%
2
3 # Every numerical result reported in the dissertation, in close to the order in which
4 # the results appear.
5
6 # 0. Pricing -- overview of functions used repeatedly,
7 # -- characteristic functions and the Carr-Madan transform
8 # 1. Data -- just focusing on how I pruned the data - not data pulling
9 # 2. Black-Scholes -- both the full model and per maturity
10 # 3. Levy models -- Meixner, VG, CGMY, NIG and GH
11 # 4. Per maturity -- the same five Levy models fitted by maturity
12 # 5. BNS -- Gamma-OU and IG-OU
13 # 6. Time change -- the twelve Levy models with stochastic time process added on
14 # 7. Moments -- variance and kurtosis of each fit
15 # 8. Volatility -- the implied volatility
16 # 9. Validation -- every model above refitted to Schoutens' S&P 500 chain
17
18 # The only inputs are k200_implied_vol_by_maturity/k200_iv_per_contract.csv, the
19 # pruned 212-contract chain, and Data.xlsx for the S&P 500 validation of Section 9.
20 # Running the whole file takes several hours.
21
22 # The libraries used throughout this dissertation
23 import time
24
25 import numpy as np
26 import pandas as pd
27 from scipy.optimize import minimize, minimize_scalar
28 from scipy.special import gamma as gamma_fn, gammaln, kv, loggamma
29 from scipy.stats import norm
30
31 # %% =====
32 # 0. PRICING MACHINERY
33 # The characteristic functions and the Carr-Madan transform. Every price computed
34 # below comes through this section.
35
36 CM_ALPHA = 0.75 # Carr-Madan damping
37 N_GRID = 4096 # quadrature points
38 V_MAX = 200.0 # upper integration bound
39
40
41 # -----
42 # 0.1 The Levy models of Chapter 6: CGMY, VG, NIG and GH.
43 CM_ALPHA = 0.75 # Carr-Madan damping (Schoutens Ch.2)
44 N_GRID = 4096
45 V_MAX = 200.0
```

```

46
47
48 # CGMY model input
49 def cgmy_char_exponent(u, C, G, M, Y):
50     return C * gamma_fn(-Y) * ((M - 1j * u) ** Y - M**Y + (G + 1j * u) ** Y - G**Y)
51
52 #Define the m drift parameter
53 def m_new_cgmy(C, G, M, Y, r, q):
54     return r - q - C * gamma_fn(-Y) * ((M - 1) ** Y - M**Y + (G + 1) ** Y - G**Y)
55
56 #Now produce the Carr and Madan, with parameters as inputs
57 def carr_madan_batch(strikes, ttms, rates, divs, C, G, M, Y, S0,
58                     alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
59
60     #Define strikes, ttm, rate and div yields - all relevant in solving equations
61     strikes = np.asarray(strikes, dtype=float)
62     ttms = np.asarray(ttms, dtype=float)
63     rates = np.asarray(rates, dtype=float)
64     divs = np.asarray(divs, dtype=float)
65     v = np.linspace(1e-10, v_max, n_grid)[: , None]
66     log_k = np.log(strikes)[None, :]
67     prices = np.empty(len(strikes))
68
69     # Denominator of psi(v) is contract-independent
70     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
71
72     #Now the actual carr-madan process - see equations for references
73     for T in np.unique(ttms):
74         sel = ttms == T
75         r, q = rates[sel][0], divs[sel][0]
76         u = v - (alpha + 1) * 1j
77         cf = np.exp(1j * u * (np.log(S0) + m_new_cgmy(C, G, M, Y, r, q) * T)
78                 + T * cgmy_char_exponent(u, C, G, M, Y)) # (N, 1)
79         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
80                             denom)
81         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
82                                     [:, 0], axis=0)
83
84     return prices
85
86 # -----
87 # Variance Gamma
88 #Same as CGMY - define the characteristic exponent
89 def vg_char_exponent(u, C, G, M):
90     return C * np.log(G * M / (G * M + (M - G) * 1j * u + u**2))
91
92 #Then define the m parameter
93 def m_new_vg(C, G, M, r, q):
94     return r - q + C * np.log((M - 1) * (G + 1) / (M * G))
95
96 #Now again define the carr and madan function process - just like CGMY
97 def carr_madan_batch_vg(strikes, ttms, rates, divs, C, G, M, S0,
98                         alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
99     strikes = np.asarray(strikes, dtype=float)
100     ttms = np.asarray(ttms, dtype=float)
101     rates = np.asarray(rates, dtype=float)
102     divs = np.asarray(divs, dtype=float)
103
104     v = np.linspace(1e-10, v_max, n_grid)[: , None]
105     log_k = np.log(strikes)[None, :]
106     prices = np.empty(len(strikes))
107     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
108
109     for T in np.unique(ttms):
110         sel = ttms == T
111         r, q = rates[sel][0], divs[sel][0]
112         u = v - (alpha + 1) * 1j
113         cf = np.exp(1j * u * (np.log(S0) + m_new_vg(C, G, M, r, q) * T)

```

```

114         + T * vg_char_exponent(u, C, G, M))
115         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
116         denom)
116         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
117        [:, 0], axis=0)
117
118     return prices
119
120 #Define the RMSE calculation for the VG process, returning 1e6 at the boundary,
121 # as a boundary condition
122 def rmse_objective_vg(params, strikes, ttms, rates, divs, mids, S0):
123     C, G, M = params
124     if C <= 1e-6 or G <= 1e-6 or M <= 1.0 + 1e-6:
125         return 1e6
126     try:
127         model = carr_madan_batch_vg(strikes, ttms, rates, divs, C, G, M, S0)
128     except Exception:
129         return 1e6
130     if not np.all(np.isfinite(model)) or np.any(model < 0):
131         return 1e6
132     return float(np.sqrt(np.mean((model - np.asarray(mids)) ** 2)))
133
134
135 # -----
136 # Normal Inverse Gaussian
137
138 #Again, same process, define characteristic exponent
139 def nig_char_exponent(u, a, b, d):
140     return -d * (np.sqrt(a**2 - (b + 1j * u) ** 2) - np.sqrt(a**2 - b**2))
141
142 #find the m drift parameter
143 def m_new_nig(a, b, d, r, q):
144     return r - q + d * (np.sqrt(a**2 - (b + 1) ** 2) - np.sqrt(a**2 - b**2))
145
146 #Define the carr and madan approach - same as before
147 def carr_madan_batch_nig(strikes, ttms, rates, divs, a, b, d, S0,
148     alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
149     strikes = np.asarray(strikes, dtype=float)
150     ttms = np.asarray(ttms, dtype=float)
151     rates = np.asarray(rates, dtype=float)
152     divs = np.asarray(divs, dtype=float)
153
154     v = np.linspace(1e-10, v_max, n_grid)[: , None]
155     log_k = np.log(strikes)[None, :]
156     prices = np.empty(len(strikes))
157     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
158
159     for T in np.unique(ttms):
160         sel = ttms == T
161         r, q = rates[sel][0], divs[sel][0]
162         u = v - (alpha + 1) * 1j
163         cf = np.exp(1j * u * (np.log(S0) + m_new_nig(a, b, d, r, q) * T)
164             + T * nig_char_exponent(u, a, b, d))
165         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
166         denom)
166         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
167        [:, 0], axis=0)
167
168     return prices
169
170 #Now the RMSE objective - note the restrictions are present too
171 def rmse_objective_nig(params, strikes, ttms, rates, divs, mids, S0):
172     a, b, d = params
173     if a <= 1e-6 or d <= 1e-6 or abs(b) >= a - 1e-6 or abs(b + 1) >= a - 1e-6:
174         return 1e6
175     try:
176         model = carr_madan_batch_nig(strikes, ttms, rates, divs, a, b, d, S0)
177     except Exception:
178         return 1e6
179     if not np.all(np.isfinite(model)) or np.any(model < 0):

```

```

180         return 1e6
181     return float(np.sqrt(np.mean((model - np.asarray(mids)) ** 2)))
182
183
184 # -----
185 # Generalized Hyperbolic (Schoutens 5.3.11). Four parameters (alpha, beta,
186 # delta, nu);
187 # Unlike the other families here, GH's characteristic function is NOT of the
188 # form exp(t * psi(u)) with psi elementary: raising it to the power t requires
189 # the whole Bessel ratio to be exponentiated, i.e. phi_t(u) = (phi_1(u))^t.
190 # That is handled below by taking t*log(phi_1(u)) with a complex logarithm.
191
192 #Define characteristic function
193 def gh_log_char(u, a, b, d, nu):
194     s_u = np.sqrt(a**2 - (b + 1j * u) ** 2)
195     s_0 = np.sqrt(a**2 - b**2)
196     ratio = kv(nu, d * s_u) / kv(nu, d * s_0)
197     val = (nu / 2) * np.log((a**2 - b**2) / (a**2 - (b + 1j * u) ** 2)) + np.log(ratio)
198     if np.ndim(val) and np.shape(val)[0] > 1:
199         val = np.real(val) + 1j * np.unwrap(np.imag(val), axis=0)
200     return val
201
202 #Now get the m_new again, like the other
203 def m_new_gh(a, b, d, nu, r, q):
204     s_1 = np.sqrt(a**2 - (b + 1) ** 2)
205     s_0 = np.sqrt(a**2 - b**2)
206     return r - q - (
207         (nu / 2) * np.log((a**2 - b**2) / (a**2 - (b + 1) ** 2))
208         + np.log(kv(nu, d * s_1) / kv(nu, d * s_0))
209     )
210
211 #Now define carr and madan, but the characteristic formulation differs by taking it to
212 #the power
213 def carr_madan_batch_gh(strikes, ttms, rates, divs, a, b, d, nu, S0,
214                         alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
215     strikes = np.asarray(strikes, dtype=float)
216     ttms = np.asarray(ttms, dtype=float)
217     rates = np.asarray(rates, dtype=float)
218     divs = np.asarray(divs, dtype=float)
219
220     v = np.linspace(1e-10, v_max, n_grid)[:, None]
221     log_k = np.log(strikes)[None, :]
222     prices = np.empty(len(strikes))
223     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
224
225     u = v - (alpha + 1) * 1j
226     log_phi1 = gh_log_char(u, a, b, d, nu) # unit-time, shared across maturities
227
228     #made it to the exponent in this section
229     for T in np.unique(ttms):
230         sel = ttms == T
231         r, q = rates[sel][0], divs[sel][0]
232         cf = np.exp(1j * u * (np.log(S0) + m_new_gh(a, b, d, nu, r, q) * T) + T *
233             log_phi1)
234         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
235             denom)
236         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
237            [:, 0], axis=0)
238
239     return prices
240
241 #Now define the RMSE objective same as the others, with parameter boundaries included as
242 #penalty
243 def rmse_objective_gh(params, strikes, ttms, rates, divs, mids, S0):
244     a, b, d, nu = params
245     if a <= 1e-6 or d <= 1e-6 or abs(b) >= a - 1e-6 or abs(b + 1) >= a - 1e-6:
246         return 1e6
247     try:
248         model = carr_madan_batch_gh(strikes, ttms, rates, divs, a, b, d, nu, S0)
249     except Exception:

```

```

245         return 1e6
246     if not np.all(np.isfinite(model)) or np.any(model < 0):
247         return 1e6
248     return float(np.sqrt(np.mean((model - np.asarray(mids)) ** 2)))
249
250
251 # Here's the CGMY objective function that was missing earlier,
252 #Also included are the boundaries as penalty
253 def rmse_objective(params, strikes, ttms, rates, divs, mids, S0):
254     C, G, M, Y = params
255     if C <= 1e-6 or G <= 1e-6 or M <= 1.0 + 1e-6 or Y >= 2.0 - 1e-6:
256         return 1e6
257     try:
258         model = carr_madan_batch(strikes, ttms, rates, divs, C, G, M, Y, S0)
259     except Exception:
260         return 1e6
261     if not np.all(np.isfinite(model)) or np.any(model < 0):
262         return 1e6
263     return float(np.sqrt(np.mean((model - np.asarray(mids)) ** 2)))
264
265
266 # -----
267 # 0.2 The BNS models Carr and Madan Approach
268
269
270 #First define characteristic function
271 def bns_gamma_char_fn(u, T, r, q, rho, lam, a, b, sigma0_sq, S0):
272     u = np.asarray(u, dtype=complex)
273
274     f1 = 1j * u * rho - 0.5 * (u**2 + 1j * u) * (1 - np.exp(-lam * T))
275     f2 = 1j * u * rho - 0.5 * (u**2 + 1j * u)
276
277     term1 = 1j * u * (np.log(S0) + (r - q - a * lam * rho / (b - rho)) * T)
278     term2 = -0.5 * (u**2 + 1j * u) * (1 - np.exp(-lam * T)) / lam * sigma0_sq
279     term3 = a / (b - f2) * (b * np.log((b - f1) / (b - 1j * u * rho)) + f2 * lam * T)
280
281     return np.exp(term1 + term2 + term3)
282
283 #Then define carr-madan - same approach
284 def carr_madan_batch_bns(strikes, ttms, rates, divs, rho, lam, a, b, sigma0_sq, S0,
285                          alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
286     strikes = np.asarray(strikes, dtype=float)
287     ttms = np.asarray(ttms, dtype=float)
288     rates = np.asarray(rates, dtype=float)
289     divs = np.asarray(divs, dtype=float)
290
291     v = np.linspace(1e-10, v_max, n_grid)[: , None]
292     log_k = np.log(strikes)[None, :]
293     prices = np.empty(len(strikes))
294     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
295
296     for T in np.unique(ttms):
297         sel = ttms == T
298         r, q = rates[sel][0], divs[sel][0]
299         cf = bns_gamma_char_fn(v - (alpha + 1) * 1j, T, r, q,
300                                rho, lam, a, b, sigma0_sq, S0)
301         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
302                                denom)
303         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
304                                     [:, 0], axis=0)
305
306     return prices
307
308 #Now RMSE
309 def rmse_objective_bns(params, strikes, ttms, rates, divs, mids, S0):
310     rho, lam, a, b, sigma0_sq = params
311     if lam <= 1e-6 or a <= 1e-6 or b <= 1e-6 or sigma0_sq <= 1e-8:
312         return 1e6
313     if rho >= b - 1e-6:
314         return 1e6

```

```

313     try:
314         model = carr_madan_batch_bns(strikes, ttms, rates, divs,
315                                     rho, lam, a, b, sigma0_sq, S0)
316     except Exception:
317         return 1e6
318     if not np.all(np.isfinite(model)) or np.any(model < 0):
319         return 1e6
320     return float(np.sqrt(np.mean((model - np.asarray(mids)) ** 2)))
321
322
323
324
325 # The IG-OU variant.
326 def bns_ig_char_fn(u, T, r, q, rho, lam, a, b, sigma0_sq, S0):
327     u = np.asarray(u, dtype=complex)
328
329     f1 = 1j * u * rho - 0.5 * (u**2 + 1j * u) * (1 - np.exp(-lam * T))
330     f2 = 1j * u * rho - 0.5 * (u**2 + 1j * u)
331
332     drift = r - q - rho * lam * a / b * (1 - 2 * rho / b**2) ** -0.5
333     term1 = 1j * u * (np.log(S0) + drift * T)
334     term2 = 0.5 * (-u**2 - 1j * u) * (1 - np.exp(-lam * T)) * sigma0_sq / lam
335     term3 = a * (np.sqrt(b**2 - 2 * f1) - np.sqrt(b**2 - 2j * u * rho))
336
337     denom = np.sqrt(2 * f2 - b**2)
338     term4 = (2 * a * f2 / denom
339             * (np.arctan(np.sqrt((b**2 - 2j * u * rho) / (2 * f2 - b**2)))
340               - np.arctan(np.sqrt((b**2 - 2 * f1) / (2 * f2 - b**2)))))
341
342     return np.exp(term1 + term2 + term3 + term4)
343
344
345 def carr_madan_batch_ig(strikes, ttms, rates, divs, rho, lam, a, b, sigma0_sq, S0,
346                        alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
347     strikes = np.asarray(strikes, dtype=float)
348     ttms = np.asarray(ttms, dtype=float)
349     rates = np.asarray(rates, dtype=float)
350     divs = np.asarray(divs, dtype=float)
351
352     v = np.linspace(1e-10, v_max, n_grid)[:, None]
353     log_k = np.log(strikes)[None, :]
354     prices = np.empty(len(strikes))
355     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
356
357     for T in np.unique(ttms):
358         sel = ttms == T
359         r, q = rates[sel][0], divs[sel][0]
360         cf = bns_ig_char_fn(v - (alpha + 1) * 1j, T, r, q,
361                             rho, lam, a, b, sigma0_sq, S0)
362         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
363                               denom)
364         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
365                                    [:, 0], axis=0)
366
367     return prices
368
369
370 def rmse_objective_ig(params, strikes, ttms, rates, divs, mids, S0):
371     rho, lam, a, b, sigma0_sq = params
372     if lam <= 1e-6 or a <= 1e-6 or b <= 1e-6 or sigma0_sq <= 1e-8:
373         return 1e6
374     if 2 * rho >= b**2 - 1e-9:
375         return 1e6
376     try:
377         model = carr_madan_batch_ig(strikes, ttms, rates, divs,
378                                     rho, lam, a, b, sigma0_sq, S0)
379     except Exception:
380         return 1e6
381     if not np.all(np.isfinite(model)) or np.any(model < 0):
382         return 1e6

```

```

381     return float(np.sqrt(np.mean((model - np.asarray(mids)) ** 2)))
382
383
384 # -----
385 # 0.3 Time changed stochastic time models, the last 12 models
386
387 #Now we first define PSI for everything, based on the definition provided
388 #This is the characteristic exponent
389 def psi_vg(u, C, G, M):
390     u = np.asarray(u, dtype=complex)
391     return C * (np.log(G * M) - np.log(G * M + (M - G) * 1j * u + u**2))
392
393
394 def psi_cgmy(u, C, G, M, Y):
395     u = np.asarray(u, dtype=complex)
396     gY = gamma_fn(-Y)
397     return C * gY * ((M - 1j * u) ** Y - M**Y + (G + 1j * u) ** Y - G**Y)
398
399
400 def psi_nig(u, alpha, beta, delta):
401     u = np.asarray(u, dtype=complex)
402     return -delta * (np.sqrt(alpha**2 - (beta + 1j * u) ** 2) - np.sqrt(alpha**2 - beta
403 **2))
404
405
406 def psi_meixner(u, alpha, beta, delta):
407     u = np.asarray(u, dtype=complex)
408     return 2 * delta * (np.log(np.cos(beta / 2))
409 - np.log(np.cosh((alpha * u - 1j * beta) / 2)))
410
411 #Note in hindsight this GH parameterisation failed
412 #We use an already defined characteristic function
413 def psi_gh(u, alpha, beta, delta, nu):
414     return gh_log_char(u, alpha, beta, delta, nu)
415
416
417 # The Levy exponents and their parameter names, for the time-changed models
418 # of Section 6.
419 LEVY_PSI = {
420     "VG": (psi_vg, ["C", "G", "M"]),
421     "CGMY": (psi_cgmy, ["C", "G", "M", "Y"]),
422     "NIG": (psi_nig, ["alpha", "beta", "delta"]),
423     "Meixner": (psi_meixner, ["alpha", "beta", "delta"]),
424     "GH": (psi_gh, ["alpha", "beta", "delta", "nu"]),
425 }
426
427
428 # ----- time changes
429 # varphi(u; t, y0) = E[exp(iu Y_t) | y0], the characteristic function of the
430 # integrated rate-of-time-change process.
431
432
433 #Now this is the time change models listed in the main text
434 def varphi_cir(u, t, kappa, eta, lam, y0=1.0):
435     u = np.asarray(u, dtype=complex)
436     gam = np.sqrt(kappa**2 - 2 * lam**2 * 1j * u)
437     ekt = np.exp(gam * t)
438     # exp(kappa^2 eta t / lam^2) * exp(2 y0 iu / (kappa + gam coth(gam t/2)))
439     # / (cosh(gam t/2) + kappa sinh(gam t/2)/gam)^{2 kappa eta / lam^2}
440     coth = (ekt + 1) / (ekt - 1)
441     num = np.exp(kappa**2 * eta * t / lam**2) * np.exp(2 * y0 * 1j * u / (kappa + gam *
442 coth))
443     cosh_h = 0.5 * (np.exp(gam * t / 2) + np.exp(-gam * t / 2))
444     sinh_h = 0.5 * (np.exp(gam * t / 2) - np.exp(-gam * t / 2))
445     den = (cosh_h + kappa * sinh_h / gam) ** (2 * kappa * eta / lam**2)
446     return num / den
447
448 # The two OU clocks. Both are integrals of a mean-reverting positive process

```

```

449 # driven by a subordinator, so the clock never runs backwards.
450 def varphi_gamma_ou(u, t, lam, a, b, y0=1.0):
451     u = np.asarray(u, dtype=complex)
452     e = 1 - np.exp(-lam * t)
453     term1 = 1j * u * y0 * e / lam
454     term2 = lam * a / (1j * u - lam * b) * (b * np.log(b / (b - 1j * u * e / lam)) - 1j
455         * u * t)
456     return np.exp(term1 + term2)
457
458 def varphi_ig_ou(u, t, lam, a, b, y0=1.0):
459     u = np.asarray(u, dtype=complex)
460     kap = -2 * b**2 * 1j * u / lam
461     e = 1 - np.exp(-lam * t)
462     s = np.sqrt(1 + kap * e)
463     A = (1 - s) / kap + (1 / np.sqrt(1 + kap)) * (
464         np.arctanh(s / np.sqrt(1 + kap)) - np.arctanh(1 / np.sqrt(1 + kap)))
465     return np.exp(1j * u * y0 * e / lam + 2 * a * 1j * u / (b * lam) * A)
466
467 # The clocks and their parameter names.
468 TIME_CHANGE = {
469     "CIR": (varphi_cir, ["kappa", "eta", "lam"]),
470     "Gamma-OU": (varphi_gamma_ou, ["lam", "a", "b"]),
471     "IG-OU": (varphi_ig_ou, ["lam", "a", "b"]),
472 }
473
474 # ----- pricing
475 def levy_sv_char_fn(u, t, r, q, psi, levy_params, varphi, tc_params, S0):
476     u = np.asarray(u, dtype=complex)
477     num = varphi(-1j * psi(u, *levy_params), t, *tc_params)
478     den = varphi(-1j * psi(-1j, *levy_params), t, *tc_params)
479     return np.exp(1j * u * ((r - q) * t + np.log(S0))) * num / den ** (1j * u)
480
481 # The Carr-Madan pass for the time-changed models, structurally the same as
482 # the pure Levy one but taking the exponent and clock as arguments.
483 def carr_madan_levy_sv(strikes, ttms, rates, divs, psi, levy_params,
484     varphi, tc_params, S0,
485     alpha=CM_ALPHA, n_grid=N_GRID, v_max=V_MAX):
486     strikes = np.asarray(strikes, dtype=float)
487     ttms = np.asarray(ttms, dtype=float)
488     rates = np.asarray(rates, dtype=float)
489     divs = np.asarray(divs, dtype=float)
490
491     v = np.linspace(1e-10, v_max, n_grid)[:, None]
492     log_k = np.log(strikes)[None, :]
493     prices = np.empty(len(strikes))
494     denom = alpha**2 + alpha - v**2 + 1j * (2 * alpha + 1) * v
495
496     for T in np.unique(ttms):
497         sel = ttms == T
498         r, q = rates[sel][0], divs[sel][0]
499         cf = levy_sv_char_fn(v - (alpha + 1) * 1j, T, r, q,
500             psi, levy_params, varphi, tc_params, S0)
501         integrand = np.real(np.exp(-1j * v * log_k[:, sel]) * np.exp(-r * T) * cf /
502             denom)
503         prices[sel] = np.exp(-alpha * log_k[0, sel]) / np.pi * np.trapezoid(integrand, v
504            [:, 0], axis=0)
505
506     return prices
507
508 #This is the objective function but I add in the boundaries by each case as if
509 #statements
510 def make_objective(levy_name, tc_name, strikes, ttms, rates, divs, mids, S0):
511     psi, levy_names = LEVY_PSI[levy_name]
512     varphi, tc_names = TIME_CHANGE[tc_name]
513     n_levy = len(levy_names)

```



```

515     mids = np.asarray(mids)
516
517     def objective(params):
518         lp, tp = params[:n_levy], params[n_levy:]
519         # --- admissibility, model by model
520         if levy_name in ("VG", "CGMY"):
521             if lp[0] <= 1e-8 or lp[1] <= 1e-8 or lp[2] <= 1e-8:
522                 return 1e6
523             if levy_name == "CGMY" and not (-1e-8 < lp[3] < 2):
524                 return 1e6
525         elif levy_name == "NIG":
526             if lp[0] <= 1e-8 or abs(lp[1]) >= lp[0] or lp[2] <= 1e-8:
527                 return 1e6
528         elif levy_name == "Meixner":
529             if lp[0] <= 1e-8 or lp[2] <= 1e-8 or abs(lp[1]) >= np.pi:
530                 return 1e6
531         elif levy_name == "GH":
532             if lp[0] <= 1e-8 or abs(lp[1]) >= lp[0] or lp[2] <= 1e-8:
533                 return 1e6
534         if np.any(np.asarray(tp) <= 1e-8):
535             return 1e6
536         try:
537             model = carr_madan_levy_sv(strikes, ttms, rates, divs,
538                                     psi, tuple(lp), varphi, tuple(tp), S0)
539         except Exception:
540             return 1e6
541         if not np.all(np.isfinite(model)) or np.any(model < 0):
542             return 1e6
543         return float(np.sqrt(np.mean((model - mids) ** 2)))
544
545     return objective, levy_names + tc_names
546
547 ##### Data related activities - I've removed how I obtain the data cause its
548     excessive
549 #I load it here, its available upon request or convert the relevant appendix to a csv
550 PRUNED_CHAIN = "k200_implied_vol_by_maturity/k200_iv_per_contract.csv"
551
552 # The Korean risk-free rate and KOSPI 200 dividend yield
553 R, Q = 0.0291, 0.0079
554
555 #Read the chain, ttm, mid, rate and div_yld are already on it
556 def load_chain():
557     o = pd.read_csv(PRUNED_CHAIN, parse_dates=["expiry_date", "snapshot_date"])
558     return o, float(o["spot_price"].iloc[0])
559
560 # We save all these results to use later
561 opts, S0 = load_chain()
562 K = opts["strike"].to_numpy()
563 T = opts["ttm"].to_numpy()
564 RATE = opts["rate"].to_numpy()
565 DIV = opts["div_yld"].to_numpy()
566 MID = opts["mid"].to_numpy()
567
568 # The seven expiries with their times to maturity, ordered from nearest to
569 # furthest. This fixes the column order of every per-maturity table.
570 MATS = [(d.strftime("%b %y"), float(g["ttm"].iloc[0]))
571         for d, g in sorted(opts.groupby("expiry_date"),
572                             key=lambda kv: kv[1]["ttm"].iloc[0])]
573
574
575
576
577
578 # The Black-Scholes price, needed here for the implied volatilities and again
579 # as the benchmark model of Section 2.
580
581 #Create the black scholes call price
582 def bs_call(K, T, r, q, sigma, S0=S0):
583     d1 = (np.log(S0 / K) + (r - q + 0.5 * sigma ** 2) * T) / (sigma * np.sqrt(T))

```

```

584     d2 = d1 - sigma * np.sqrt(T)
585     return S0 * np.exp(-q * T) * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
586
587 #Create the vega thats needed to compute volatility later on
588 def bs_vega(K_, T_, r, q, sigma, spot):
589     d1 = (np.log(spot / K_) + (r - q + 0.5 * sigma ** 2) * T_) / (sigma * np.sqrt(T_))
590     return spot * np.exp(-q * T_) * np.sqrt(T_) * norm.pdf(d1)
591
592
593 #: Starting value for the per-contract Newton-Raphson iteration.
594 IV_SEED = 0.66
595
596 #: The same, for the S&P 500 chain of Section 9, whose volatility is a quarter
597 #: of the KOSPI's. This is Schoutens' own 0.20, which converges there.
598 SP_IV_SEED = 0.20
599
600 #Now calc implied volatility, I've added the newton raphson algo in there too
601 def implied_vol(mkt, K_, T_, r, q, sigma0, spot, tol=1e-9, max_iter=100):
602     #The algorithm
603     sigma = sigma0
604     for _ in range(max_iter):
605         diff = bs_call(K_, T_, r, q, sigma, S0=spot) - mkt
606         if abs(diff) < tol:
607             return sigma
608         v = bs_vega(K_, T_, r, q, sigma, spot)
609         # The degenerate cases the sigma_0 = 0.20 run kept reaching.
610         if v < 1e-12 or sigma <= 0 or not np.isfinite(sigma):
611             return np.nan
612         sigma = sigma - diff / v
613     return np.nan
614
615 #Now for the entire chain
616 def chain_implied_vols(o, spot, sigma0=IV_SEED):
617     return np.array([implied_vol(m, k_, t_, r_, q_, sigma0, spot)
618                     for m, k_, t_, r_, q_ in zip(o["mid"], o["strike"],
619                                                  o["ttm"], o["rate"],
620                                                  o["div_yld"])]])
621
622
623 #Convergence has succeeded
624 opts["iv"] = chain_implied_vols(opts, S0)
625 print(f"Implied volatility: {int(opts['iv'].notna().sum())}/{len(opts)} "
626       f"contracts converged")
627 print(opts.groupby("expiry_date")["iv"]
628       .agg(["count", "mean", "median", "min", "max"]).round(4).to_string())
629
630
631 #Create the four error measures
632 def measures(mid, model):
633     e = np.abs(mid - model)
634     return dict(APE=100 * e.mean() / mid.mean(),
635               AAE=float(e.mean()),
636               RMSE=float(np.sqrt(np.mean((mid - model) ** 2))),
637               ARPE=100 * float(np.mean(e / mid)))
638
639
640 # %% =====
641 # 2. BLACK-SCHOLES
642
643 #Actually fit the black-scholes model - done via the newton raphson algorithm
644 def fit_bs(o, spot=None, sigma0=IV_SEED, tol=1e-12, max_iter=100):
645
646     #Define fixed parameters
647     spot = S0 if spot is None else spot
648     K_ = o["strike"].to_numpy()
649     T_ = o["ttm"].to_numpy()
650     r_ = o["rate"].to_numpy()
651     q_ = o["div_yld"].to_numpy()
652     mid = o["mid"].to_numpy()
653

```

```

654 def g(s):
655     p = bs_call(K_, T_, r_, q_, s, S0=spot)
656     return float(np.mean((p - mid) * bs_vega(K_, T_, r_, q_, s, spot)))
657
658 #This helps with boundary conditions and actually making the algo
659 #But note the max iterations
660 sigma, h = sigma0, 1e-6
661 for _ in range(max_iter):
662     gs = g(sigma)
663     if abs(gs) < tol:
664         break
665     d = (g(sigma + h) - g(sigma - h)) / (2 * h)
666     if d == 0 or not np.isfinite(d):
667         break
668     step = gs / d
669     sigma -= step
670     if sigma <= 0 or not np.isfinite(sigma):
671         raise RuntimeError(f"Black-Scholes fit diverged from {sigma0:.4f}")
672     if abs(step) < 1e-14:
673         break
674 return sigma
675
676
677 # One volatility fitted across all 212 contracts at once. This is the benchmark
678 # every later model is measured against.
679 SIGMA_BS = fit_bs(opts)
680 bs_prices = bs_call(K, T, RATE, DIV, SIGMA_BS)
681 print(f"Black-Scholes, one sigma: {SIGMA_BS:.10f}")
682 print(" ", measures(MID, bs_prices))
683
684 # The same fit repeated expiry by expiry, giving seven volatilities rather than
685 # one. The pooled errors are recomputed over the union of the seven fits, so
686 # they are directly comparable with the whole-chain figures above.
687 bs_per_mat, fitted = [], np.empty(len(opts))
688 for expiry, g in opts.groupby("expiry_date"):
689     s = fit_bs(g)
690     p = bs_call(g["strike"].to_numpy(), g["ttm"].to_numpy(),
691                 g["rate"].to_numpy(), g["div_yld"].to_numpy(), s)
692     fitted[g.index.to_numpy()] = p
693     bs_per_mat.append(dict(expiry=expiry.strftime("%b %y"), n=len(g),
694                           ttm=float(g["ttm"].iloc[0]), sigma=s,
695                           **measures(g["mid"].to_numpy(), p)))
696
697 # Ordered by maturity, this is the term structure reported in the results.
698 bs_per_mat = pd.DataFrame(bs_per_mat).sort_values("ttm")
699 print("\nBlack-Scholes per maturity:")
700 print(bs_per_mat.round(4).to_string(index=False))
701 print(" pooled:", measures(MID, fitted))
702
703 # %% =====
704 # 3. LEVY MODELS ACROSS THE CHAIN
705
706
707 #Now the levy models
708 def meixner_density(x, a, b, d, m):
709     z = (x - m) / a
710     log_c = 2 * d * np.log(2 * np.cos(b / 2)) - np.log(2 * a * np.pi) - gammaln(2 * d)
711     return np.exp(np.clip(log_c + b * z + 2 * np.real(loggamma(d + 1j * z)),
712                          -745, 700))
713
714 #This gets the meixner batch correctly, we manage to do this because it has the closed
715 #form density
716 #This one was harder, so took many attempts and solution from AI
717 #The others are more consistent cause I can use the Carr-Madan approach
718 def meixner_batch(K, T, r, q, a, b, d, S0=None, n=1200):
719     K = np.asarray(K, float); T = np.asarray(T, float)
720     m = r - q - 2 * d * (np.log(np.cos(b / 2)) - np.log(np.cos((a + b) / 2)))
721     spot = S0 if S0 is None else S0_
722     lo = np.log(K / spot); hi = lo + 12 * a * np.sqrt(d * T) + 3

```

```

723 t = np.linspace(0.0, 1.0, n)[None, :]
724 x = lo[:, None] + (hi - lo)[:, None] * t
725 dens = meixner_density(x, a, b, (d * T)[:, None], (m * T)[:, None])
726 return np.exp(-r * T) * np.trapezoid((spot * np.exp(x) - K[:, None]) * dens,
727                                     x, axis=1)
728
729
730 # Admissibility conditions, enforced by returning a penalty rather than by a
731 # constrained solver.
732 def ok_vg(p): return p[0] > 1e-8 and p[1] > 1e-8 and p[2] > 1.0
733 def ok_cgmy(p): return p[0] > 1e-8 and p[1] > 1e-8 and p[2] > 1.0 and p[3] < 2
734 def ok_nig(p): return p[0] > 1e-8 and abs(p[1]) < p[0] and p[2] > 1e-8
735 def ok_gh(p): return p[0] > 1e-8 and abs(p[1]) < p[0] and p[2] > 1e-8
736 def ok_mx(p): return (p[0] > 1e-4 and p[2] > 1e-4 and abs(p[1]) < np.pi
737                     and abs(p[0] + p[1]) < np.pi)
738
739
740 # Starting points
741 LEVY = {
742     "Meixner": dict(price=meixner_batch, ok=ok_mx, starts=[
743         [0.10, -0.10, 1.00], [0.50, -1.00, 0.50], [0.20, 0.50, 2.00],
744         [1.00, -2.00, 0.20], [0.05, -0.50, 3.00], [0.30, -1.50, 0.35],
745         [0.80, 0.00, 1.00], [1.50, -1.00, 0.50], [2.00, -2.50, 0.30],
746         [1.00, -0.50, 2.00]]),
747     "VG": dict(price=carr_madan_batch_vg, ok=ok_vg, starts=[
748         [1.5, 6.0, 15.0], [5.0, 20.0, 30.0], [0.5, 2.0, 5.0],
749         [20.0, 40.0, 60.0], [1.0, 0.3, 1.6], [50.0, 90.0, 150.0]]),
750     "CGMY": dict(price=carr_madan_batch, ok=ok_cgmy, starts=[
751         [0.02, 0.08, 7.6, 1.29], [1.0, 5.0, 15.0, 0.5],
752         [3.0, 60.0, 130.0, 1.36], [0.5, 2.0, 8.0, 1.0],
753         [0.1, 1.0, 20.0, 1.5], [10.0, 30.0, 60.0, 0.2]]),
754     "NIG": dict(price=carr_madan_batch_nig, ok=ok_nig, starts=[
755         [6.2, -3.9, 0.16], [20.0, -5.0, 0.5], [2.0, -0.5, 0.3],
756         [50.0, -20.0, 1.0], [100.0, 10.0, 50.0], [10.0, 0.0, 0.2]]),
757     "GH": dict(price=carr_madan_batch_gh, ok=ok_gh, starts=[
758         [3.8, -3.8, 0.24, -1.76], [10.0, -3.0, 0.5, -2.0],
759         [36.5, 3.1, 18.8, -83.7], [5.0, 0.0, 1.0, -1.0],
760         [20.0, -10.0, 2.0, -5.0], [2.0, -1.0, 0.3, 1.0]]),
761 }
762
763 #Now fit each levy model
764 def fit_levy(spec, o, spot=None, maxiter=4000):
765
766     #Define spot price
767     spot = S0 if spot is None else spot
768
769     # Extracted once, outside the objective, since the optimiser evaluates it
770     # thousands of times.
771     k, t = o["strike"].to_numpy(), o["ttm"].to_numpy()
772     r, q, mid = (o["rate"].to_numpy(), o["div_yld"].to_numpy(),
773                 o["mid"].to_numpy())
774
775     # The objective
776     def obj(p):
777         if not spec["ok"](p):
778             return 1e6
779         try:
780             m = spec["price"](k, t, r, q, *p, spot)
781         except Exception:
782             return 1e6
783         if not np.all(np.isfinite(m)):
784             return 1e6
785         return float(np.sqrt(np.mean((m - mid) ** 2)))
786
787     # Every start is run to convergence and only the best kept,
788     best = None
789     for x0 in spec["starts"]:
790         res = minimize(obj, x0=np.array(x0, float), method="Nelder-Mead",
791                       options={"xatol": 1e-6, "fatol": 1e-6,
792                               "maxiter": maxiter, "maxfev": maxiter})

```

```

793         if res.fun < 1e5 and (best is None or res.fun < best.fun):
794             best = res
795         return best
796
797
798     print("\nLevy models across the chain:")
799     # The five whole-chain fits. Each parameter vector is kept, since Section 4
800     # reuses it as the leading starting point for the per-maturity calibrations.
801     levy_fits = {}
802     for name, spec in LEVY.items():
803         best = fit_levy(spec, opts)
804         levy_fits[name] = best.x
805         p = spec["price"](K, T, RATE, DIV, *best.x, S0)
806         print(f"    {name:<8} {np.round(best.x, 4)} {measures(MID, p)}")
807
808     # %% =====
809     # 3b. RECOVERED LOG-RETURN DENSITIES
810     # =====
811     # The risk-neutral density of  $\log(S_T/S_0)$  under each whole-chain Levy fit,
812     # recovered from the same characteristic function that priced it. Only the
813     # Meixner has a density in closed form; for the rest the density comes from
814     # Fourier inversion of  $\phi_T(u) = \exp(T \psi_X(u) + i u m T)$ ,
815     #
816     # 
$$f(x) = (1/2\pi) \int_{-\infty}^{\infty} e^{-iux} \phi(u) du$$

817     # 
$$= (1/\pi) \int_0^{\infty} \text{Re}[e^{-iux} \phi(u)] du,$$

818     #
819     # the second form holding because  $f$  is real valued, so  $\phi(-u) = \text{conj}(\phi(u))$ ,
820     # and the two halves of the line contribute conjugate values.
821     #
822     # No figure is produced here. The densities are summarised by their first four
823     # moments, obtained by integrating against the recovered curve.
824
825     #This was somewhat complex and follows closely from the appendix, in the relevant
826     #section
827
828     #Get the drift parameter for the respective levy model
829     def levy_drift(name, params):
830         if name == "Meixner":
831             a, b, d = params
832             return r_ - q_ - 2 * d * (np.log(np.cos(b / 2))
833                                     - np.log(np.cos((a + b) / 2)))
834         return {"VG": m_new_vg, "CGMY": m_new_cgmy, "NIG": m_new_nig,
835                 "GH": m_new_gh}[name](*params, r_, q_)
836
837     #Get the respective log return density, by fourier inversion
838     def logreturn_pdf(x, T_, psi, params, drift, n_u=2 ** 15):
839         u = np.linspace(0.0, 60.0 / (SIGMA_BS * np.sqrt(T_)), n_u)
840         cf = np.exp(T_ * psi(u, *params) + 1j * u * drift * T_)
841         bad = ~np.isfinite(cf)
842         if bad.any():
843             cut = int(np.argmax(bad))
844             u, cf = u[:cut], cf[:cut]
845         integ = np.real(np.exp(-1j * u[None, :] * np.asarray(x)[: , None]) * cf)
846         return np.trapezoid(integ, u, axis=1) / np.pi
847
848     #Now get the moments
849     def density_moments(T_, name, params):
850         drift = levy_drift(name, params)
851         centre = (r_ - q_ - 0.5 * SIGMA_BS ** 2) * T_
852         half = 6 * SIGMA_BS * np.sqrt(T_)
853         x = np.linspace(centre - half, centre + half, 1500)
854
855         #Meixner is unique cause we have the closed form density
856         if name == "Meixner":
857             a, b, d = params
858             f = meixner_density(x, a, b, d * T_, drift * T_)
859         else:
860             f = logreturn_pdf(x, T_, LEVY_PSI[name][0], params, drift)
861

```

```

862 # Normalising by the integrated mass rather than assuming it is one keeps
863 # the moments meaningful when the inversion has only partly converged.
864 mass = np.trapezoid(f, x)
865 mu = np.trapezoid(x * f, x) / mass
866 c2, c3, c4 = (np.trapezoid((x - mu) ** k * f, x) / mass for k in (2, 3, 4))
867 return dict(mass=mass, mean=mu, sd=np.sqrt(c2),
868             skew=c3 / c2 ** 1.5, exkurt=c4 / c2 ** 2 - 3.0)
869
870
871 print("\nRecovered log-return densities, front month and longest maturity:")
872 # The two maturities the density figures are drawn at. The GH fit is included
873 # deliberately: its density is the one that fails to recover, and the mass
874 # column is the diagnostic, since it does not reach one.
875 r_, q_ = R, Q
876 dens_rows = [dict(expiry=lab, model=name,
877                  **density_moments(t_, name, tuple(params)))
878              for lab, t_ in (MATS[0], MATS[-1])
879              for name, params in levy_fits.items()]
880 print(pd.DataFrame(dens_rows).round(6).to_string(index=False))
881
882 # %% =====
883 # 4. THE SAME LEVY MODELS, FITTED MATURITY BY MATURITY
884 # =====
885 # Each model is refitted to the contracts of one expiry at a time. The first
886 # starting point is the whole-chain fit above, which is already in the right
887 # region, should speed it up; three dispersed points guard against it being a poor one
888 # for a single
889 # expiry.
890 print("\nPer-maturity calibrations (RMSE):")
891 # Black-Scholes and all five Levy models, refitted to each expiry in turn.
892 per_mat_rows = []
893
894 #Now go through expiry in a for loop for efficiency
895 for expiry, g in sorted(opts.groupby("expiry_date"),
896                             key=lambda kv: kv[1]["ttm"].iloc[0]):
897     lab = expiry.strftime("%b %y")
898     s = fit_bs(g)
899     p = bs_call(g["strike"].to_numpy(), g["ttm"].to_numpy(),
900                g["rate"].to_numpy(), g["div_yld"].to_numpy(), s)
901     per_mat_rows.append(dict(model="Black-Scholes", expiry=lab, n=len(g),
902                             params=f"sigma={s:.4f}",
903                             **measures(g["mid"].to_numpy(), p)))
904
905     #For each levy item now,
906     for name, spec in LEVY.items():
907         sub = dict(spec)
908         sub["starts"] = [list(levy_fits[name])] + spec["starts"][:3]
909
910         #Run respective fit levy
911         best = fit_levy(sub, g, maxiter=1500)
912         if best is None:
913             continue
914         m = spec["price"](g["strike"].to_numpy(), g["ttm"].to_numpy(),
915                          g["rate"].to_numpy(), g["div_yld"].to_numpy(),
916                          *best.x, S0)
917         per_mat_rows.append(dict(model=name, expiry=lab, n=len(g),
918                                 params=", ".join(f"{v:.4f}" for v in best.x),
919                                 **measures(g["mid"].to_numpy(), m)))
920
921 # RMSE by model and expiry: the per-maturity table of the results.
922 per_mat = pd.DataFrame(per_mat_rows)
923 print(per_mat.pivot(index="model", columns="expiry", values="RMSE")
924       .reindex(["Black-Scholes"] + list(LEVY))[m for m, _ in MATS]
925       .round(4).to_string())
926
927 # %% =====
928 # 5. BNS
929 # =====
930 # The first start of each is Schoutens' Table 7.1 estimate, the rest dispersed
931 # around it with a wider spread in sigma_0^2 for the higher-volatility chain.

```

```

931 BNS_STARTS = {
932     "Gamma-OU": [(-1.2606, 0.5783, 1.4338, 11.6641, 0.01450),
933                  (-1.0, 1.0, 1.5, 10.0, 0.05), (-2.0, 2.0, 1.0, 5.0, 0.20),
934                  (-0.5, 5.0, 2.0, 20.0, 0.40), (-5.0, 8.0, 3.0, 30.0, 0.65),
935                  (-0.2, 0.5, 0.5, 2.0, 0.70), (-10.0, 10.0, 5.0, 50.0, 0.80),
936                  (-1.0, 3.0, 0.3, 1.0, 0.30)],
937     "IG-OU": [(-0.1926, 0.0636, 6.2410, 0.7995, 0.01560),
938               (-0.5, 1.0, 3.0, 1.0, 0.05), (-1.0, 3.0, 1.0, 2.0, 0.30),
939               (-0.3, 8.0, 0.1, 2.0, 0.85), (-2.0, 5.0, 0.5, 1.5, 0.50),
940               (-0.1, 0.5, 5.0, 0.8, 0.20), (-3.0, 10.0, 0.05, 3.0, 0.75),
941               (-0.7, 2.0, 2.0, 0.5, 0.40)],
942 }
943 # Each model paired with its objective and its pricer.
944 BNS_MODELS = {"Gamma-OU": (rmse_objective_bns, carr_madan_batch_bns),
945               "IG-OU": (rmse_objective_ig, carr_madan_batch_ig)}
946
947 print("\nBNS models:")
948 # Both models fitted from all eight starts.
949 bns_fits, bns_starts_rows = {}, []
950
951 #Go through each specific model in BNS models - note we called the function in that term
952 for name, (obj, pricer) in BNS_MODELS.items():
953     best = None
954
955     #Find the best
956     for i, x0 in enumerate(BNS_STARTS[name], 1):
957         res = minimize(obj, x0=np.array(x0, float),
958                       args=(K, T, RATE, DIV, MID, S0), method="Nelder-Mead",
959                       options={"xatol": 1e-6, "fatol": 1e-6,
960                               "maxiter": 4000, "maxfev": 4000})
961
962         #Adding in start rows
963         bns_starts_rows.append(dict(model=name, start=i,
964                                     **dict(zip(["rho", "lam", "a", "b", "s0"],
965                                                 res.x)), RMSE=res.fun))
966
967         #Now find the best
968         if best is None or res.fun < best.fun:
969             best = res
970
971         #we remembered to save the fits here
972         bns_fits[name] = best.x
973         p = pricer(K, T, RATE, DIV, *best.x, S0)
974         print(f"    BNS {name:<9} {np.round(best.x, 4)} {measures(MID, p)}")
975
976 # The full set of terminal vectors, showing how many starts agree on the
977 # optimum and how far the rest drift from it.
978 print("\n    terminal values from every start:")
979 print(pd.DataFrame(bns_starts_rows).round(4).to_string(index=False))
980
981
982
983
984 # %% =====
985 # 6. STOCHASTIC TIME CHANGE
986 # =====
987 # The twelve combinations of four Levy processes with three clocks.
988 MEIXNER_STARTS = [
989     [0.1231, -0.5875, 3.3588], [0.1108, -0.9858, 3.6288],
990     [0.0890, -1.1323, 5.0262], [0.3, -0.5, 1.5],
991     [0.05, -1.5, 8.0], [0.2, -0.8, 2.5],
992     [0.15, -2.0, 4.0], [0.5, -0.3, 1.0],
993 ]
994 VG_STARTS = [
995     [11.9896, 25.8523, 35.5344], [11.4838, 23.2880, 40.1291],
996     [14.9248, 26.1529, 50.4425], [5.0, 10.0, 20.0],
997     [20.0, 40.0, 60.0], [1.0, 5.0, 10.0],
998     [8.0, 15.0, 30.0], [30.0, 50.0, 80.0],
999 ]
1000 CGMY_STARTS = [

```

```

1001     [0.0074, 0.1025, 11.3940, 1.6765], [0.0415, 3.9134, 30.6322, 1.3664],
1002     [0.0672, 6.1316, 44.7448, 1.2911], [0.5, 5.0, 15.0, 0.5],
1003     [0.01, 1.0, 5.0, 1.0], [0.1, 2.0, 20.0, 1.5],
1004     [0.005, 0.5, 8.0, 1.8], [0.2, 8.0, 35.0, 0.8],
1005 ]
1006 NIG_STARTS = [
1007     [18.4815, -4.8412, 0.4685], [29.4722, -15.9048, 0.5071],
1008     [29.1553, -13.9331, 0.5600], [10.0, -2.0, 1.0],
1009     [40.0, -20.0, 0.3], [15.0, -8.0, 0.7],
1010     [50.0, -25.0, 0.4], [8.0, -1.0, 1.5],
1011 ]
1012
1013 # Clock starting points, shared across the Levy families.
1014 TC_STARTS = {
1015     "CIR": [[0.5391, 1.5746, 1.8772], [0.5145, 0.9029, 1.3750],
1016             [0.6020, 1.5560, 1.9992], [0.3881, 1.4012, 1.3612],
1017             [0.6, 1.5, 1.0], [1.0, 1.0, 1.0],
1018             [0.4, 1.4, 1.36], [1.5, 0.8, 2.5]],
1019     "Gamma-OU": [[0.6252, 0.4239, 0.5962], [1.2517, 0.5841, 0.6282],
1020                  [0.8826, 0.5945, 0.8524], [1.0740, 0.3573, 0.6143],
1021                  [1.1729, 0.5914, 0.6558], [1.0, 1.0, 1.0],
1022                  [0.5, 0.3, 0.4], [2.0, 0.8, 1.2]],
1023     "IG-OU": [[1.1559, 0.6496, 0.8572], [1.2801, 0.6615, 0.8104],
1024               [1.0622, 0.6092, 0.9999], [1.2190, 0.6564, 0.8266],
1025               [1.1315, 0.8993, 0.6168], [1.0, 1.0, 1.0],
1026               [0.6, 0.4, 0.5], [2.0, 1.0, 1.5]],
1027 }
1028
1029 # The starting points keyed by Levy process, so the loop below can pair each
1030 # with every clock.
1031 SV_STARTS = {"Meixner": MEIXNER_STARTS, "VG": VG_STARTS,
1032              "CGMY": CGMY_STARTS, "NIG": NIG_STARTS}
1033
1034 # All eight starts of each list are used
1035 N_SV_STARTS = 8
1036
1037 #Now calibrate the stochastic volatility models
1038 def calibrate_sv(chain_tag, o, spot, levy_name, tc_name, levy_starts,
1039                 n_starts=N_SV_STARTS, maxiter=3000):
1040
1041     #Set the basics again
1042     k = o["strike"].to_numpy()
1043     t_ = o["ttm"].to_numpy()
1044     r = o["rate"].to_numpy()
1045     q = o["div_yld"].to_numpy()
1046     mid = o["mid"].to_numpy()
1047
1048     #State the objective functions
1049     objective, pnames = make_objective(levy_name, tc_name, k, t_, r, q, mid, spot)
1050
1051     #Denote the starts
1052     starts = [lp + tp for lp, tp
1053               in zip(levy_starts, TC_STARTS[tc_name])][:n_starts]
1054
1055     t0 = time.time()
1056
1057     #Now noting the relevant objective function, minimise it
1058     results = [minimize(objective, x0=np.array(x0, dtype=float),
1059                        method="Nelder-Mead",
1060                        options={"xatol": 1e-5, "fatol": 1e-5,
1061                                "maxiter": maxiter, "maxfev": maxiter})
1062                for x0 in starts]
1063
1064     #And get the best results
1065     results.sort(key=lambda res: res.fun)
1066     best = results[0]
1067
1068     #How many agree?
1069     n_agree = sum(1 for res in results if res.fun < best.fun * 1.01)
1070

```



```

1071 # The fitted prices, and the error measures taken from them.
1072 psi, _ = LEVY_PSI[levy_name]
1073 varphi, _ = TIME_CHANGE[tc_name]
1074 n_levy = len(LEVY_PSI[levy_name][1])
1075 lp, tp = tuple(best.x[:n_levy]), tuple(best.x[n_levy:])
1076 model = carr_madan_levy_sv(k, t_, r, q, psi, lp, varphi, tp, spot)
1077
1078 #Some extra stats from the chain - get RMSE
1079 stats = dict(chain=chain_tag, levy=levy_name, time_change=tc_name,
1080             n_contracts=len(o), n_params=len(best.x),
1081             **measures(mid, model),
1082             n_agree=n_agree, n_starts=len(starts),
1083             seconds=time.time() - t0)
1084 stats["RMSE"] = best.fun
1085 stats.update(dict(zip(pnames, best.x)))
1086 return stats
1087
1088
1089 # Every Levy process against every clock: four times three fits.
1090 #Now do it ALL
1091 sv_rows = []
1092 for levy_name, starts in SV_STARTS.items():
1093     for clock in ("CIR", "Gamma-OU", "IG-OU"):
1094         stats = calibrate_sv("k200", opts, S0, levy_name, clock, starts)
1095         sv_rows.append(stats)
1096         print(f" {levy_name}-{clock}: RMSE {stats['RMSE']:.4f}")
1097
1098 # Parameters and error measures for all twelve, the goodness-of-fit table.
1099 sv = pd.DataFrame(sv_rows)
1100
1101 # %% =====
1102 # 7. MOMENTS OF THE STOCHASTIC VOLATILITY FITS
1103 # =====
1104 # Variance and excess kurtosis of the risk-neutral log return, read off the
1105 # characteristic function used for pricing.
1106
1107 #Testing Ladder
1108 H_LADDER = [3e-3, 5e-3, 1e-2, 2e-2, 3e-2, 5e-2, 8e-2, 1.2e-1]
1109
1110 #Tolerance
1111 TOL = 0.02
1112
1113 #Define the cumulants
1114 def cumulants(T_, cf, h):
1115     K0 = 0.0 + 0.0j
1116     k2 = -np.real((cf(h) - 2 * K0 + cf(-h)) / h ** 2)
1117     k4 = np.real((cf(2 * h) - 4 * cf(h) + 6 * K0
1118                 - 4 * cf(-h) + cf(-2 * h)) / h ** 4)
1119     return k2, k4
1120
1121 #Now define the actual moments for the stochastic volatility models using those
1122 #cumulants
1123 def sv_moments(T_, levy_name, clock, lp, tp):
1124     # The Levy exponent and the clock's Laplace exponent for this pairing.
1125
1126     #Get the characteristic and characteristic exponent functions
1127     psi, _ = LEVY_PSI[levy_name]
1128     varphi, _ = TIME_CHANGE[clock]
1129
1130     # The cumulant generating function of the log return. (Recentered)
1131     def logphi(u):
1132         v = levy_sv_char_fn(np.array([u], dtype=complex), T_, R, Q,
1133                             psi, lp, varphi, tp, S0)[0]
1134         return np.log(v * np.exp(-1j * u * np.log(S0)))
1135
1136     # One estimate per step size, skipping any that fail outright.
1137     est = []
1138     for h in H_LADDER:
1139         try:
1140             k2, k4 = cumulants(T_, logphi, h)

```

```

1140     except Exception:
1141         continue
1142     if k2 > 0:
1143         est.append((np.sqrt(k2), k4 / k2 ** 2))
1144
1145     # The first pair of consecutive steps to agree. Returning NaN when no pair
1146     def settle(i):
1147         for j in range(len(est) - 1):
1148             a, b = est[j][i], est[j + 1][i]
1149             if abs(a - b) <= TOL * max(1.0, abs(a)):
1150                 return a
1151         return np.nan
1152
1153     return settle(0), settle(1)
1154
1155
1156 # Both moments for all twelve models at all seven maturities.
1157 mom_rows = []
1158 for _, r in sv.iterrows():
1159     levy_name, clock = r["levy"], r["time_change"]
1160     lp = [r[c] for c in LEVY_PSI[levy_name][1]]
1161     tp = [r[c] for c in TIME_CHANGE[clock][1]]
1162     for lab, t_ in MATS:
1163         sd, ek = sv_moments(t_, levy_name, clock, lp, tp)
1164         mom_rows.append(dict(model=f"{levy_name}-{clock}", expiry=lab,
1165                             ttm=t_, sd=sd, exkurt=ek, rmse=r["RMSE"]))
1166
1167 # Excess kurtosis by maturity, the kurtosis table of the results.
1168 moments = pd.DataFrame(mom_rows)
1169 print(moments.pivot(index="expiry", columns="model", values="exkurt")
1170       .reindex([m for m, _ in MATS]).round(3).to_string())
1171
1172 # %% =====
1173 # 8. IMPLIED VOLATILITY TERM STRUCTURE
1174 # =====
1175 # The annualised volatility sqrt(Var(log(S_T/S_0)) / T) of each fit
1176
1177 #Define volatility moments that we need
1178 vol = moments.assign(vol=lambda d: d["sd"] / np.sqrt(d["ttm"]))
1179
1180 #Print
1181 print(vol.pivot(index="expiry", columns="model", values="vol")
1182       .reindex([m for m, _ in MATS]).round(4).to_string())
1183
1184 #Now actually get the levy unit moments
1185 def levy_unit_moments(name, p):
1186     if name == "Meixner":
1187         a, b, d = p
1188         return a * d * np.tan(b / 2), a ** 2 * d / (2 * np.cos(b / 2) ** 2)
1189     if name == "VG":
1190         C, G, M = p
1191         return C * (G - M) / (M * G), C * (G ** 2 + M ** 2) / (M * G) ** 2
1192     if name == "NIG":
1193         a, b, d = p
1194         return (d * b / np.sqrt(a ** 2 - b ** 2),
1195                 a ** 2 * d * (a ** 2 - b ** 2) ** -1.5)
1196     if name == "CGMY":
1197         C, G, M, Y = p
1198         if Y >= 2:
1199             return np.nan, np.nan
1200         return (C * (M ** (Y - 1) - G ** (Y - 1)) * gamma_fn(1 - Y),
1201                 C * (M ** (Y - 2) + G ** (Y - 2)) * gamma_fn(2 - Y))
1202     raise ValueError(name)
1203
1204
1205 # The two ends of each term structure. sigma(0) is fixed by the Levy process
1206 # alone; sigma(inf) adds the variance the clock itself contributes, through
1207 # the stationary rate of time change plus the (E[X_1])^2 Var(Y_T) term of the
1208 # law of total variance. The half-life is log(2) over the reversion rate.
1209 print("\nShort- and long-maturity limits:")

```

```

1210 for _, r in sv.iterrows():
1211     levy_name, clock = r["levy"], r["time_change"]
1212     EX, VX = levy_unit_moments(levy_name, [r[c] for c in LEVY_PSI[levy_name][1]])
1213     if not np.isfinite(VX):
1214         print(f" {levy_name}-{clock}: variance not finite")
1215         continue
1216     # eta plays the role of a/b and kappa that of lambda, so the two branches
1217     # are the same formula with the clock's own parameter names.
1218     if clock == "CIR":
1219         k, eta, lam = r["kappa"], r["eta"], r["lam"]
1220         ratio = eta * (1 + (EX ** 2 / VX) * lam ** 2 / k ** 2)
1221         rate = k
1222     else:
1223         lam, a, b = r["lam"], r["a"], r["b"]
1224         power = 2 if clock == "Gamma-OU" else 3
1225         ratio = a / b + (EX ** 2 / VX) * (2 * a / (lam * b ** power))
1226         rate = lam
1227     print(f" {levy_name}-{clock:<9} sigma(0) {np.sqrt(VX):.4f} "
1228           f"sigma(inf) {np.sqrt(VX * ratio):.4f} "
1229           f"half-life {365 * np.log(2) / rate:.1f} days")
1230
1231 # %% =====
1232 # 9. VALIDATION ON THE S&P 500 (2002) CHAIN
1233 # This was all about comparing to schoutens data
1234
1235 # Load his SP500 chain - I have it saved
1236 def load_sp500():
1237     o = pd.read_excel("Data.xlsx", sheet_name="Schoutens Data", skiprows=7)
1238     return o.reset_index(drop=True), float(o["spot"].iloc[0])
1239
1240
1241 # The same columns as the KOSPI chain, so every routine below is called
1242 # unchanged - using the functions we already defined
1243 sp, SP_S0 = load_sp500()
1244 spK, spT = sp["strike"].to_numpy(), sp["ttm"].to_numpy()
1245 spR, spQ = sp["rate"].to_numpy(), sp["div_yld"].to_numpy()
1246 spMID = sp["mid"].to_numpy()
1247
1248 # Black-Scholes.
1249 sp_sigma = fit_bs(sp, SP_S0, sigma0=SP_IV_SEED)
1250 sp_bs = bs_call(spK, spT, spR, spQ, sp_sigma, S0=SP_S0)
1251 print(f"\nS&P 500 Black-Scholes: sigma = {sp_sigma:.4f}")
1252 print(" ", measures(spMID, sp_bs))
1253
1254 # Levy models. The starting points are Schoutens' Table 6.1 estimates, since
1255 # the object here is to recover his optimum rather than to search widely.
1256 SP500_LEVY_STARTS = {
1257     "Meixner": [0.3977, -1.4940, 0.3462],
1258     "VG": [1.3, 6.0, 15.0],
1259     "CGMY": [0.0244, 0.0765, 7.5515, 1.2945],
1260     "NIG": [6.1882, -3.8941, 0.1622],
1261     "GH": [1.0, -5.0, 0.5, -5.0],
1262 }
1263
1264 # The five Levy models, fitted from Schoutens' own estimate plus three of the
1265 # dispersed starts as a check that his optimum is the better one.
1266 print("\nS&P 500 Levy models:")
1267 sp_levy_fits = {}
1268 for name, spec in LEVY.items():
1269     sub = dict(spec)
1270     sub["starts"] = [SP500_LEVY_STARTS[name]] + spec["starts"][:3]
1271     best = fit_levy(sub, sp, SP_S0)
1272     if best is None:
1273         print(f" {name:<8} no admissible fit")
1274         continue
1275     sp_levy_fits[name] = best.x
1276     p = spec["price"](spK, spT, spR, spQ, *best.x, SP_S0)
1277     print(f" {name:<8} {np.round(best.x, 4)} {measures(spMID, p)}")
1278
1279 # BNS, from the same starting points as Section 5.

```

```

1280 print("\nS&P 500 BNS models:")
1281 for name, (obj, pricer) in BNS_MODELS.items():
1282     best = None
1283     for x0 in BNS_STARTS[name]:
1284         res = minimize(obj, x0=np.array(x0, float),
1285                        args=(spK, spT, spR, spQ, spMID, SP_S0),
1286                        method="Nelder-Mead",
1287                        options={"xatol": 1e-6, "fatol": 1e-6,
1288                               "maxiter": 4000, "maxfev": 4000})
1289         if best is None or res.fun < best.fun:
1290             best = res
1291     p = pricer(spK, spT, spR, spQ, *best.x, SP_S0)
1292     print(f"    BNS {name:<9} {np.round(best.x, 4)} {measures(spMID, p)}")
1293
1294 # The twelve time-changed models, through the same runner as Section 6.
1295 print("\nS&P 500 stochastic time-change models:")
1296 for levy_name, starts in SV_STARTS.items():
1297     for clock in ("CIR", "Gamma-OU", "IG-OU"):
1298         stats = calibrate_sv("sp500", sp, SP_S0, levy_name, clock, starts)
1299         print(f"    {levy_name}-{clock}: RMSE {stats['RMSE']:.4f}")

```

Listing 1: Every numerical result in the dissertation, in a single script.

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